



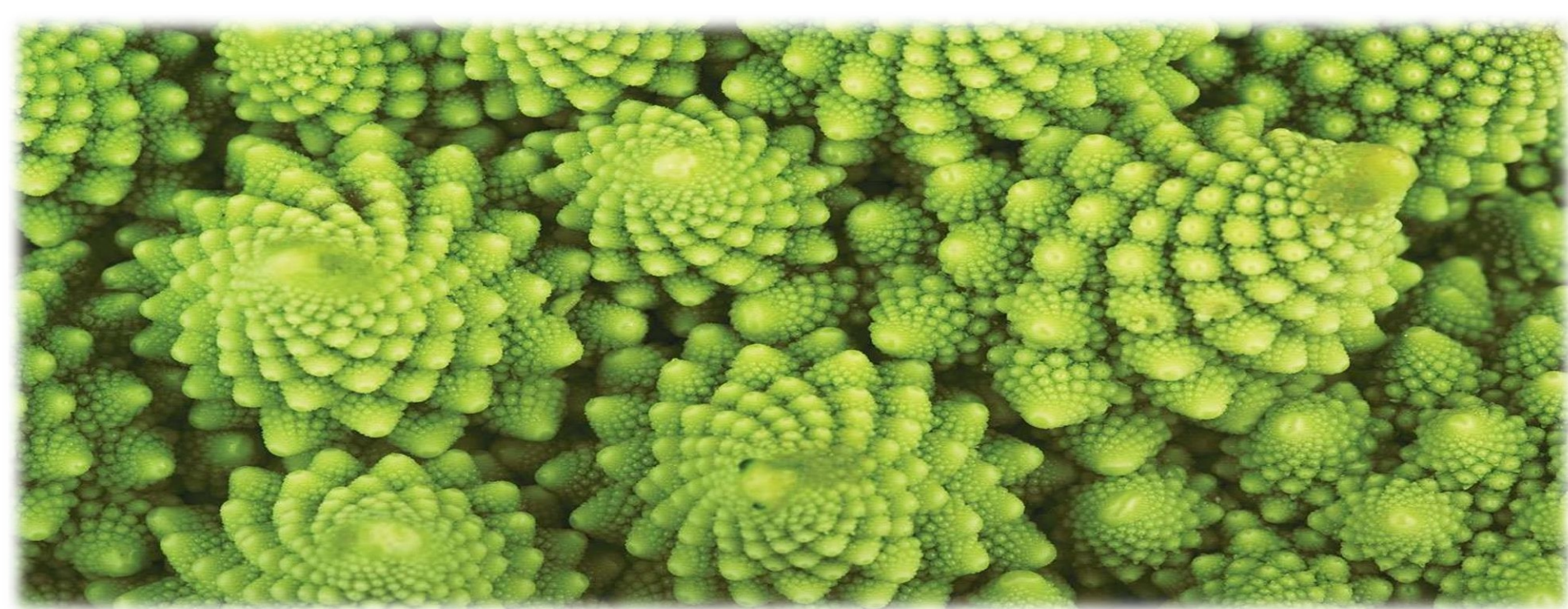
PLANT ANATOMY & PHYSIOLOGY

By

Dr. Mona Fatin Syazwanee binti Mohamed Ghazali

Are plant computers?

- A head of Romanesco, an edible relative of broccoli.
- Its smaller buds resembles in miniature of the entire vegetable.
- It looks as if it is generated by a computer because the growth pattern follows a repetitive sequence of instructions – subjected to natural selection.



- No two plants are alike
- To some people
 - The fanwort (*Cabomba caroliniana*) is an **intrusive weed**, but to others it is an **attractive aquarium plant**
- This plant exhibits plasticity
 - The ability to alter itself in response to its environment





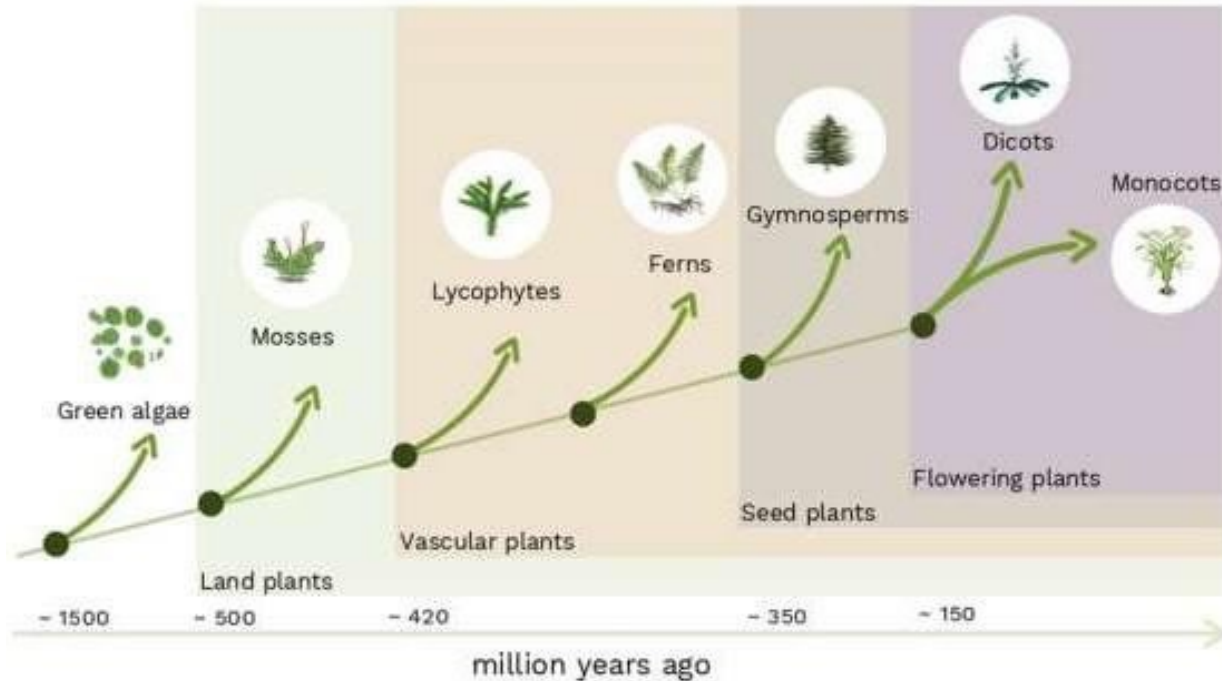
Just drop the stems in the tank and the stems will float near the water surface. Because the stems are closer to the light source, they can grow faster floating as opposed to planted.



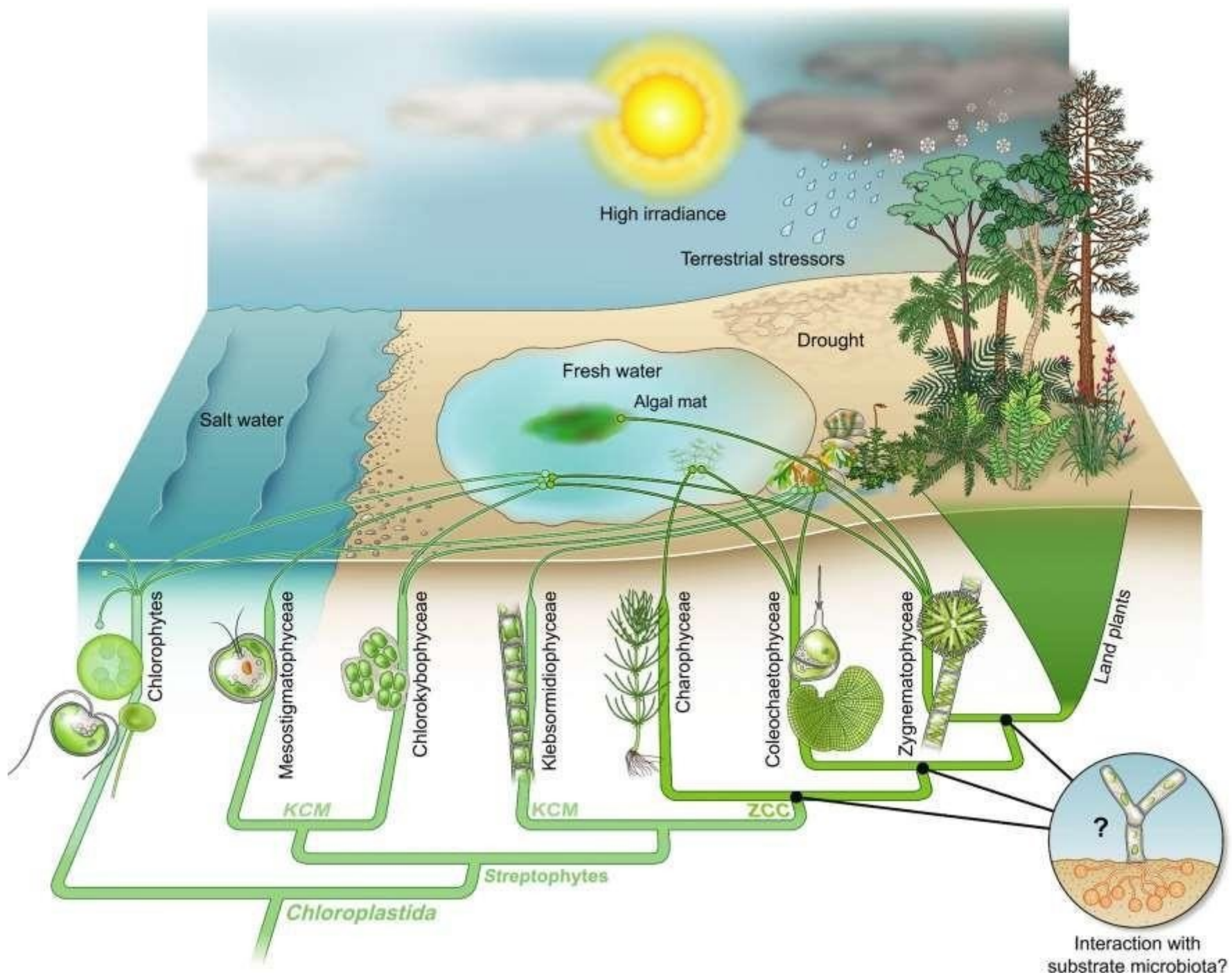
Michigan inland lake infested with *Cabomba caroliniana* Carolina fanwort (in foreground).

Photo courtesy of Jane Herbert.

Hierarchy of organs, tissues and cells

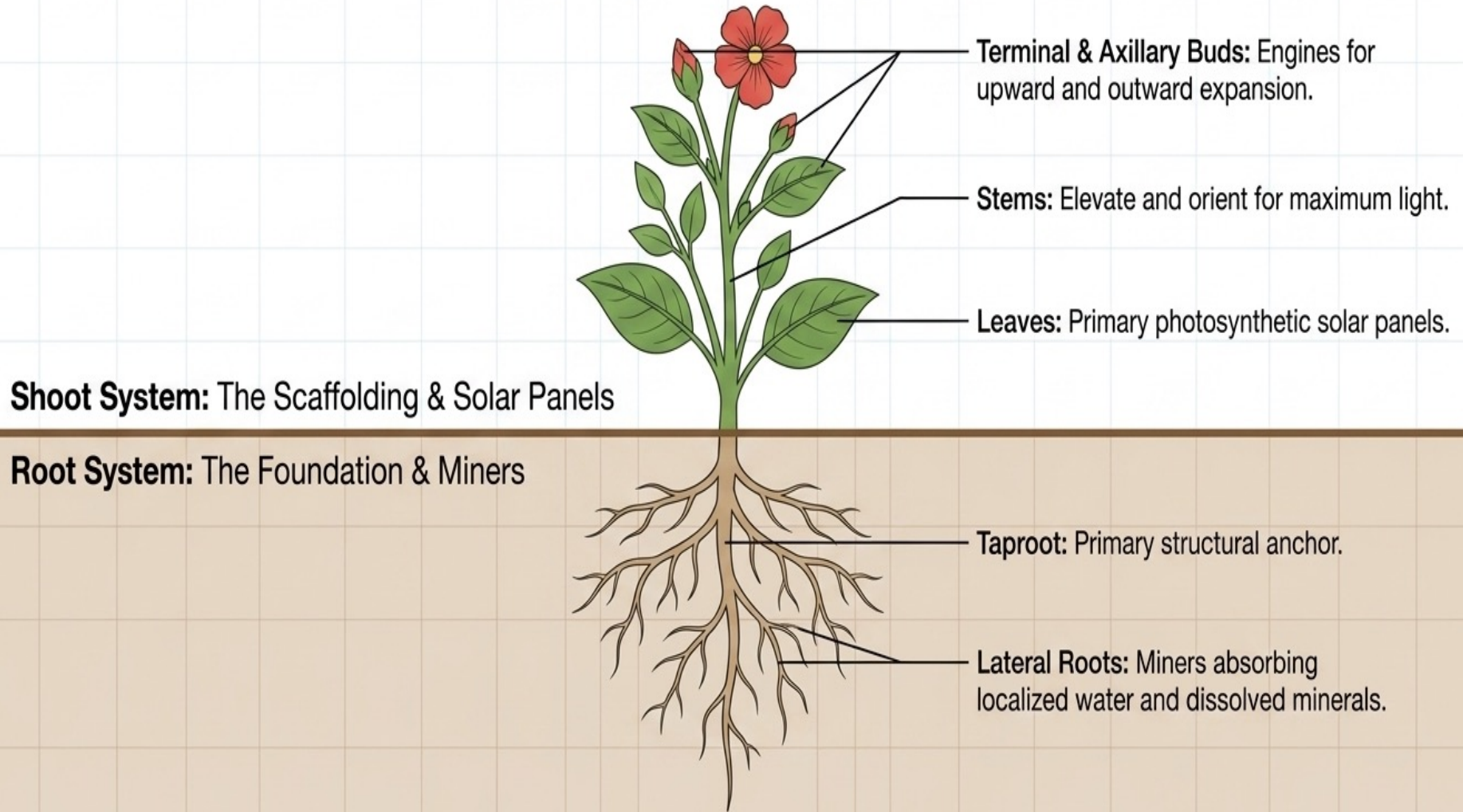


- Plants, like multicellular animals
 - Have **organs** composed of different **tissues**, which are in turn composed of **cells**
- The basic morphology of **vascular plants**
 - Reflects their evolutionary history as terrestrial organisms that draw nutrients from two very different environments:
 - below-ground and above-ground



The Modular Machine: Core Architecture

Conquering two distinct environments: dry air and dark soil.



The Organ Modification Matrix

Form follows function: reprogramming baseline organs for environmental survival.

Roots



Pneumatophores:
Snorkels for aeration in waterlogged soil.



Storage Roots:
Carbohydrate reserves (e.g., carrots, sweet potatoes).

Stems



Tubers:
Enlarged underground rhizomes for food storage.



Rhizomes:
Horizontal underground stems for propagation.

Leaves



Tendrils:
Modified leaves for clinging and structural support.



Spines:
Defense and extreme water retention.

Roots

- A root
 - Is an **organ** that **anchors** the vascular plant
 - Absorbs **minerals and water**
 - Often **stores organic nutrients (starchy roots and tubers)**



Roots

- In most plants
 - The absorption of water and minerals occurs near the **root tips**, where vast numbers of **tiny root hairs increase the surface area** of the root



Modified Roots



(a) Props roots

- Modified adventitious roots that support the aerial branchial species (**grow downward**). The prop roots are aerial roots that are found above the ground.



(b) Storage roots

- Primarily function to store carbohydrates **underground** as reserves for perennial species.



(c) “Strangling” aerial roots

- A type of roots where it **grows from the stem** of the plant, i.e. above the ground. This type of root absorbs water directly from the air.



(d) Buttress roots

- Roots that **grows out of the tree trunk**. Large, wide roots on all sides of a shallowly rooted tree. Found in nutrient-poor tropical forest soils

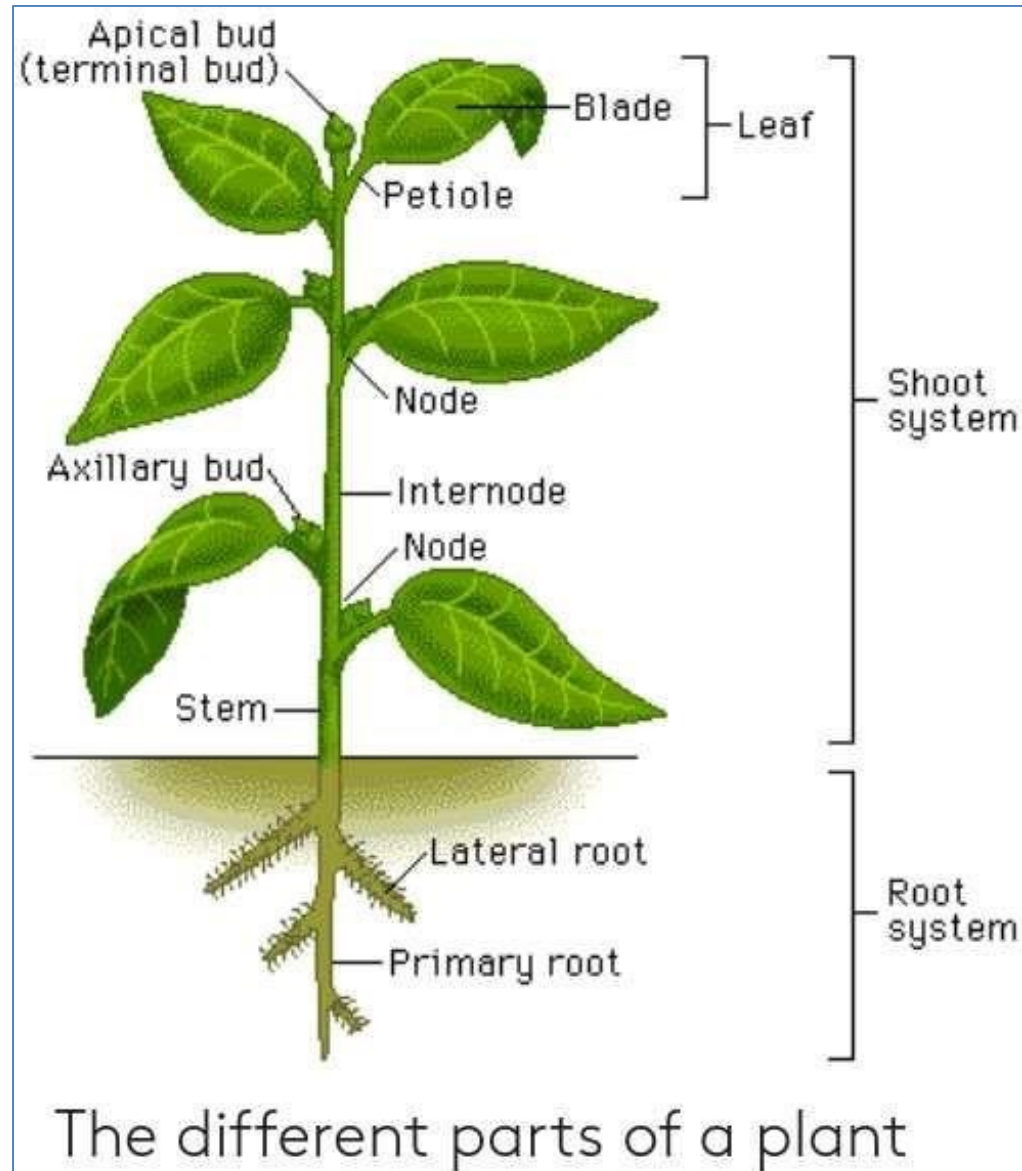


(e) Pneumatophores

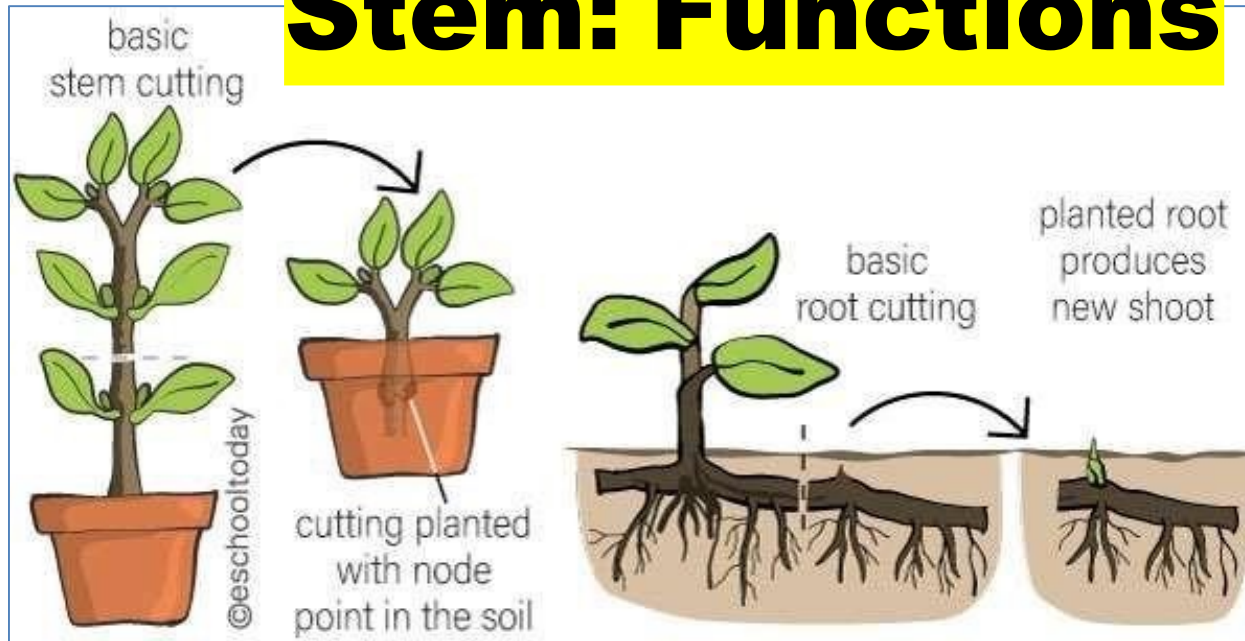
- Specialized root structure that **grow out from the water surface** and facilitate the aeration necessary for root respiration in hydrophytic trees.

Stem: Structure

- A stem is an **organ** consisting of
 - An alternating system of nodes, the points at which **leaves are attached**
 - **Internodes**, the stem segments between nodes
- An **axillary bud**
 - Is a structure that has the potential to form a **lateral shoot**, or **branch**
- A **terminal/apical bud**
 - Is located near the **shoot tip** and causes **elongation of a young shoot**



Stem: Functions



Elongate and orient the shoot to **maximize photosynthesis**

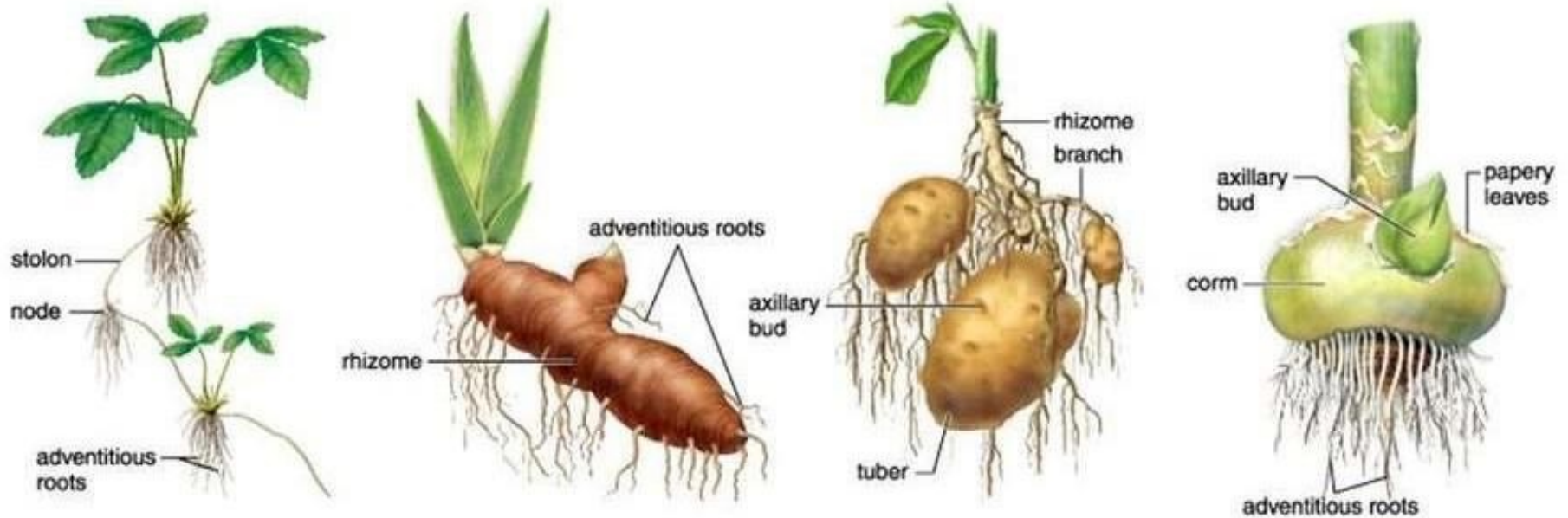
Elevate **reproductive structures**

Modified stem

- **Above ground, underground, or aerial**, enable plants to survive in particular habitats and environments.
- **Food storage**
- **Asexual** reproduction
- **Bulbs, corms, rhizomes, tubers**

Modified stems - examples

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a. Stolon



b. Rhizome



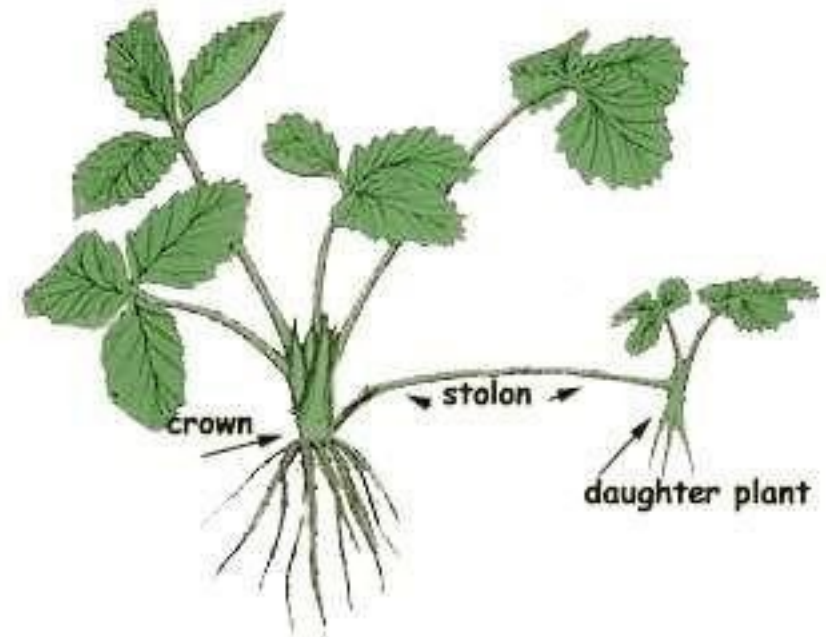
c. Tuber



d. Corm

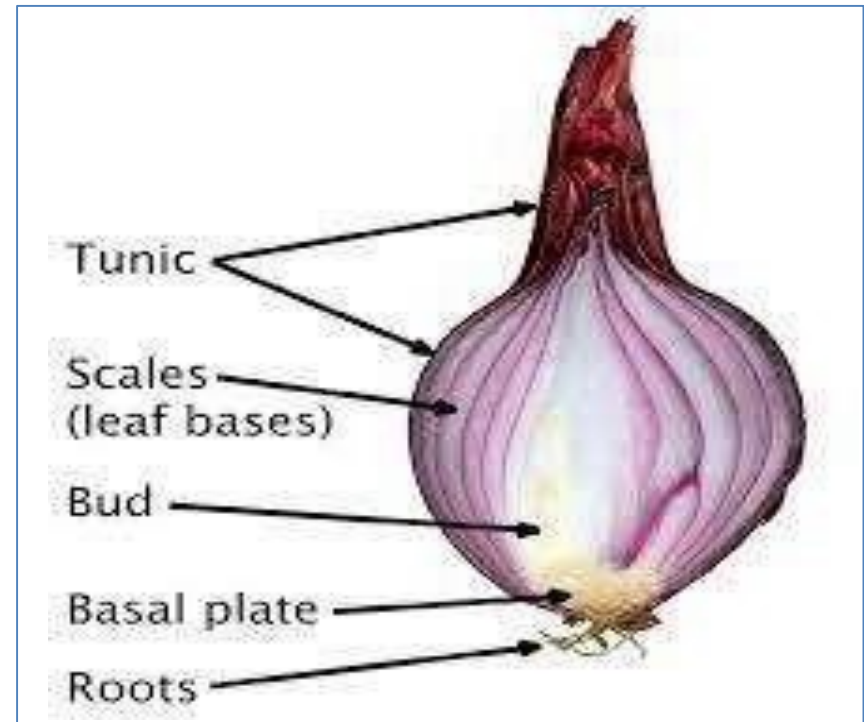
(a) Stolon

- Stolon are **horizontal stems** that grow along the **surface**.
- These “runners” enable a plant to reproduce **asexually**, as **plantlets** form at nodes along each runner.



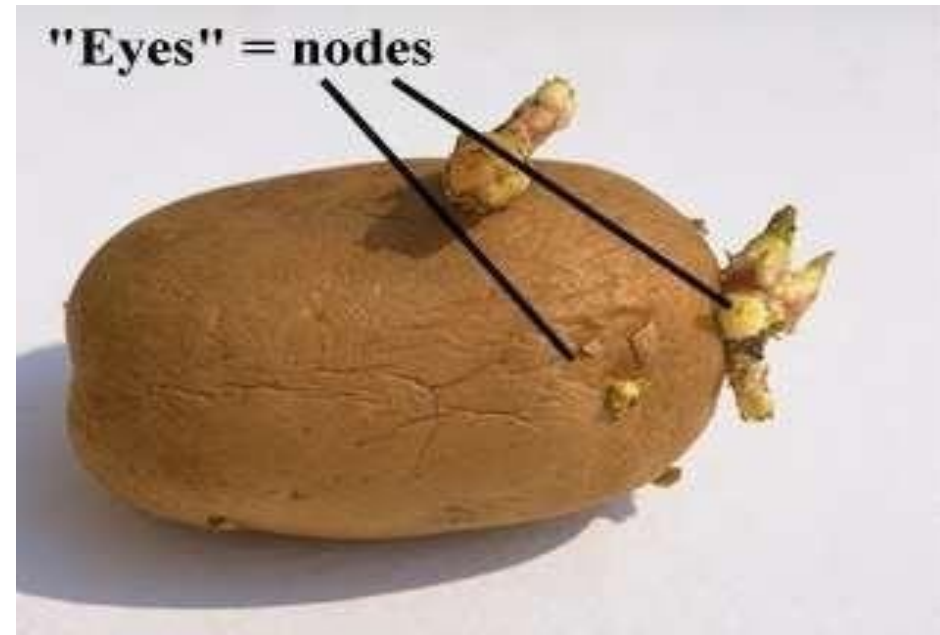
(b) Bulbs

- Are **vertical underground shoots** consisting mostly of the enlarged bases of leaves that **store food**.
- Many **layers of modified leaves** attached to the short stem by slicing an onion bulb lengthwise.



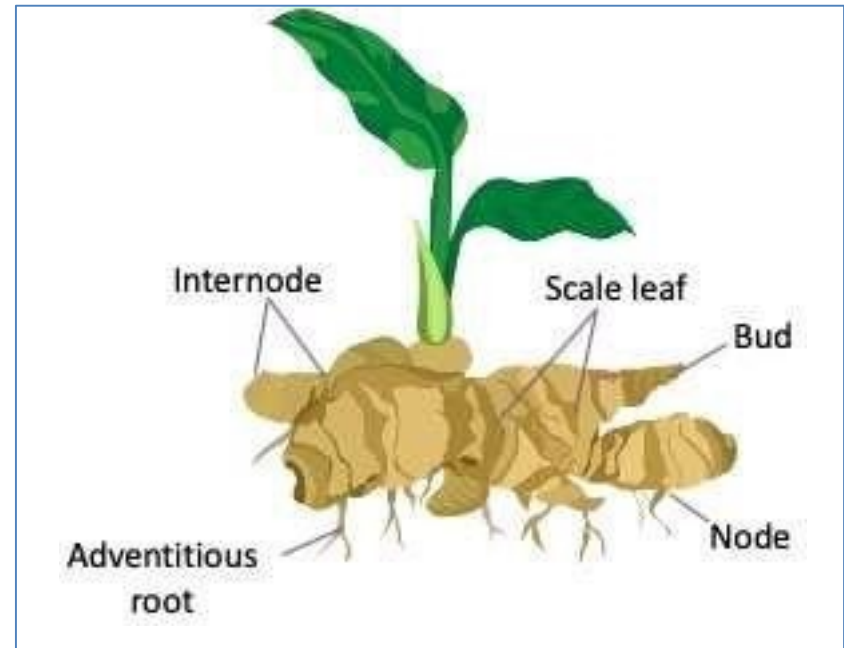
(c) Tubers

- Are **enlarged ends of rhizomes** specialized for **storing food**.
- The “eyes” arranged in a **spiral pattern** around a potato are clusters of **axillary buds** that mark the **nodes**.



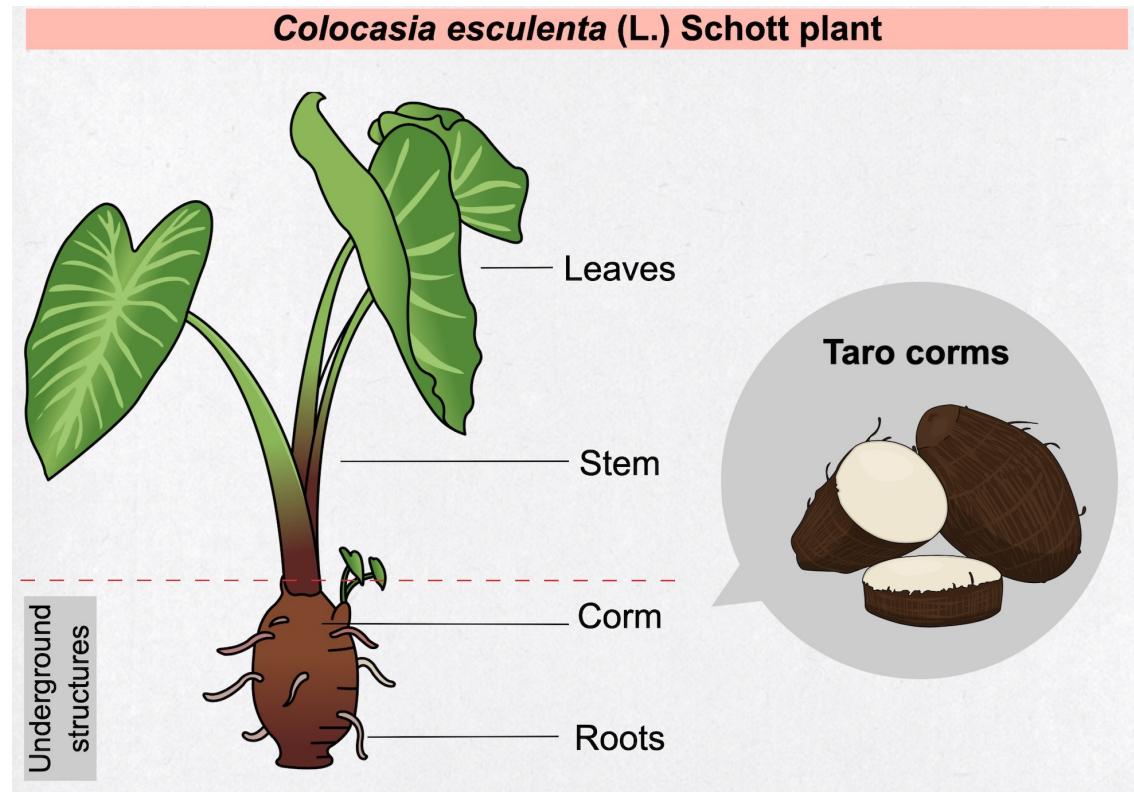
(d) Rhizomes

- The edible base of this ginger plant is an example of a rhizome
- A **horizontal stem** that grows just **below the surface** or **emerges** and grows along the surface.



(e) Corms

- A swollen, solid underground stem specialized for food storage and asexual reproduction, enabling survival in harsh seasons.



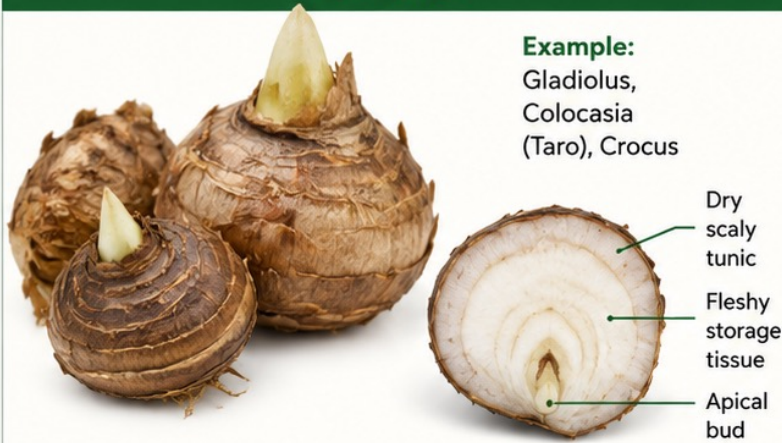
Solid Stem Body: Consists entirely of compact, starch-storing stem tissue (unlike layered bulbs).

- **Papery Leaves:** Thin, dry, reduced leaves wrapping the outside for protection.
- **Axillary Buds:** Sprout on nodes to grow into new shoots.
- **Adventitious Roots:** Fibrous root cluster growing from the basal plate to absorb water.
- **Real-World Examples:** Taro (keladi), Gladiolus, Crocus, and Water Chestnut.

CORMS vs TUBERS

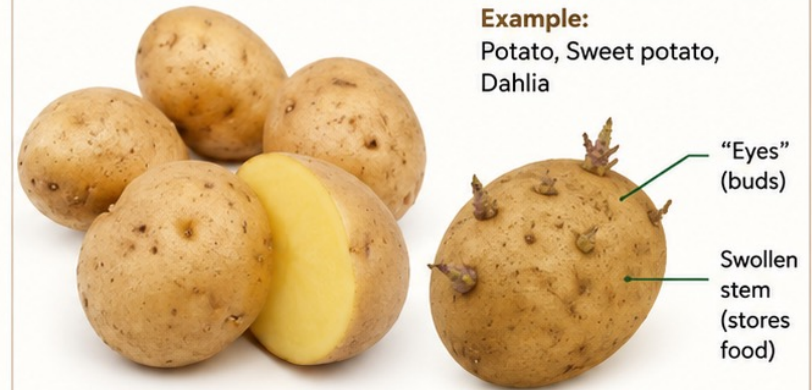
Both are underground storage organs that help plants survive unfavorable conditions.

CORMS



 What it is	A short, vertical, swollen underground stem (solid).
 Structure	Solid and compact with a central pith; covered by dry, papery scales (tunic).
 Buds	Has an apical bud at the top and small lateral buds called cormels (in some plants).
 Roots	Adventitious roots arise from the lower part of the corm.
 Examples	Gladiolus, Colocasia (Taro), Crocus, Amorphophallus, Freesia
 Function	Store food and help in perennation (survive unfavorable conditions).

TUBERS



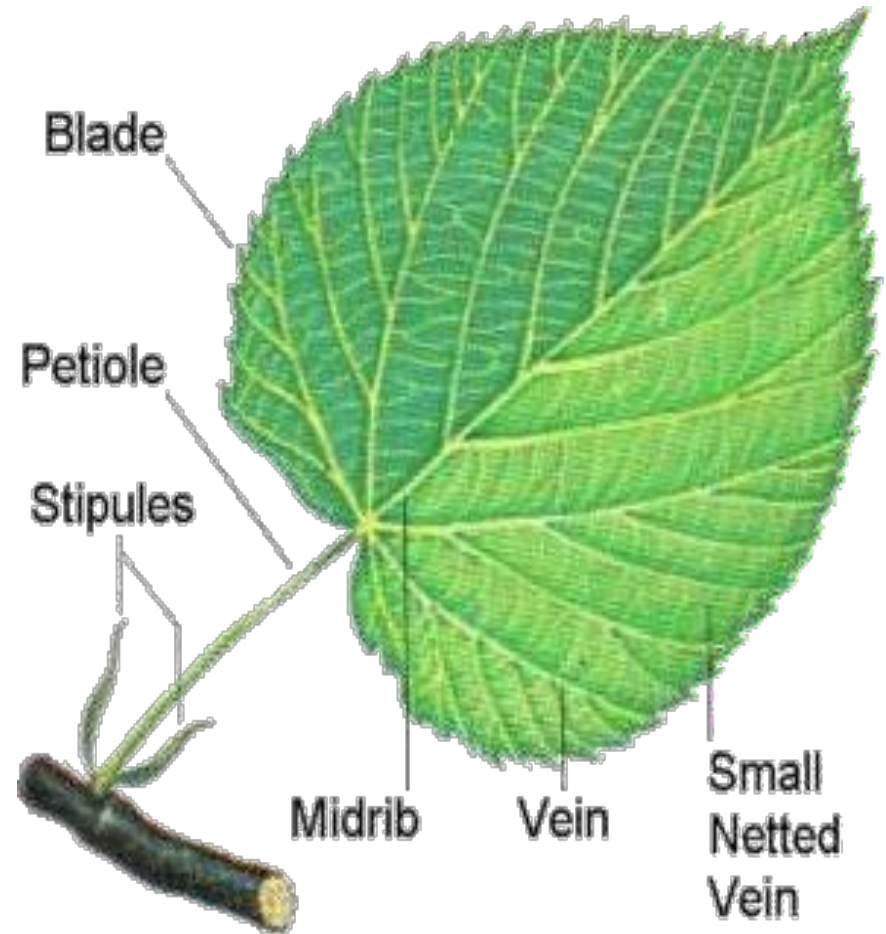
 What it is	A swollen underground stem (usually lateral stem).
 Structure	Fleshy and usually irregular in shape; no distinct tunic.
 Buds	Has many buds or "eyes" on the surface.
 Roots	Adventitious roots arise from the "eyes" or nodes.
 Examples	Potato, Sweet potato, Dahlia, Jerusalem artichoke
 Function	Store food and help in vegetative propagation.

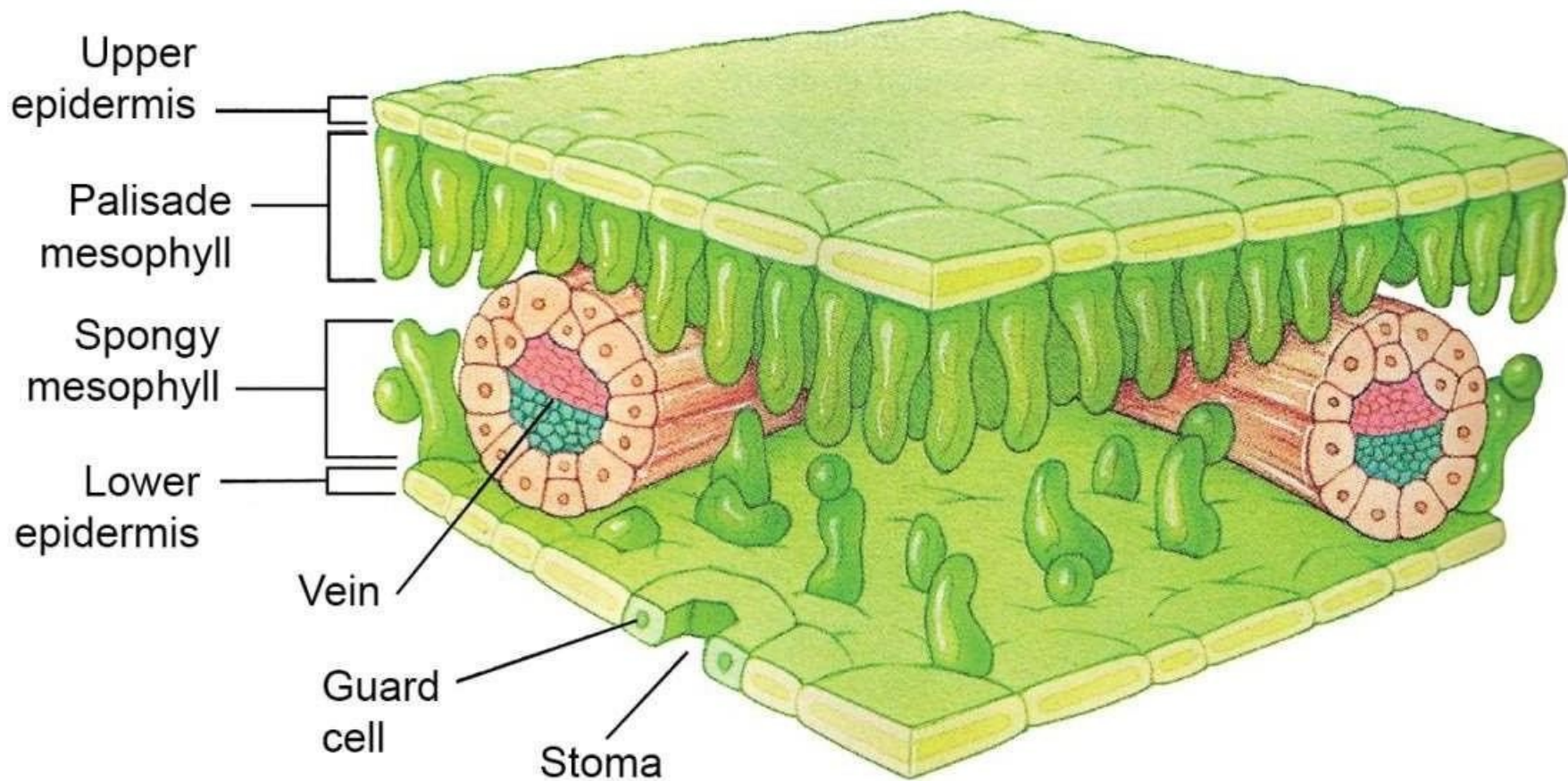


Key difference: Corms are solid, vertical stems with a tunic and apical bud, while tubers are swollen (usually lateral) stems with "eyes" (buds) and no tunic.

Leaves: Structure

- The leaf
 - Is the **main photosynthetic organ** of most vascular plants
- Leaves generally consist of
 - A flattened **blade** and a stalk (**petiole**)
 - The **petiole**, which joins the leaf to a node of the stem





Evolutionary Adaptations of Leaves



Spines



Tendrils



Reproductive leaves



Storage leaves

Stem

Storage leaves

(a) Tendrils



- Help the pea plant clings to a **support**
- After "lassoed" a support, a tendril forms a **coil**,
- Bringing it **closer to the support**
- Are typically **modified leaves**, but some can be **modified stems** (grapevines)

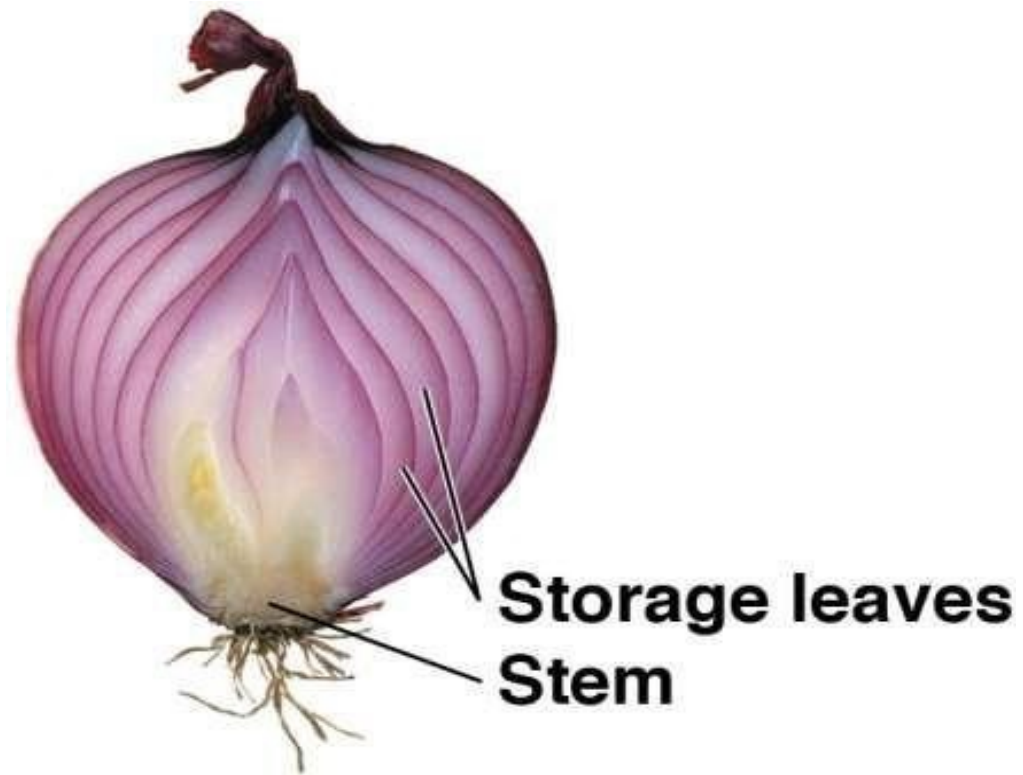
(b) Spines



- Spines of a cacti (prickly pear) are actually leaves.
- Photosynthesis is carried out by fleshy green stems

(c) Storage leaves

- **Bulbs**, such as this cut onion, have a **short underground stem** and modified leaves that **store food**.



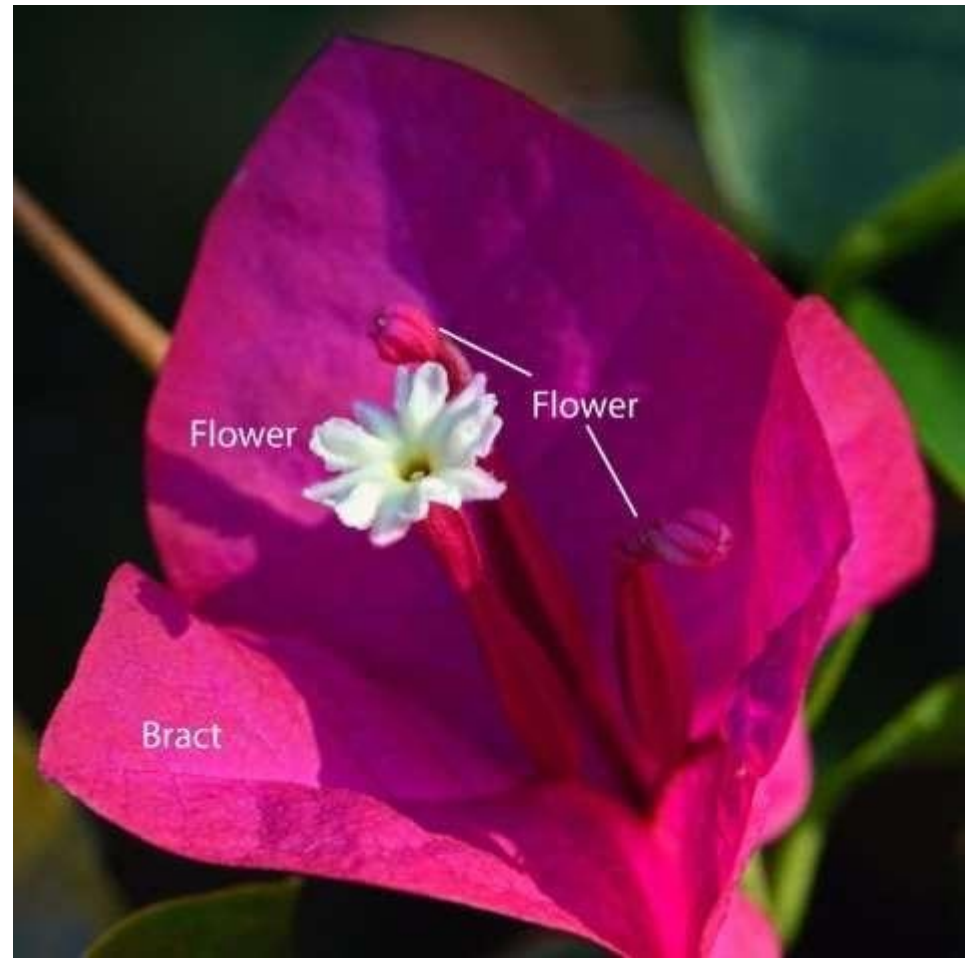
Storage leaves

(d) Reproductive leaves

- Leaves of **succulents**, such as *Kalanchoe daigremontiana*, produce **adventitious plantlets**
- Which fall off the leaf and **take root in the soil**



(e) Bracts



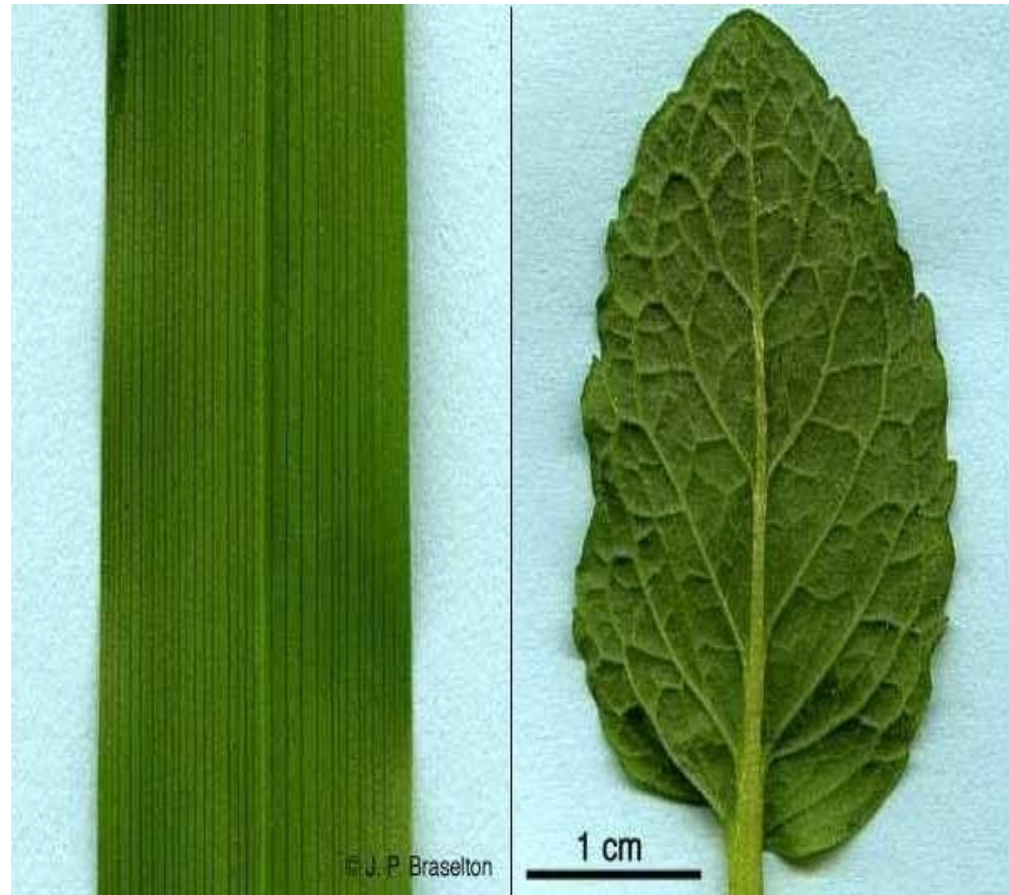
- Red parts of the **poinsettia** are often mistaken for petals but are actually **modified leaves** called bracts
- **surround a group of flowers**
- Such brightly colored leaves attract **pollinators**.

Leaf: Structure

- **Monocots and dicots**
 - Differ in the **arrangement of veins**, the **vascular tissue** of leaves
- Most monocots
 - Have **parallel veins**
- Most dicots
 - Have **branching veins**

monocots

dicots

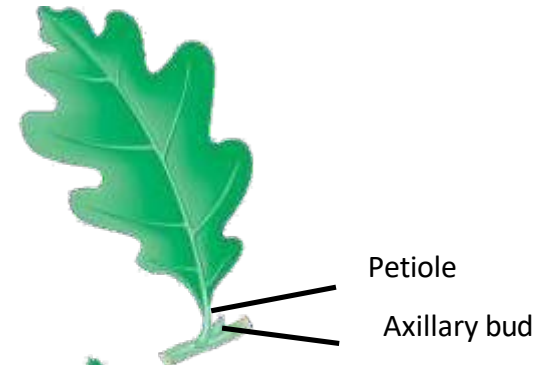


Leaves: structure

- In classifying angiosperms
 - Taxonomists may use leaf morphology as a criterion

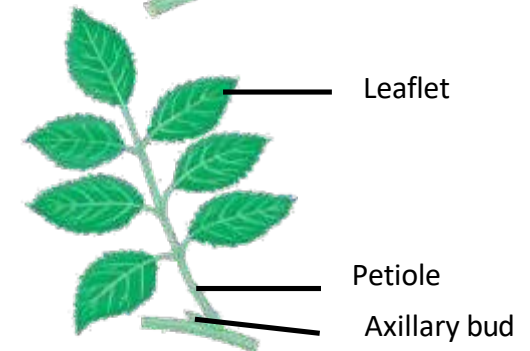
(a) Simple leaf

- A simple leaf is a **single, undivided blade**.
- Some simple leaves are **deeply lobed**, as in an oak leaf.



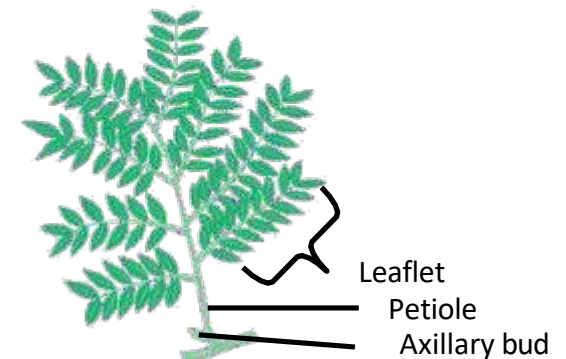
(b) Compound leaf

- In a compound leaf, the blade consists of **multiple leaflets**. Notice that a leaflet has **no axillary bud** at its base.



(c) Doubly compound leaf

- In a doubly compound leaf, each leaflet is **divided into smaller leaflets**.



Video

- Tomato plant growth timelapse

<https://www.youtube.com/watch?v=LICDb8nM5rs>

The Three Tissue Systems: **Dermal**, **Vascular**, and **Ground**

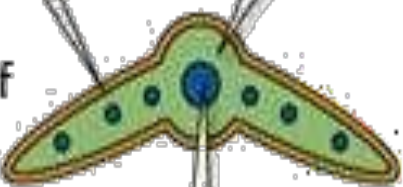
The **dermal tissue** system is the outer covering of the plants

The **ground tissue system** carries out photosynthesis, stores photosynthesis products, and help support the plants

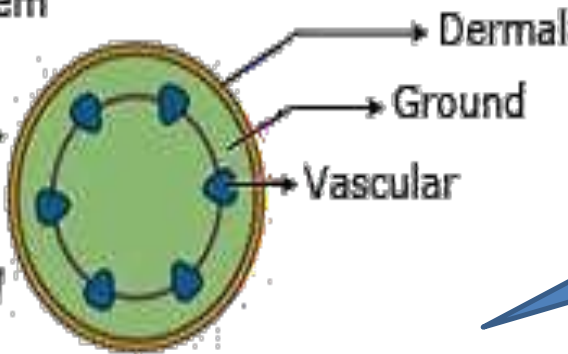
The **vascular tissue** system conducts water and solutes through out the plant.

Each plant organ (root, stem, leaves) has **dermal**, **vascular** and **ground tissue**

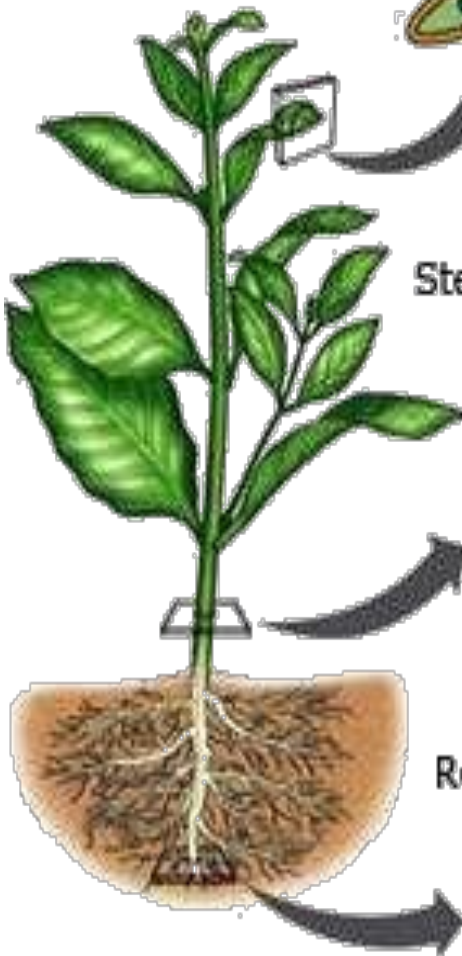
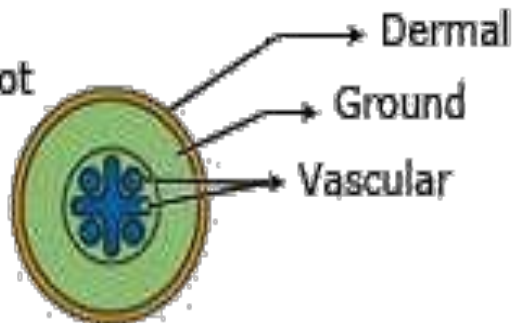
Leaf



Stem



Root



Plant tissue systems

- The **dermal** tissue system
 - Consists of the **epidermis** and **periderm** (bark)
- The **vascular** tissue system
 - Carries out **long-distance transport** of materials between roots and shoots
 - Consists of two tissues, **xylem** and **phloem**
- The **ground** tissue system
 - Includes various cells specialized for functions such as **storage**, **photosynthesis**, and **support**

The Three Tissue Systems

A continuous, unbroken network spanning every organ

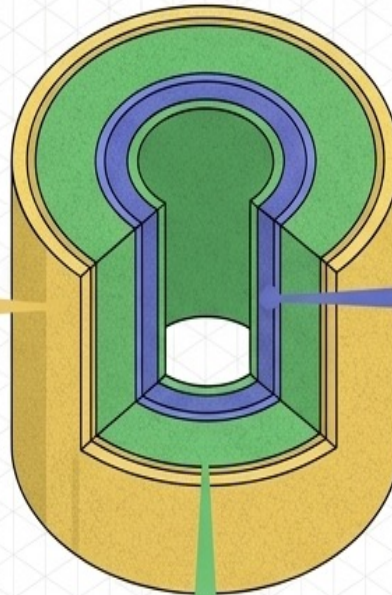
Dermal Tissue (The Skin)



Outer protective covering.

First line of defense against damage and pathogens.

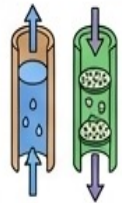
Modified forms include guard cells (vents) and trichomes (hairs).



Vascular Tissue (The Plumbing)

Executes long-distance transport between roots and shoots.

Moves water up and sugars down.

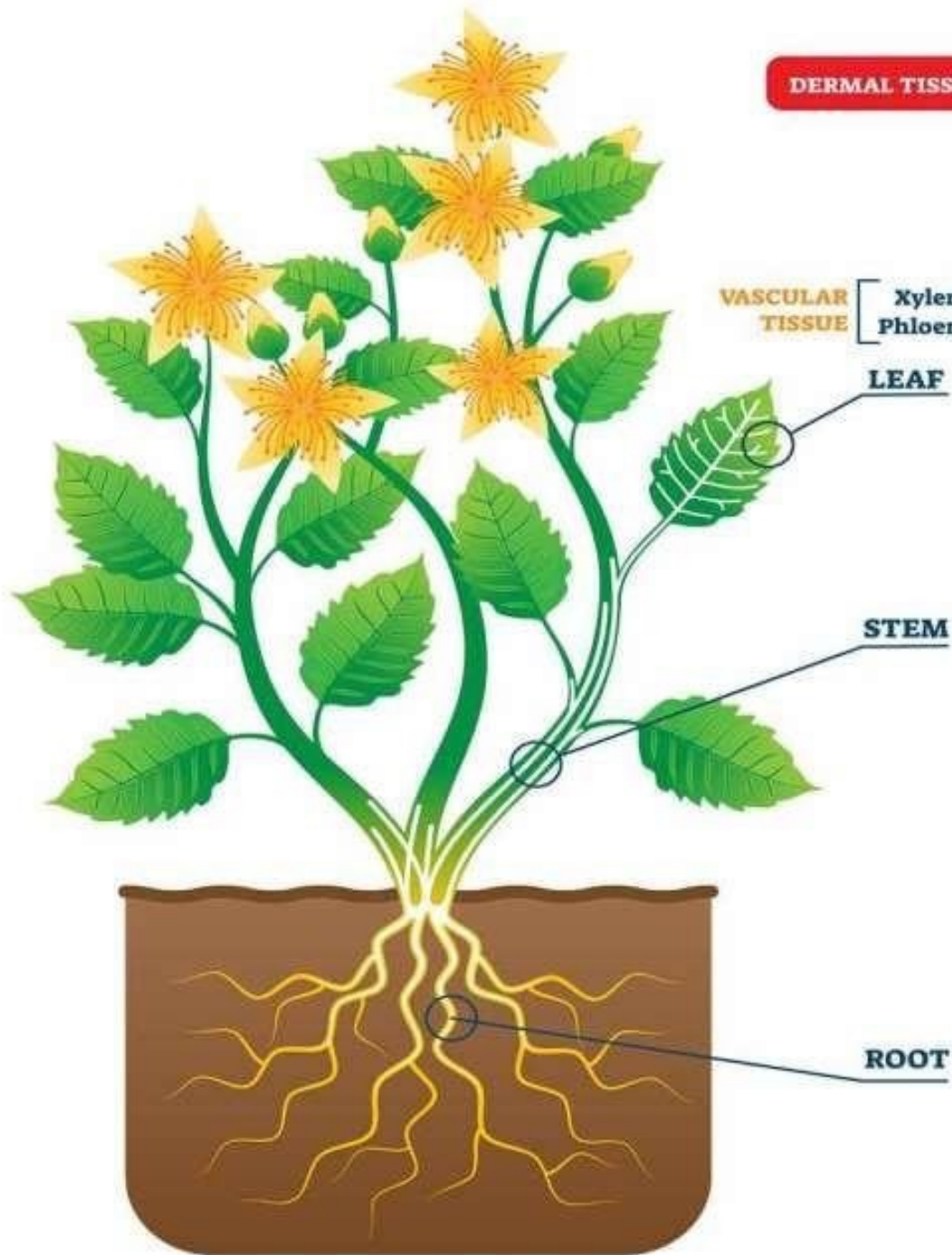


Ground Tissue (The Matrix / Insulation)



Fills the interior.

Specialized for storage, structural support, and photosynthesis (e.g., Parenchyma, Collenchyma, Sclerenchyma).



DERMAL TISSUE

VASCULAR TISSUE

GROUND TISSUE

**VASCULAR
TISSUE**

Xylem
Phloem

LEAF

STEM

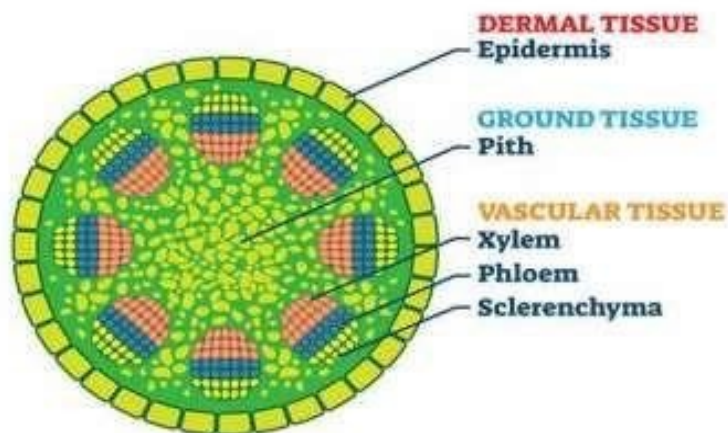
ROOT



DERMAL TISSUE
Upper Epidermis

GROUND TISSUE
Mesophyll

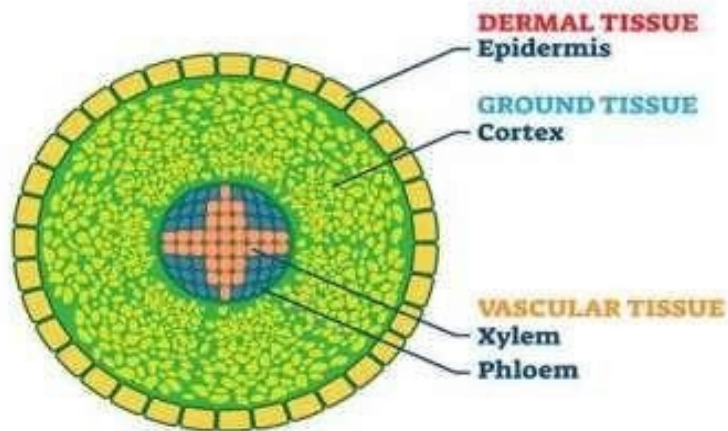
DERMAL TISSUE
Lower Epidermis



DERMAL TISSUE
Epidermis

GROUND TISSUE
Pith

VASCULAR TISSUE
Xylem
Phloem
Sclerenchyma

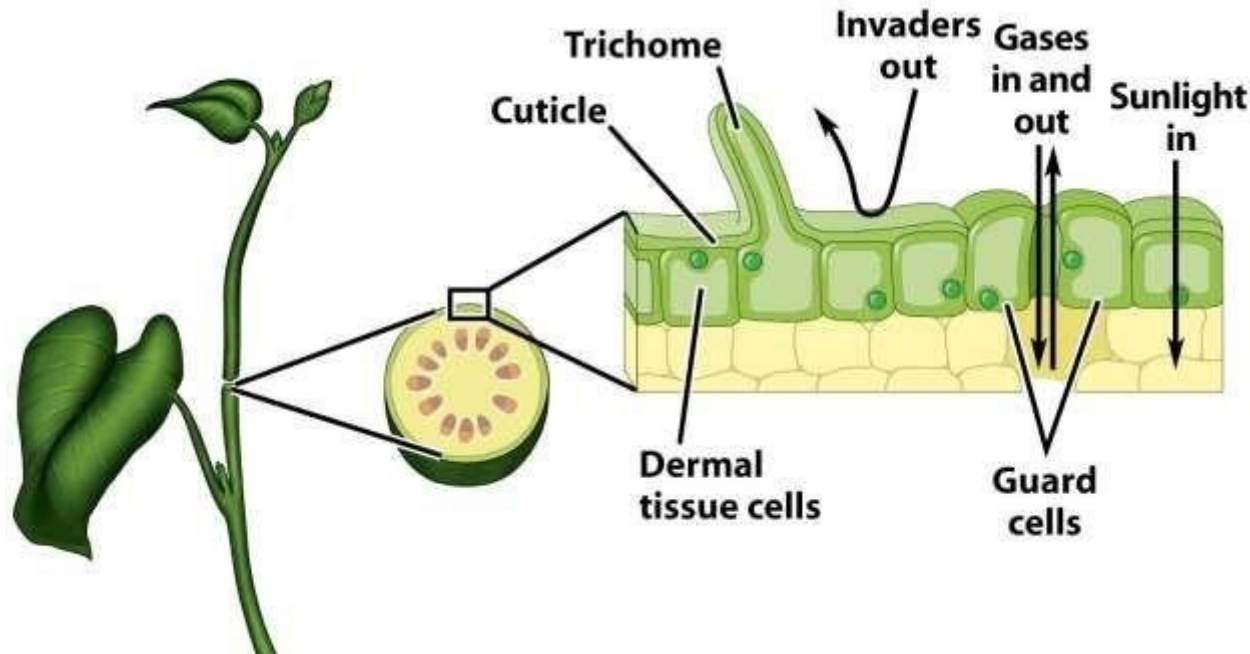


DERMAL TISSUE
Epidermis

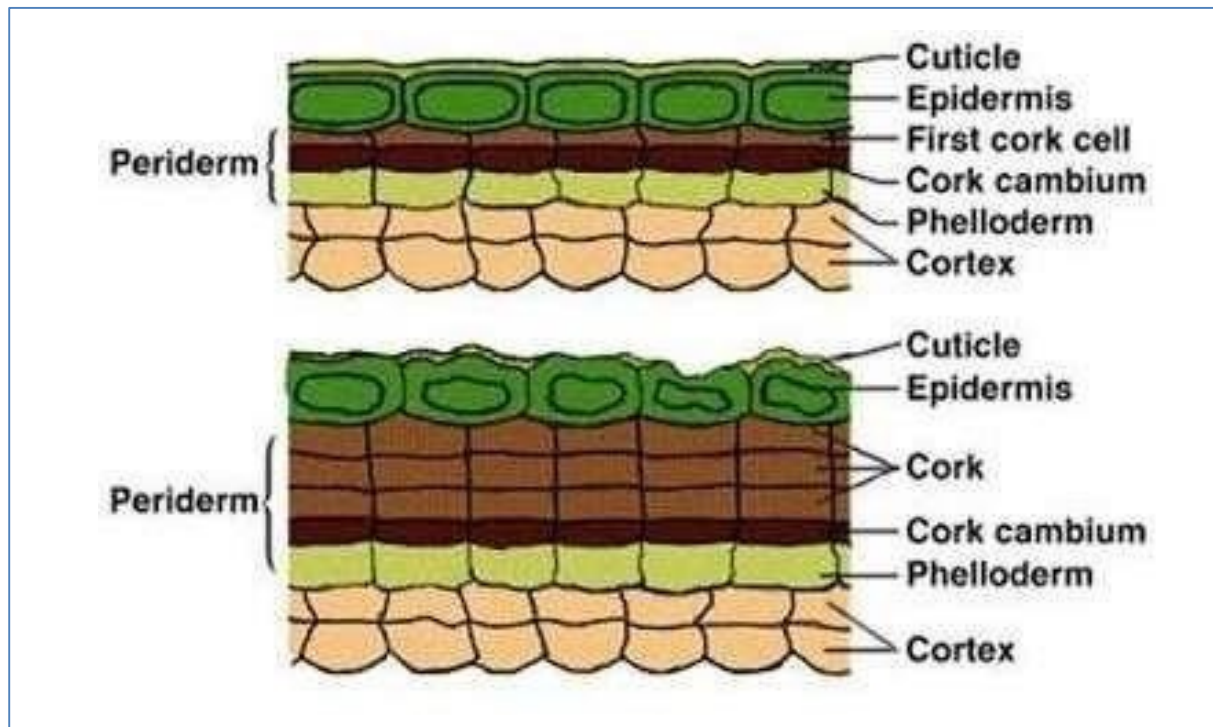
GROUND TISSUE
Cortex

VASCULAR TISSUE
Xylem
Phloem

Dermal Tissue



- **Outer protective** covering the whole plant
- **First line defense** against physical **damage** + **pathogens**
- **Non-woody plants**: usually a single tissue called **epidermis** (a layer of tightly packed cells)
- **Woody plants**: **periderm** replace epidermis in **older regions of stems and roots**.
- Modified cells e.g: **guard cells**, **root hairs**, cells that produce a **waxy cuticle** (prevent water loss)



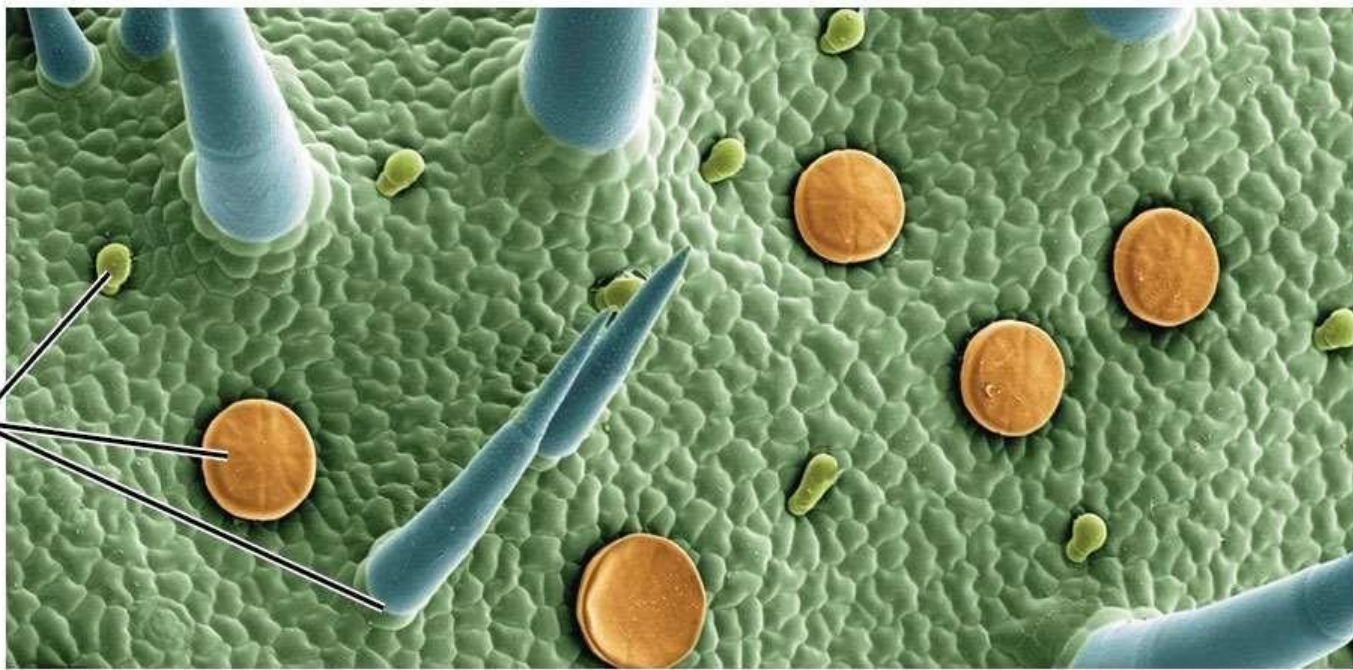
Addition of protecting plant from water loss and disease:

(a) **Guard cells** : involved in gaseous exchange

(b) **Trichomes**: found in shoots, consists of outgrowths

- Desert species: hairlike trichome **reduce water loss** + **reflect excess light**
- Some defend against **insects** through
 - **Shapes** that **hinder movement**
 - **Glands** that secrete **sticky fluid**

Trichomes



300 μm

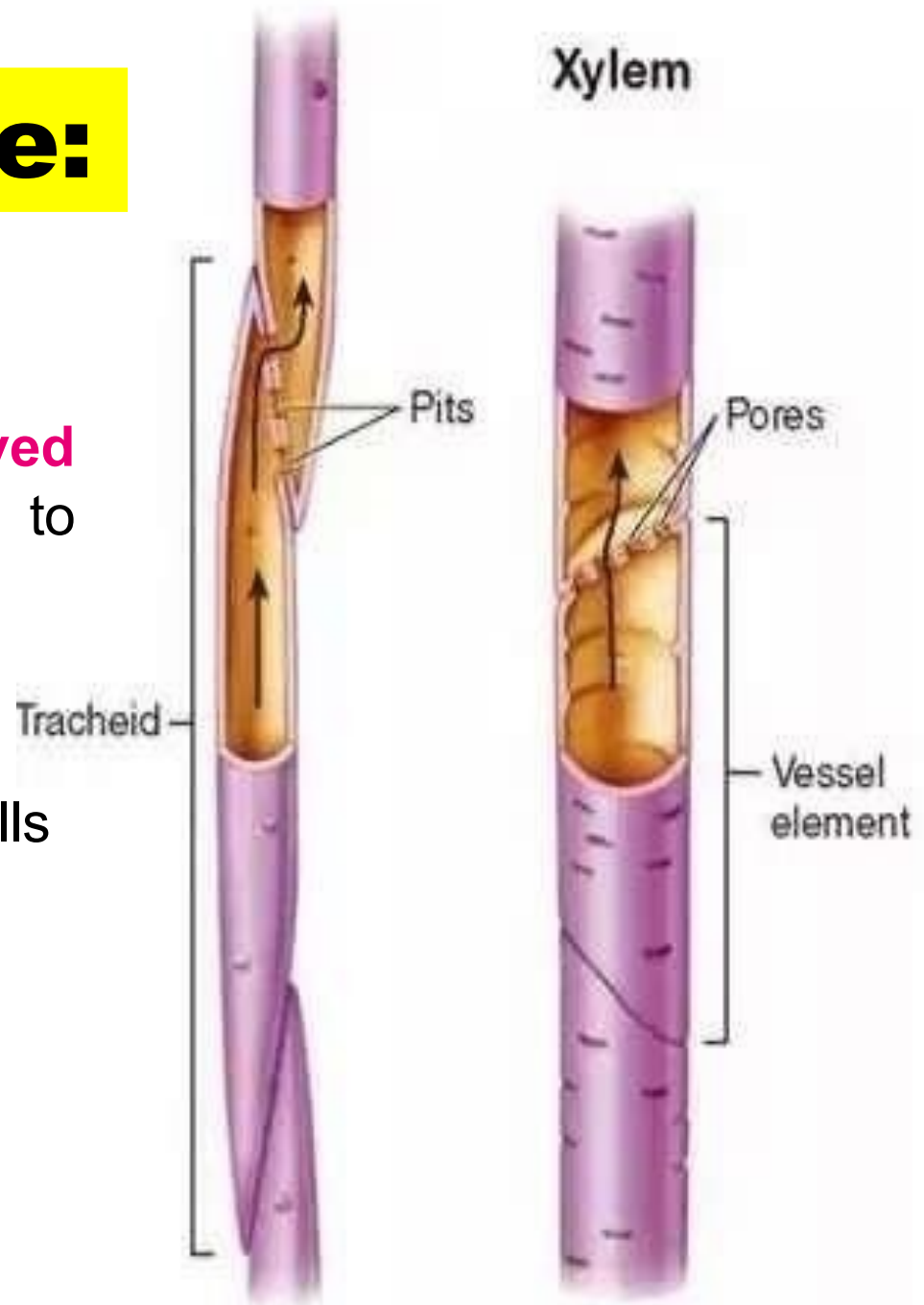
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- Three types of trichomes found on the surface of **marjoram** (*Origanum majorana*) – common name oregano
- **Spear like** trichomes help hinder the movement of crawling insects
- Other two trichomes **secrete oils and other chemicals** involved in defense

Vascular Tissue:

Xylem

- **Conducts water and dissolved minerals** upward from roots to shoots
- Provide **mechanical support**
- 2 types of Vascular tissues:
 - **tracheids**: long , tapered cells
 - **vessel elements**: wider, shorter
- Diameter as large as 0.7 mm
- **Secondary wall** of xylem hardened with **lignin**



Try this at home!



Water-Conducting Cells of Xylem

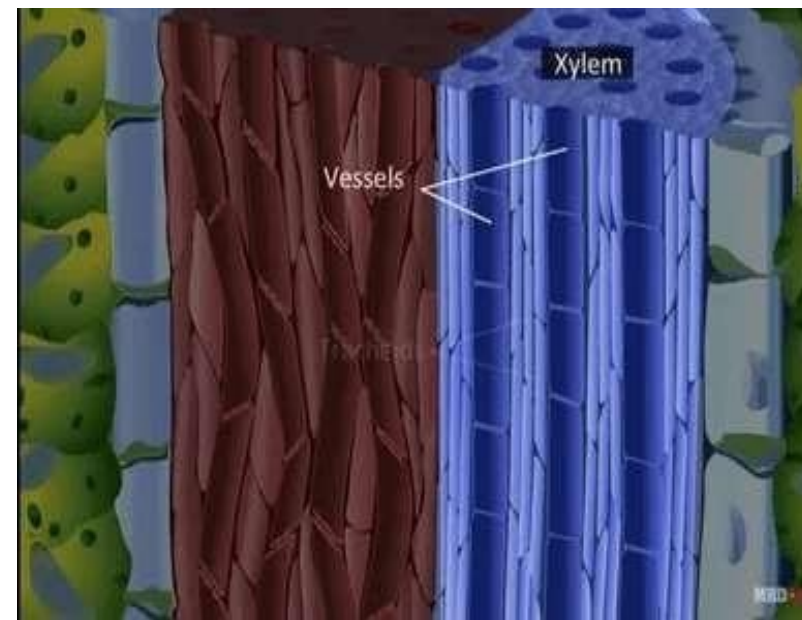
Two types of water conducting (dead) cells:

a) Tracheids

- Occur in **all vascular plants**
- **Long, thin cells** with **tapered ends**
- Water moves from cell to cell through **pits** (no need to cross thick secondary walls)

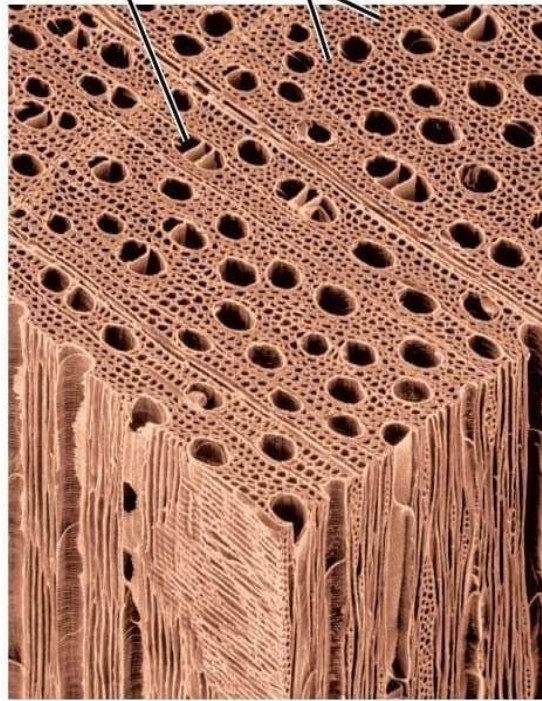
b) Vessel elements

- **Most angiosperms**, few gymnosperms, few seedless vascular plants
- **Wider, shorter, thinner walled, less tapered**
- Aligned end to end, forming **long pipes**
- End walls have **perforation plates**, enable water to flow freely



Water-Conducting Cells of the Xylem

Vessel Tracheids



100 μm

- Both secondary walls are hardened with lignin – provides support, prevent collapse under tension of water transport

Tracheids and vessels
(colorized SEM)

Perforation
plate

Vessel
element

Vessel elements, with
perforated end walls

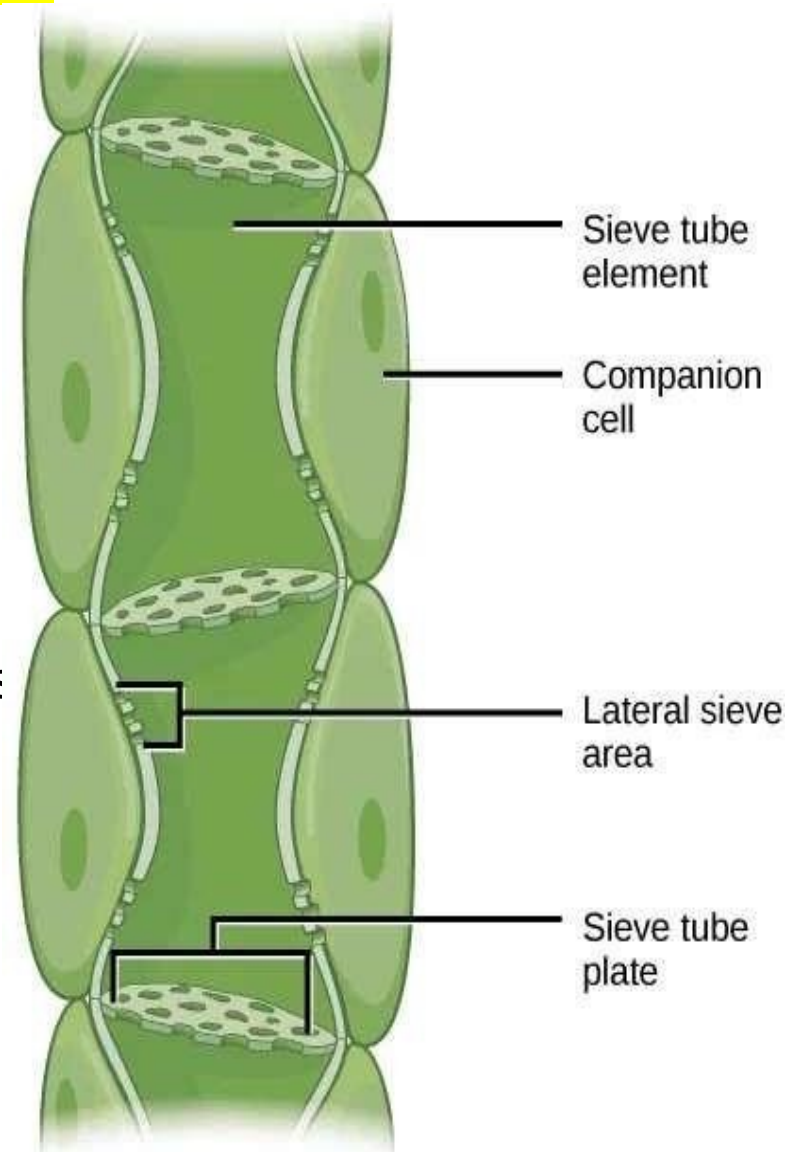
Pits

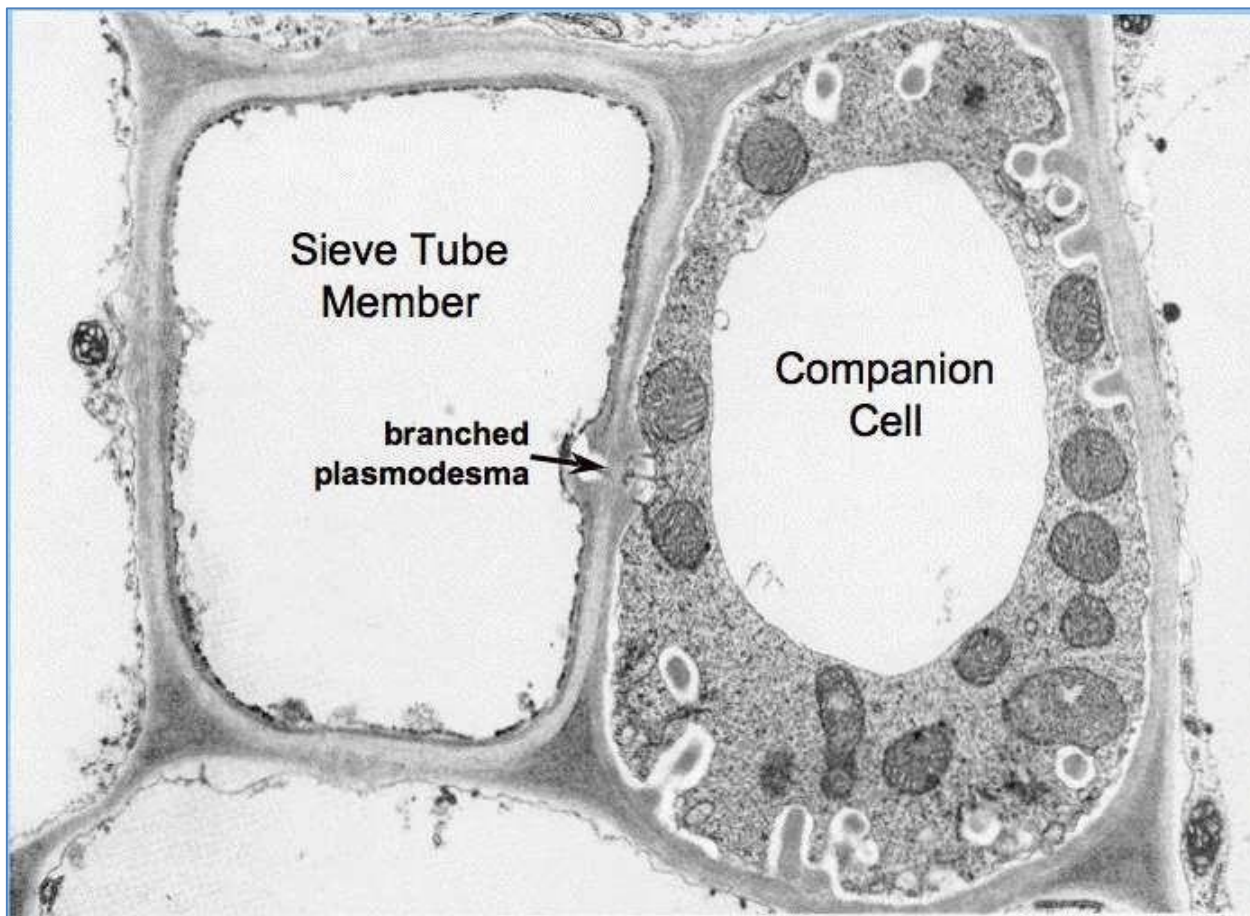
Pits

Tracheids

Sugar-conducting Cells of the Phloem

- **Alive cells** at functional maturity
 - **Sugars** + **other nutrients** are transported through sieve tube elements/members
- a) **Sieve tubes**
- **Lack nucleus, ribosomes**, a distinct **vacuole**, **cytoskeleton** elements
 - **This enables nutrient to pass more easily!**
 - End walls between sieve tube elements (**sieve plates**) have **pores** to facilitate the **flow of fluid** from cell to cell

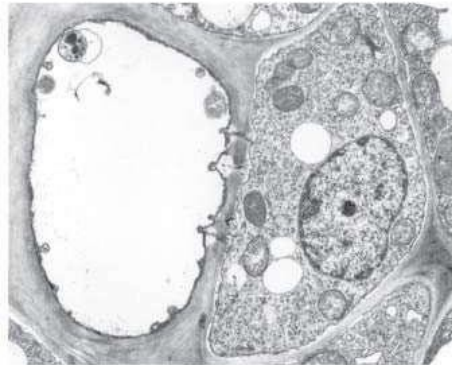




b) Companion cell

- **Non-conducting** cells
- Connected to sieve tube elements by **plasmodesmata**
- Nucleus + ribosomes serve for the cell itself and for the adjacent sieve tubes elements – **to load/unload sugars**

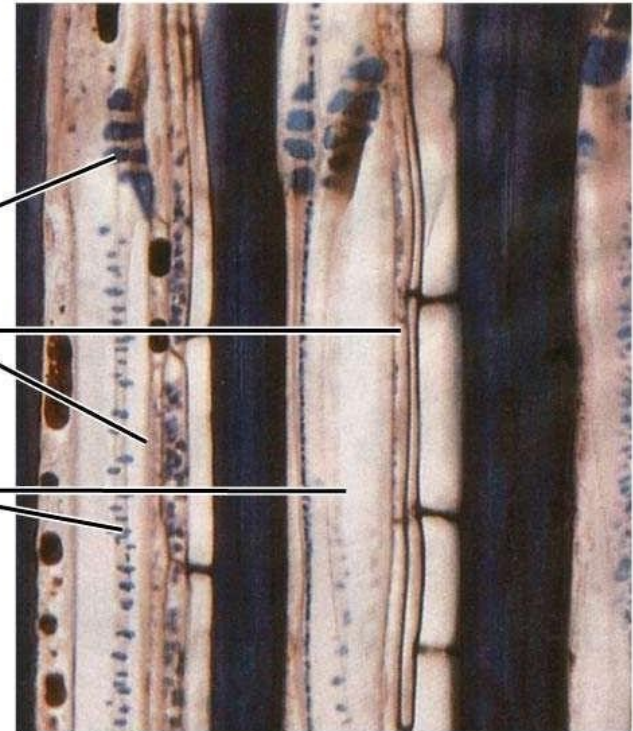
Sugar-Conducting Cells of the Xylem



Sieve-tube element and companion cell cross section (TEM)

3 μm

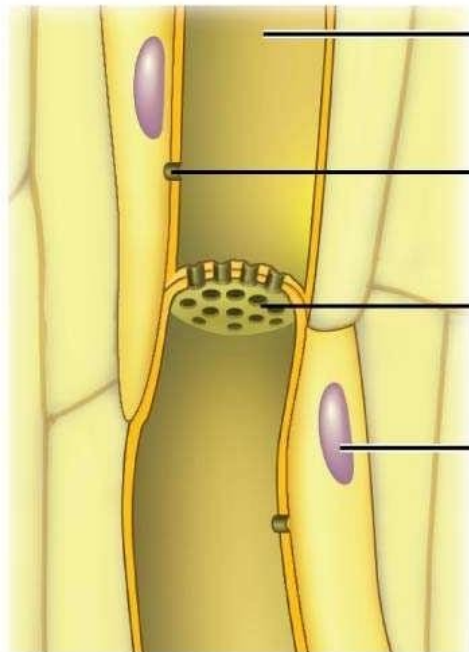
Sieve-tube elements: longitudinal view (LM)



Sieve plate

Companion cells

Sieve-tube elements

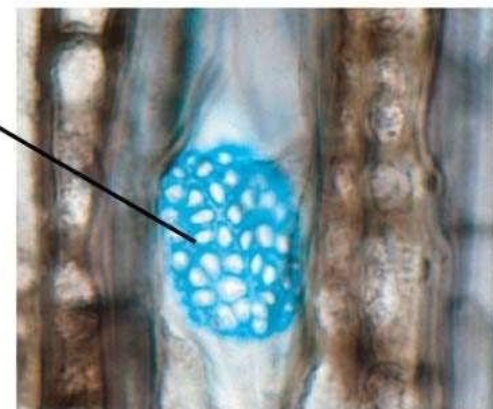


Plasmodesma

Sieve plate

Nucleus of companion cell

Sieve-tube elements: longitudinal view



30 μm

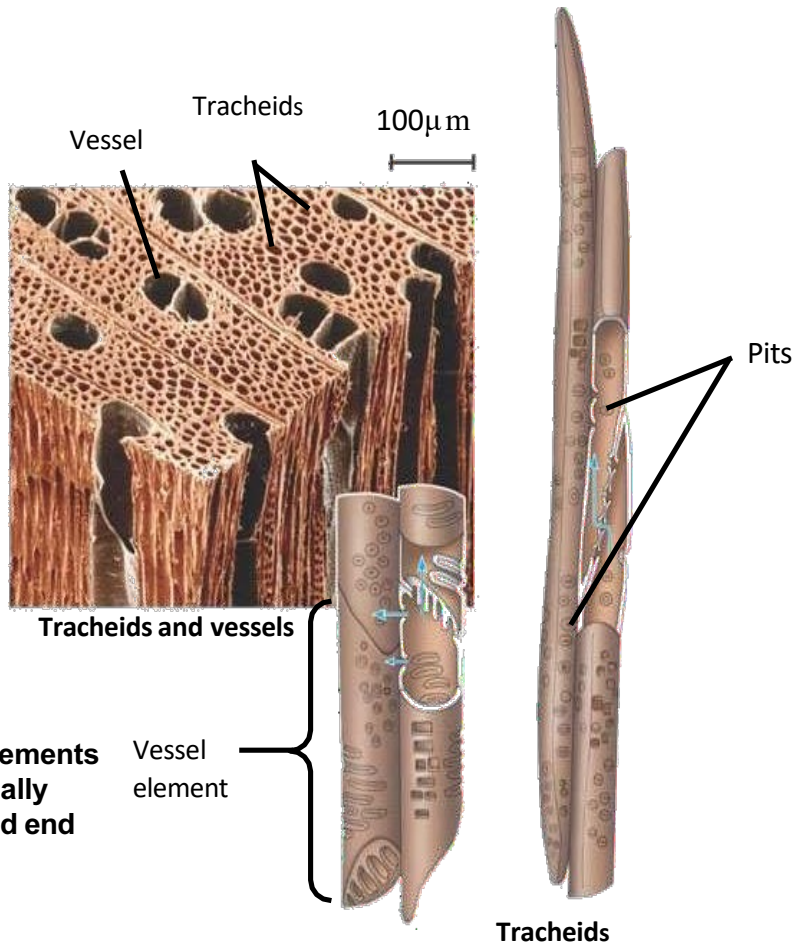
15 μm

Sieve plate with pores (LM)

Sieve elements	Companion cell
• End walls are perforated • Allows cytoplasmic connections between vertically stacked cells • No nucleus, ribosome & vacuoles • Depend on companion cells	• Located beside sieve tube • Compact protoplasma & mitochondria • Supply sugar & amino acid to sieve elements • Supply ATP for transport of sugar & amino acid

Xylem vs Phloem

WATER-CONDUCTING CELLS OF THE XYLEM



SUGAR-CONDUCTING CELLS OF THE PHLOEM

Sieve-tube members:
longitudinal view

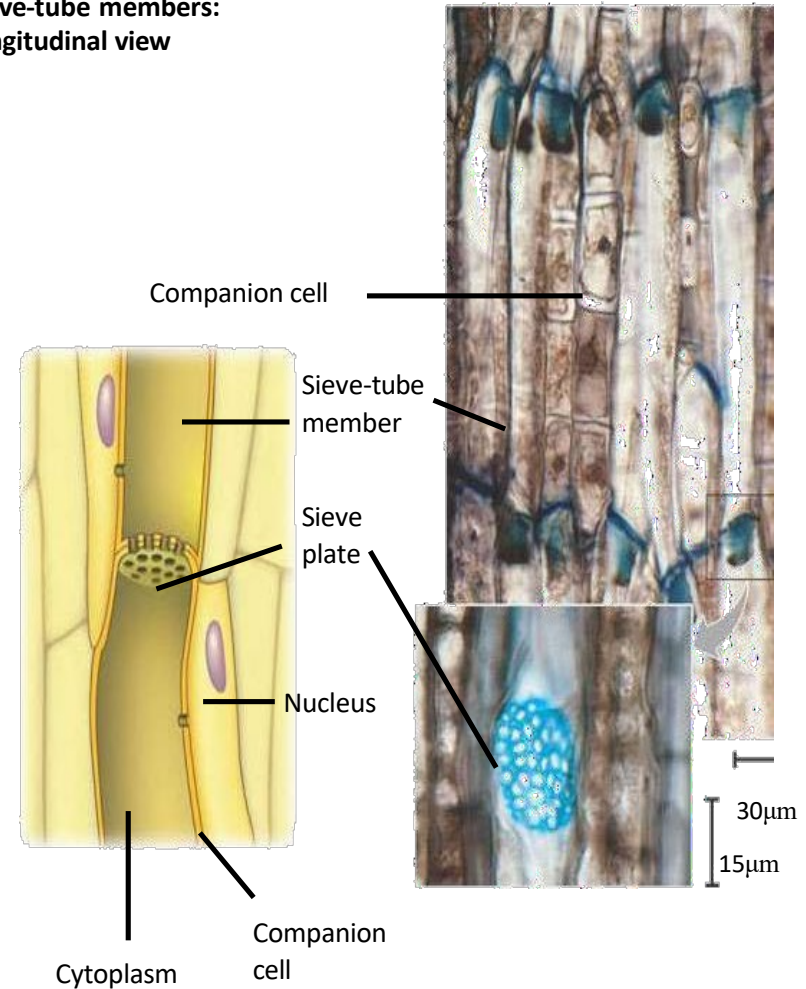


Figure. 35.10

The Transport Matrix: Xylem vs. Phloem

	XYLEM	PHLOEM
Cargo	Water & Dissolved Minerals	Sugars & Organic Nutrients
Directionality	Unidirectional (Roots to Shoots upward flow)	Bidirectional (Source-to-Sink flow)
Cell Vitality at Maturity	Dead at maturity (Forms hollow, rigid tubes)	Alive at maturity (Active transport channels)
Structural Reinforcement	Thick secondary walls hardened with Lignin	Thinner cell walls, reliant on companion cells
Primary Cell Types	Tracheids & Vessel Elements	Sieve-tube elements & Companion cells

Why phloem cells need to be alive but not xylem cells?

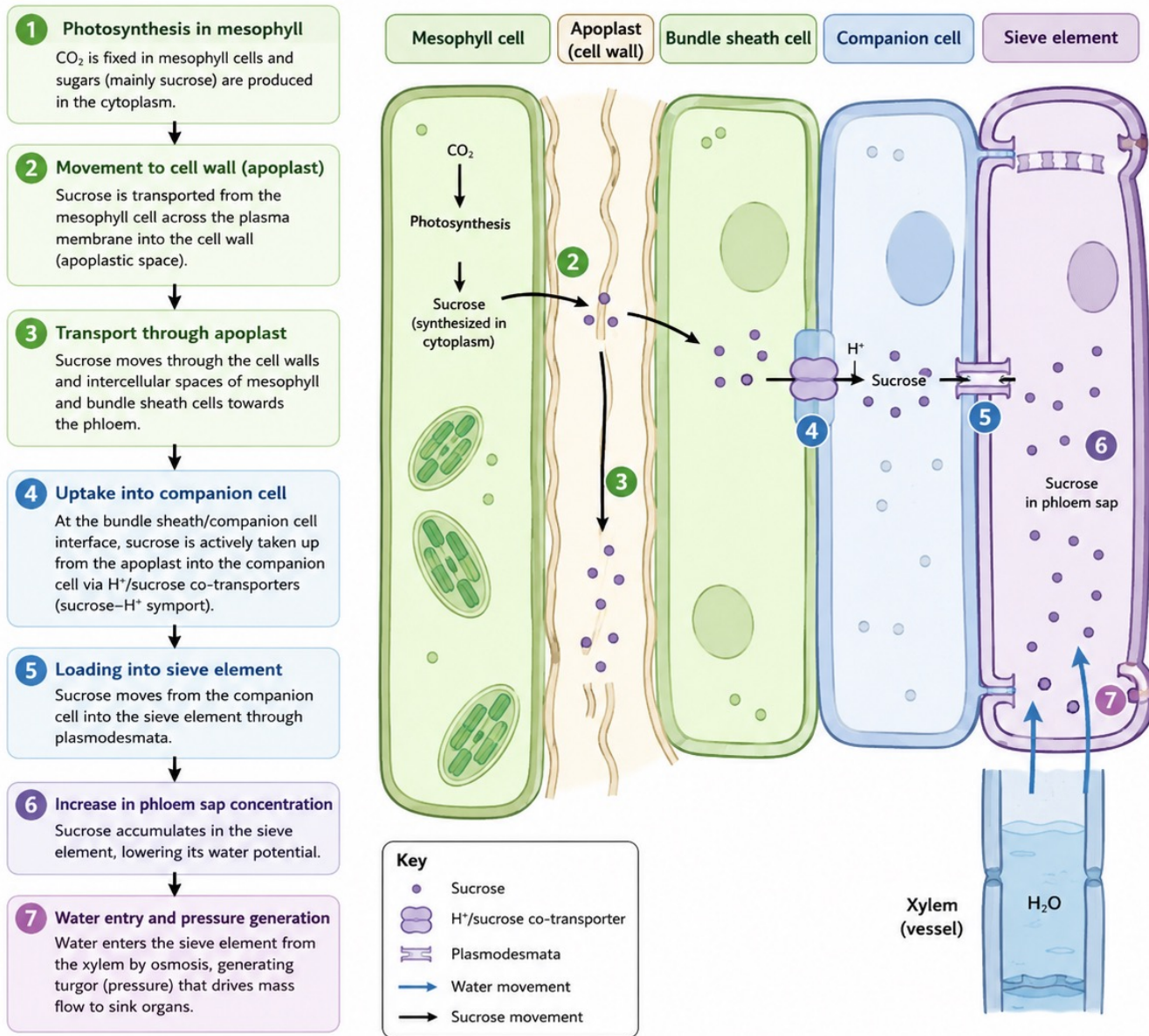


Because sugars get loaded actively into the phloem tubes and this **requires energy**, that's why it's called as an **active transport** and that's why these cells need to stay alive.



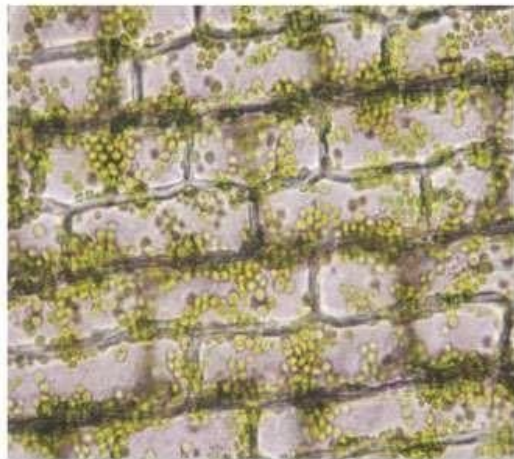
Pathway of Sugar Loading from Leaf to Phloem

Apoplastic (Sieve Element–Companion Cell) Pathway



Ground Tissues

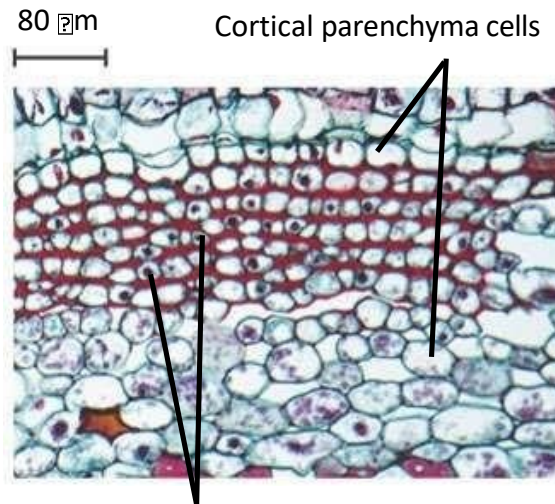
PARENCHYMA CELLS



Parenchyma cells

60 μm

COLLENCHYMA CELLS

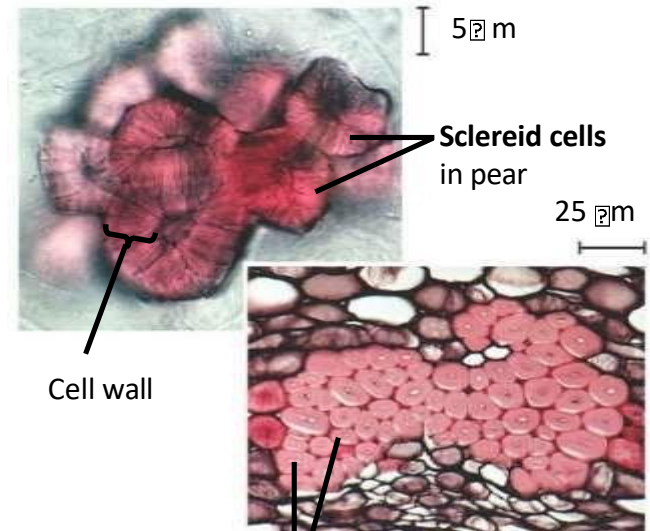


80 μm

Cortical parenchyma cells

Collenchyma cells

SCLERENCHYMA CELLS

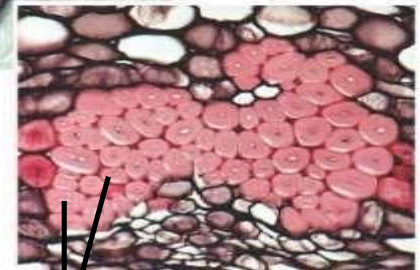


5 μm

Sclereid cells
in pear

25 μm

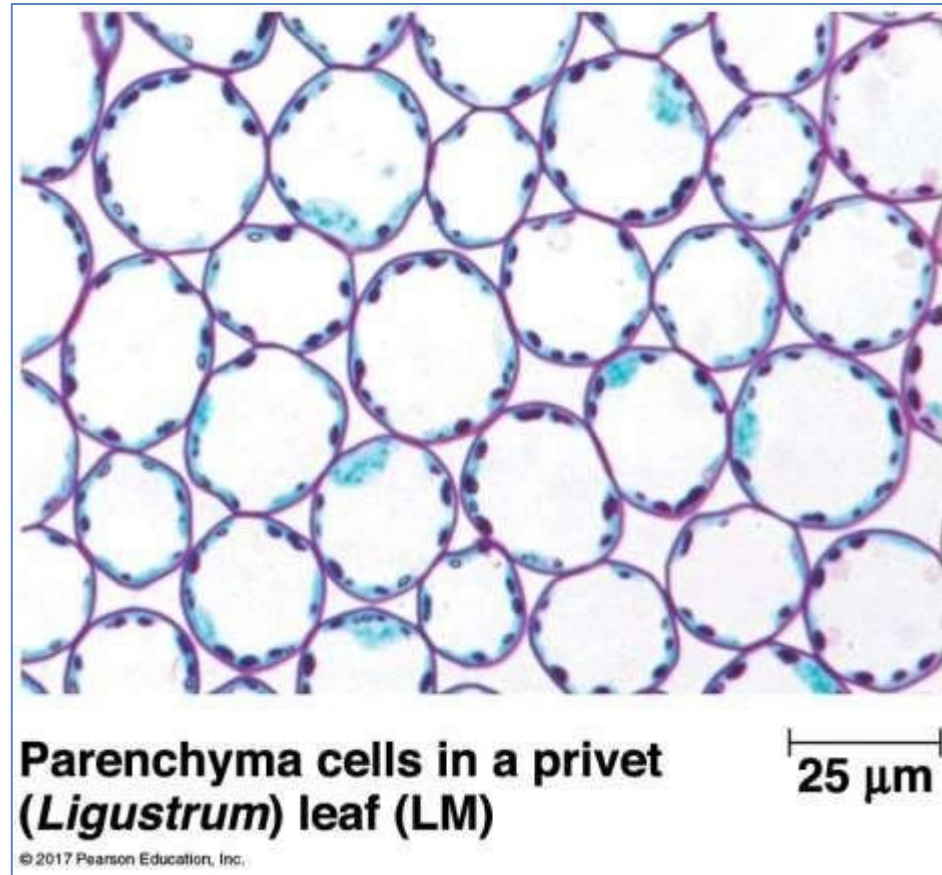
Cell wall



Fiber cells

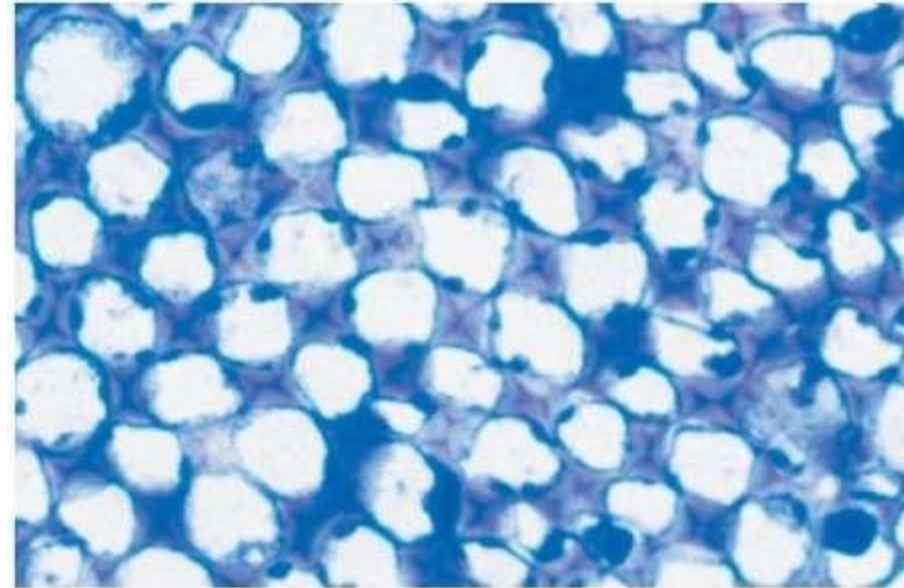
Parenchyma

- **Large, thin-walled, large central vacuole** (when matured)
- Found in all of the organs of plant
- Perform most of metabolic functions – **synthesizing and storing organic products**
- **Mesophyll (palisad and spongy) – contain chloroplasts**
- **Stems and roots** – contain colorless plastids (**amyloplasts**) to store starch
- **Fruits – fleshy tissues** composed mainly parenchyma
- **Function: food storage & photosynthesis**



Collenchyma

- **Thick walls** (unevenly thickened)
- Function: **provide mechanical support** for plants
- Found in areas that are **growing rapidly** and need to be **strengthened**
- Below epidermis of **petiole/stalk** of leaves.
- Occurs as bundles of cells beneath the epidermis



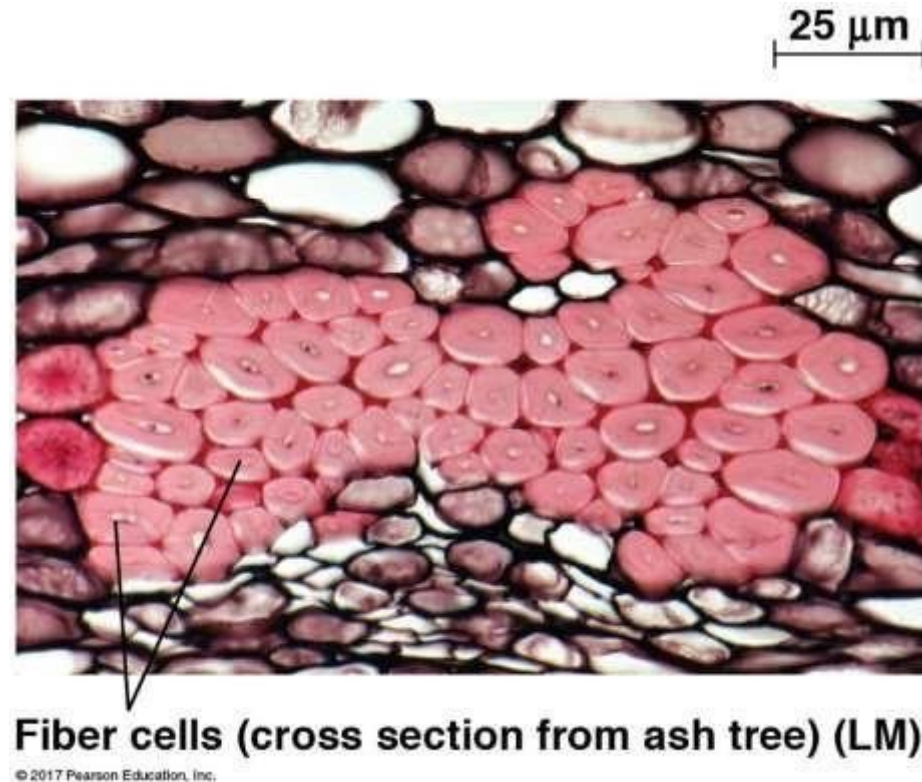
**Collenchyma cells
(in *Helianthus* stem) (LM)**

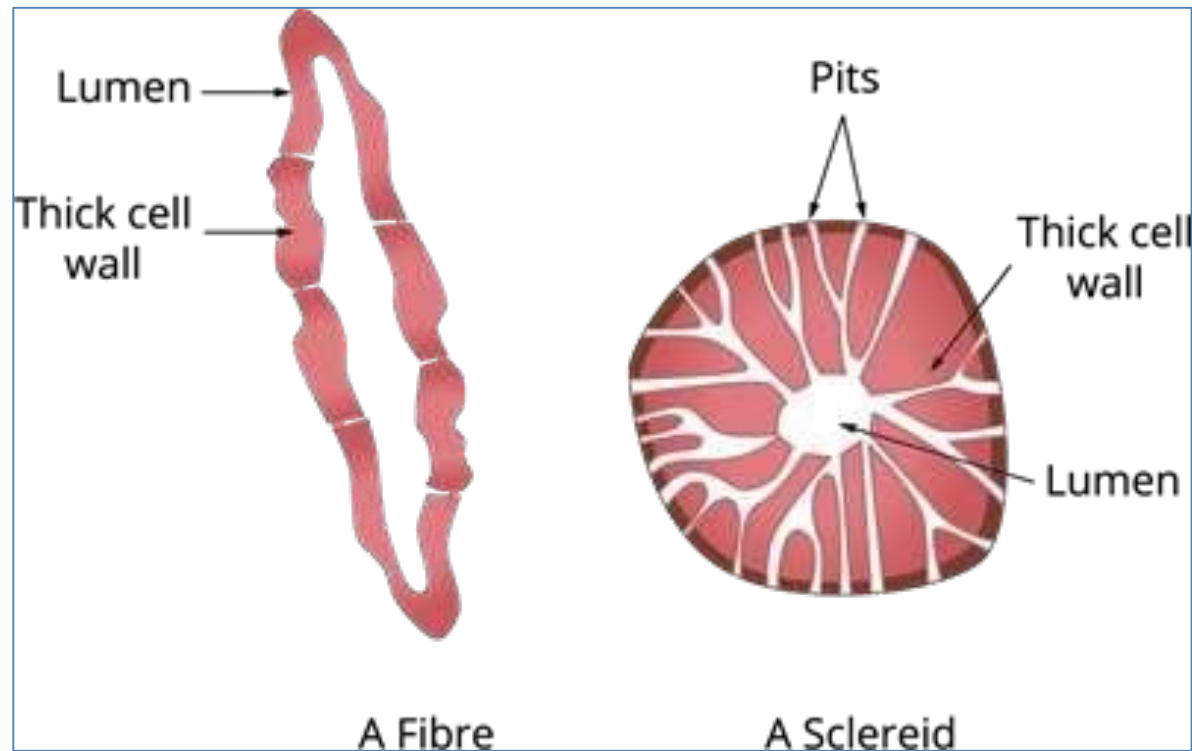
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5 μ m

Sclerenchyma

- **Thick** 2° cell walls in **uniform layer**
- **More rigid** than collenchyma
- Hardened with **lignin**
- **Protoplast die** after cell wall is fully formed
- **Mature cells cannot elongate**, occur at parts that have stopped growing
- Function: **provide mechanical support**
- Found in stems, leaf veins, **hard outer covering of seeds & nuts**





- Two types of sclerenchyma cells:
 - a) Sclereids** : boxier, irregular shape, thick lignified 2° cell walls
 - Hardness to nutshells + seed coats, gritty texture of pears
 - b) Fibers** : grouped in strands, long, slender, tapered.
 - Hemp fibers to make rope, weave linen

Meristems

- Plants can keep growing because they have **undifferentiated tissues** called **meristems**
- containing cells that can divide, leading to new cells, later elongate and differentiate
- Plant growth :
 - a) **Indeterminate growth**: Growth occurs throughout the plant's life
 - b) **Determinate growth**: Stop growing after reaching certain period



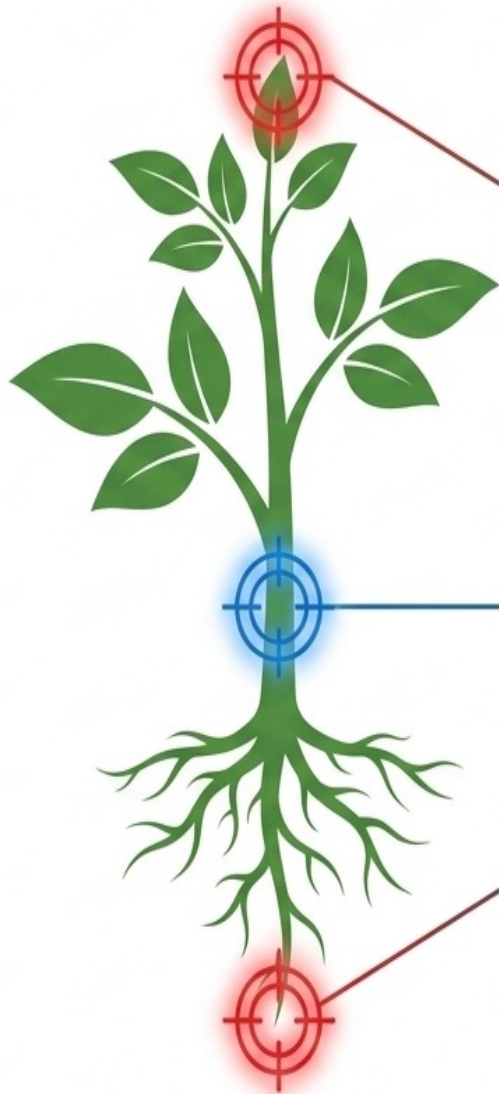
INDETERMINATE vs DETERMINATE TOMATOES

What's The Difference?



GetBusyGardening.com

Meristems: The Engines of Growth



Apical Meristems (Vertical Vectors)

Located at the tips of shoots and roots.
Responsible for Primary Growth (lengthening).
Generates new leaves, flowers, and pushes roots deeper.

Lateral Meristems (Horizontal Vectors)

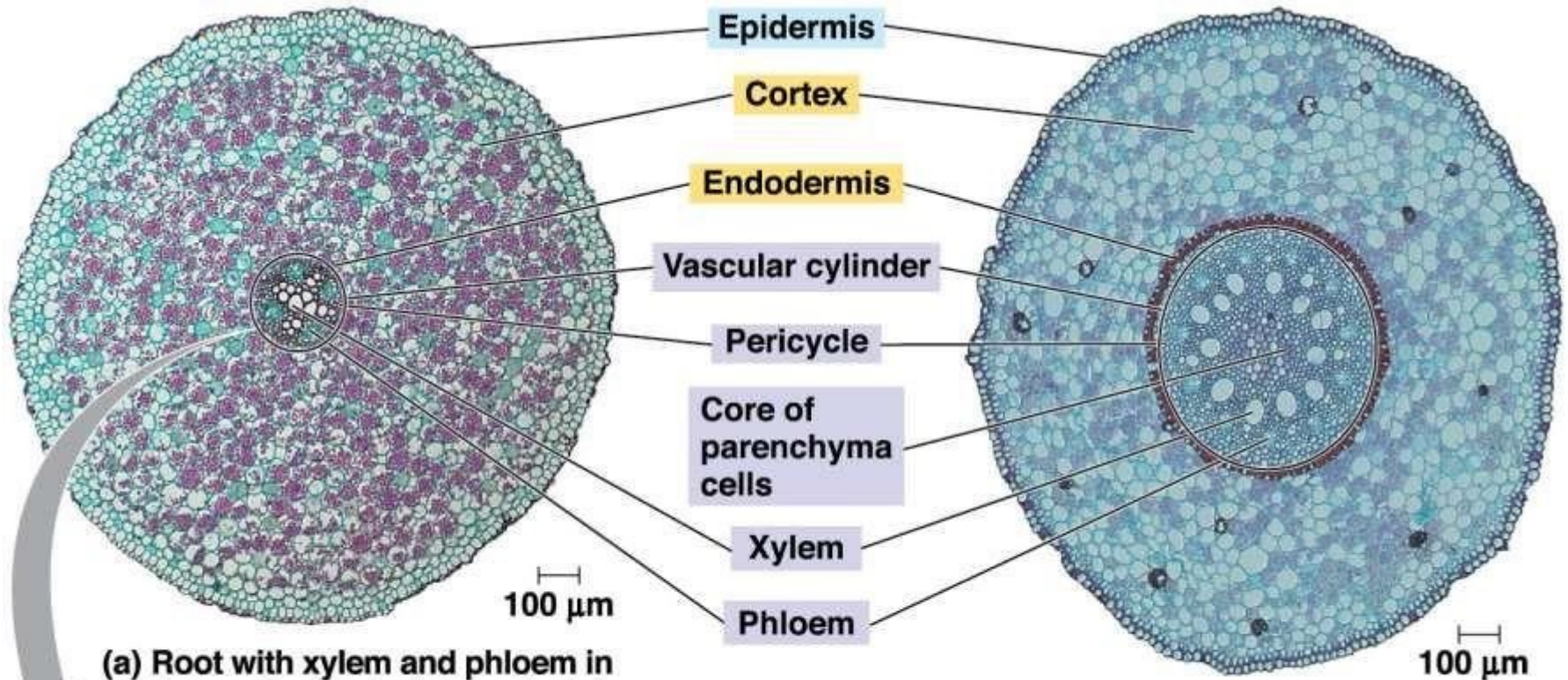
Located in the cambium cylinders.
Responsible for Secondary Growth (widening).
Adds thickness to woody plants, extruding wood and bark.

Roots

- Function:
 - **Absorb nutrients** from soils
 - **Anchor** plant
 - **Store food**
- Structure:
 - **Epidermis**: covers the entire surface of roots for absorption, root hairs
 - **Cortex**: have parenchyma cells with plastids for food storage
 - Give rise to mature ground tissue
 - Mostly parenchyma cells in roots
 - Storing carbohydrates
 - Transport water and salts from root hair to root centre
 - Allows extracellular diffusion of water, minerals, oxygen from root hairs inwards (large spaces between cells)

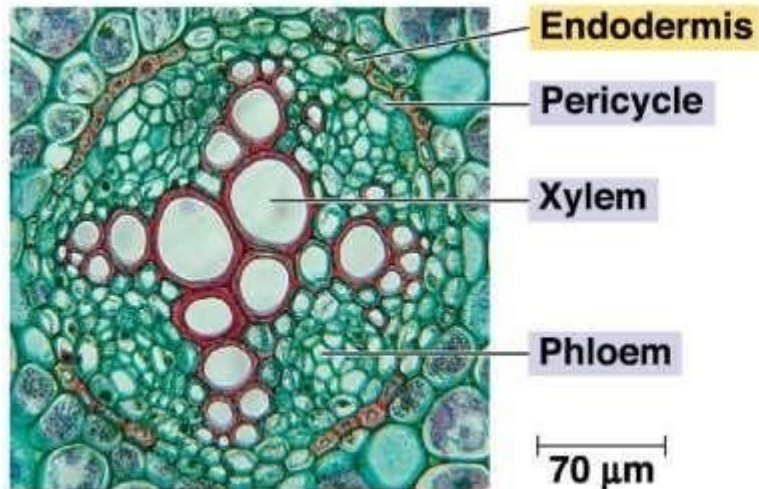
Roots

- **Endoderm**: innermost layer of cortex, a cylinder one cell thick, forming boundary with vascular cylinder. Surrounds vascular cylinder, each cell wrapped with Casparian strip, a continuous band of suberin (waxy materials).
- **Function**: select minerals that enter vascular cylinder & body of plant.
- **Stele/vascular cylinder**: have vascular bundle tissues surrounded by pericycle, from which lateral roots arise

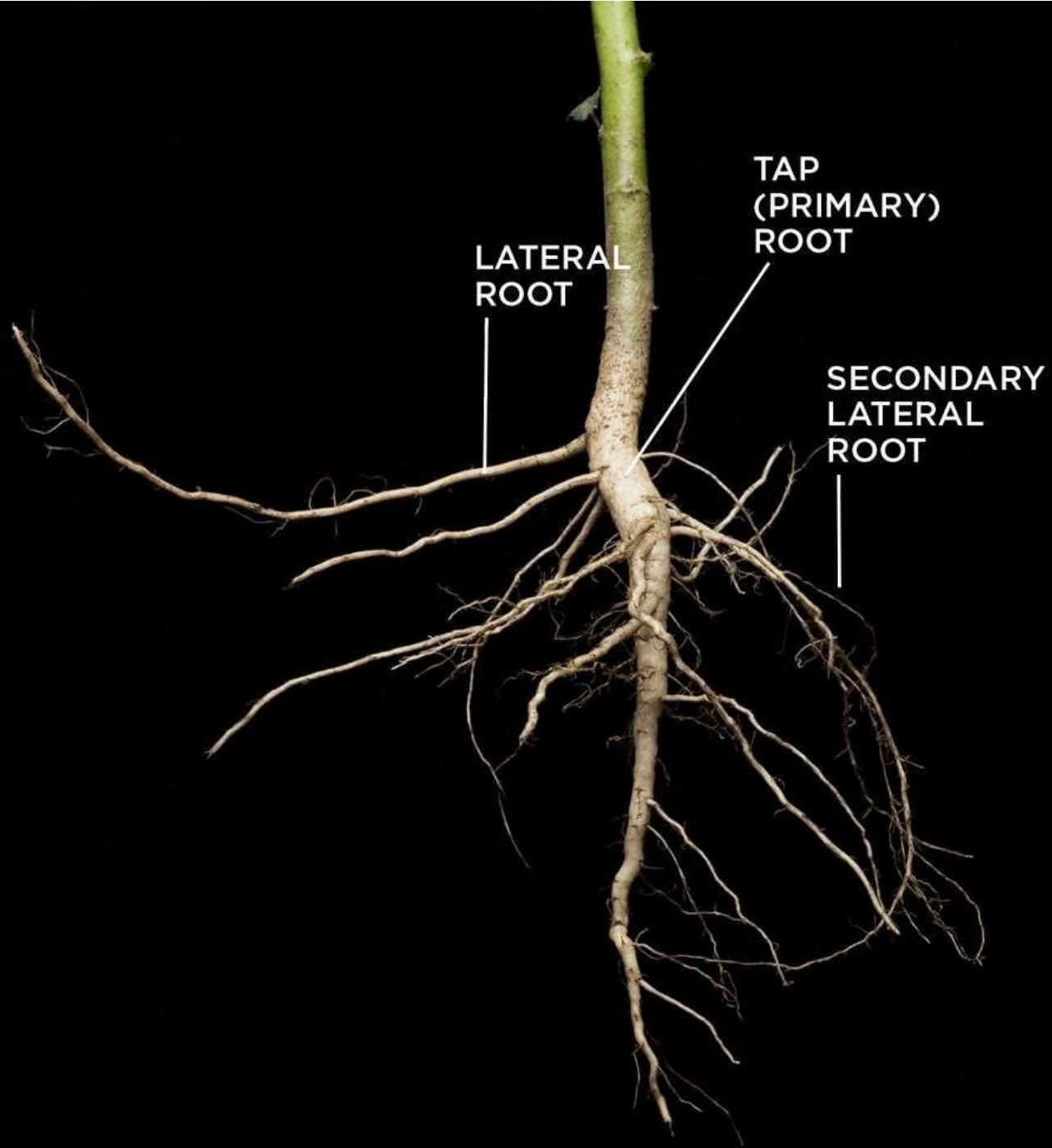


(a) Root with xylem and phloem in the center (typical of eudicots).

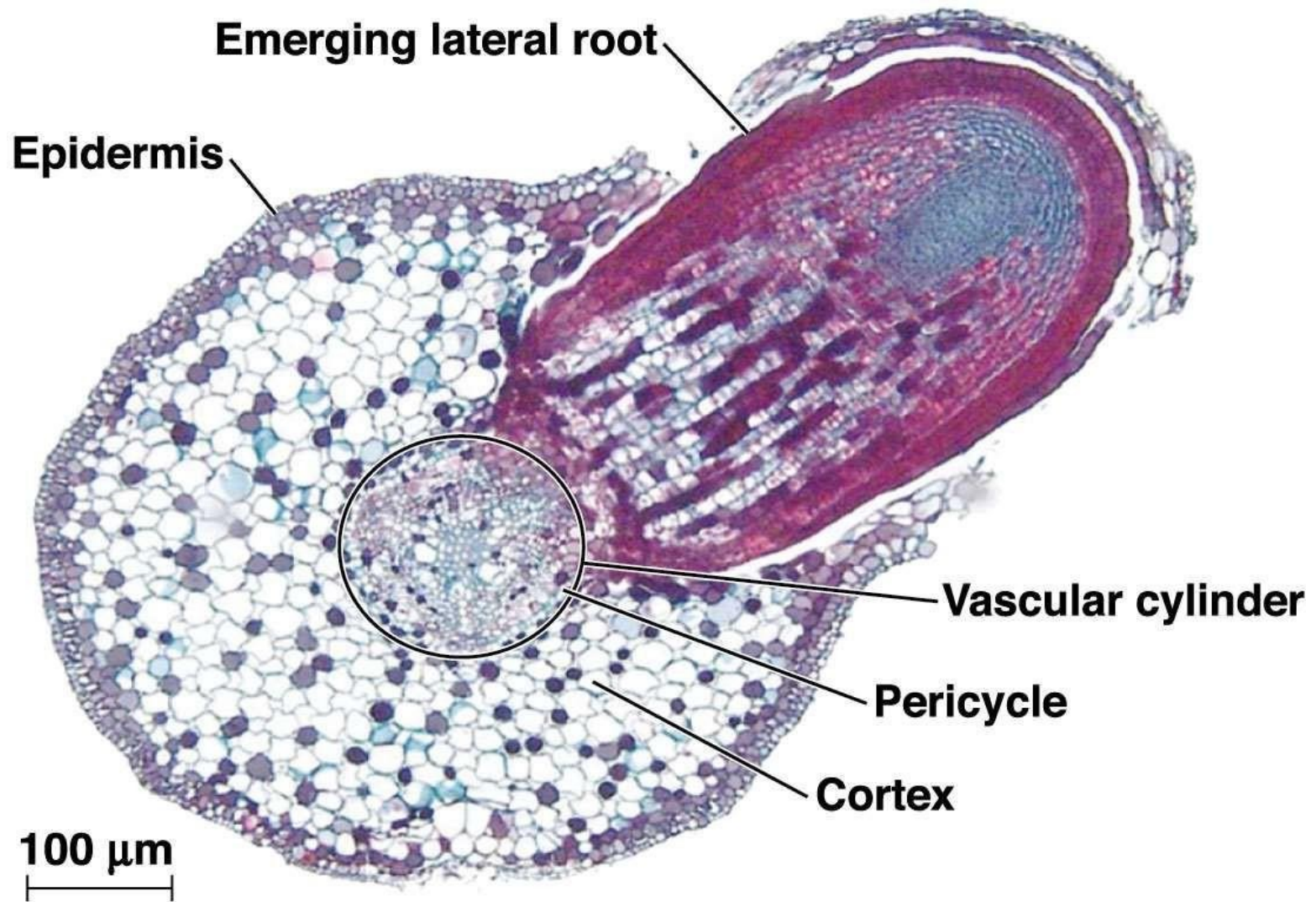
(b) Root with parenchyma in the center (typical of monocots).



Dermal
 Ground
 Vascular



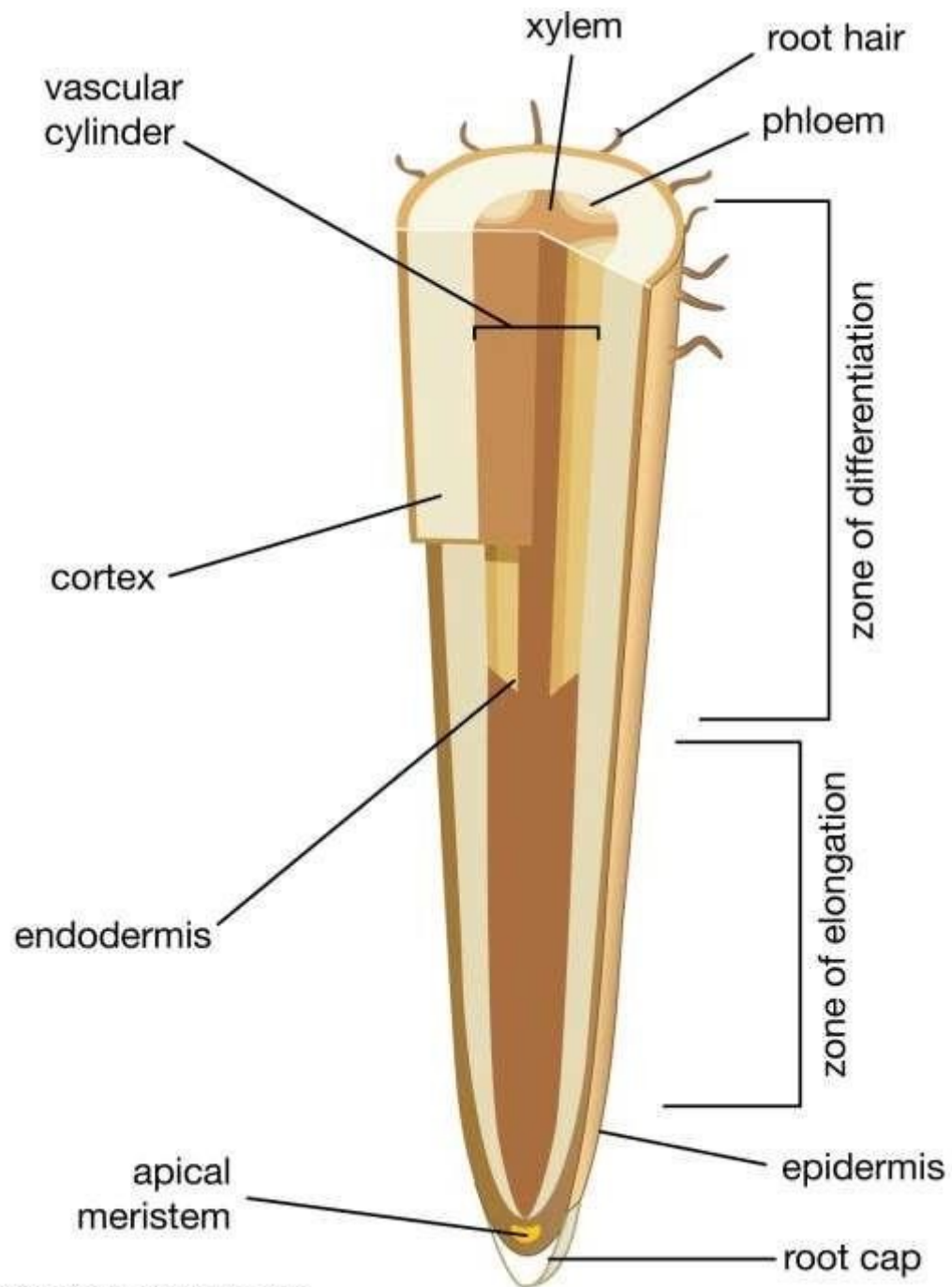
- **Formation of lateral root:** Lateral root originates in pericycle (outermost layer of vascular cylinder of root).



Primary growth lengthens roots and shoots

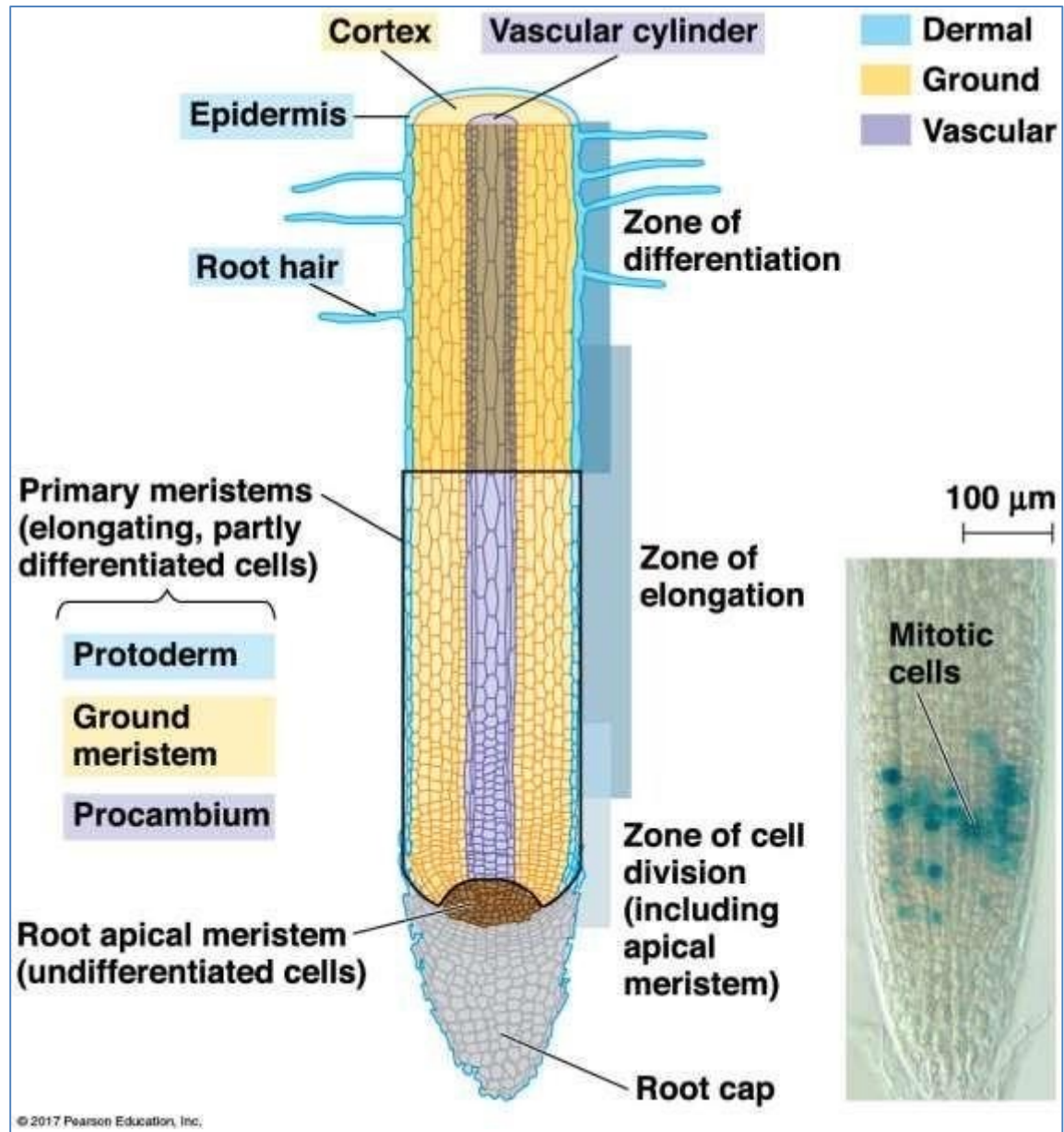
- Primary growth arises directly from cells produced by **apical meristems**
- **Herbaceous plants**: entire plant consists of primary growth
- **Woody plants**: recently formed parts (non-woody area) represents primary growth
- **Shoots and roots lengthens** as a result of cells derived from apical meristems.





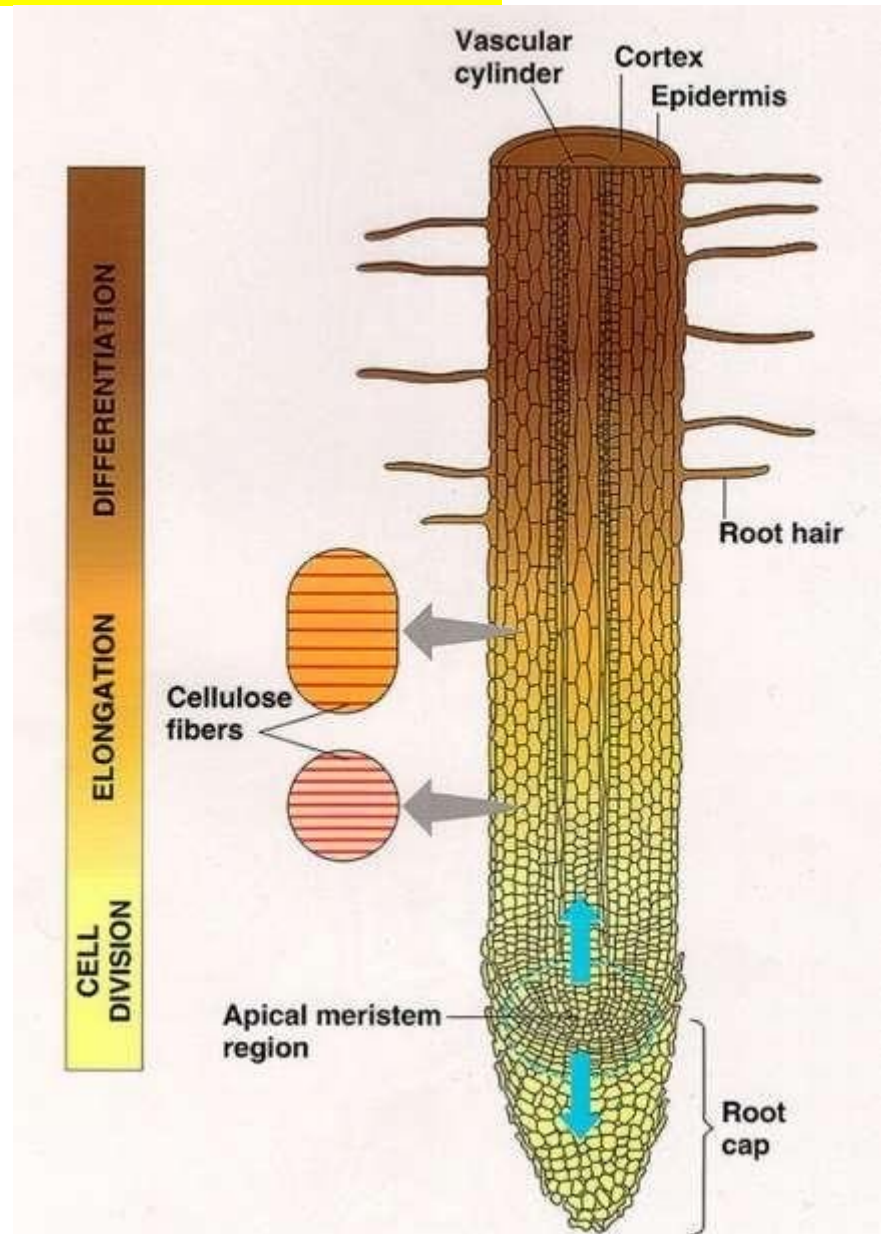
Primary Growth of Roots

- The root tip is covered by a **root cap**, which **protects** the delicate **apical meristem** as the root pushes through soil during **primary growth**



Primary Growth of Roots

- Produces the **epidermis**, **ground tissue**, and **vascular tissue**
- a) **Zone of cell division**: meristem cells actively dividing to **produce new cells** grow into soil
- b) **Zone of elongation**: cells elongate & responsible for **pushing root cap downward** deeper into soil
- c) **Zone of differentiation/maturation**: cells undergo **specialization** into **protoderm** (epidermis), ground **meristem** (cortex), **procambium** (1° xylem & phloem)



After growing carrots for one season, a gardener decides that the **carrots are too small**. Since carrots are biennials, the gardener **leaves the crop in the ground for a second year**, thinking the carrots will grow larger. Is this a good idea?

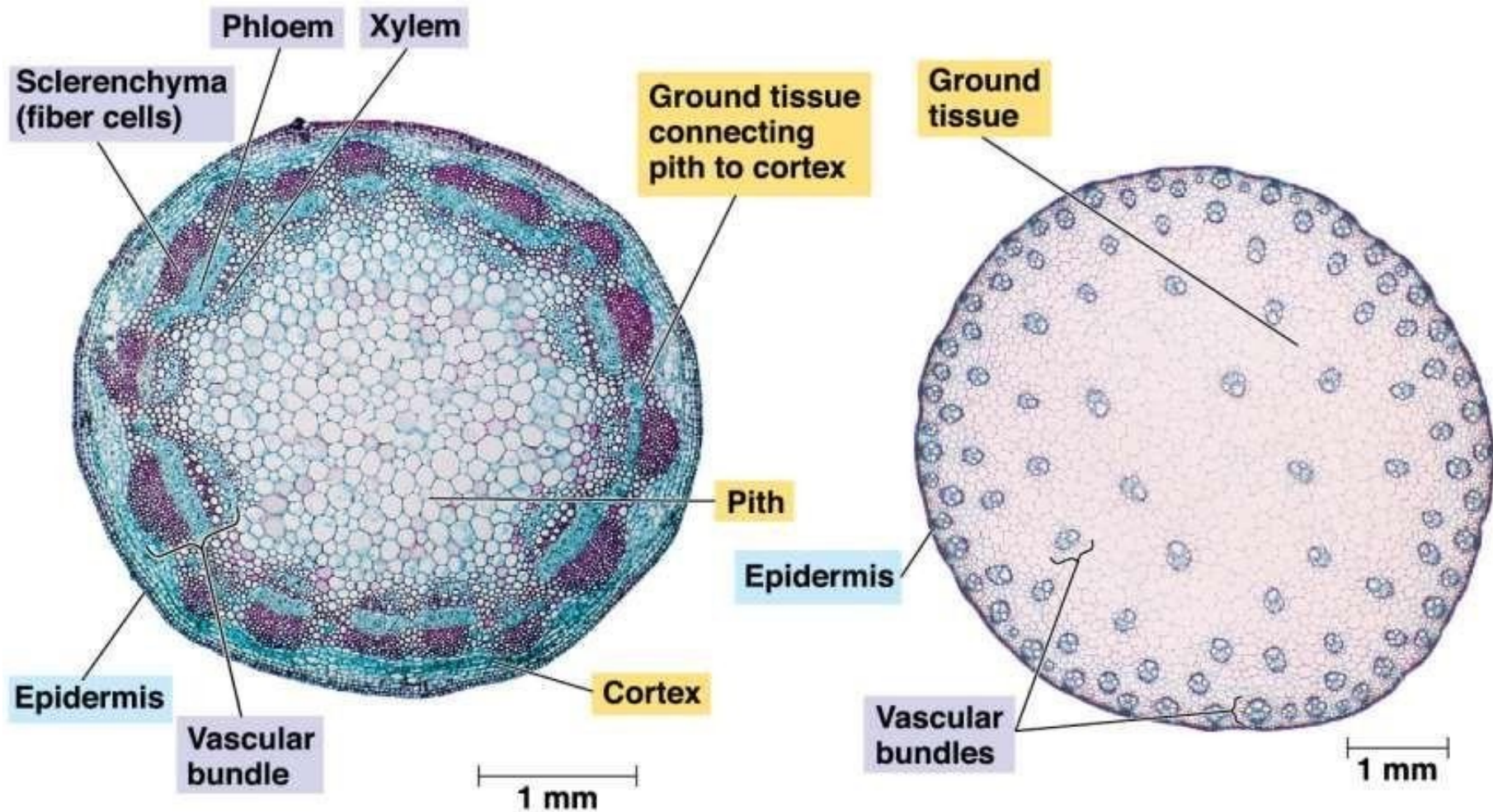
- No
- The carrot roots will probably be **smaller** at the end of the second year because the food stored in the roots will be **used to produce flowers, fruits and seeds**.



Stems: Tissue Organization

- Covered by an **epidermis** (one cell thick)
- Covered with **waxy cuticle** to prevent water loss
- Specialised epidermal cells in stem: **guard cells + trichome**
- **Vascular tissue:**
 - Runs the length of a stem in vascular bundles.
 - Lateral shoots develop from **axillary bud** meristems on stem's surface
- **Ground tissue:**
 - Mostly **parenchyma cells**
 - Collenchyma cells beneath epidermis **strengthen stems during primary growth**
 - Sclerenchyma cells (fiber) provide **support** in stem parts which no longer elongating

Stems: Tissue Organization

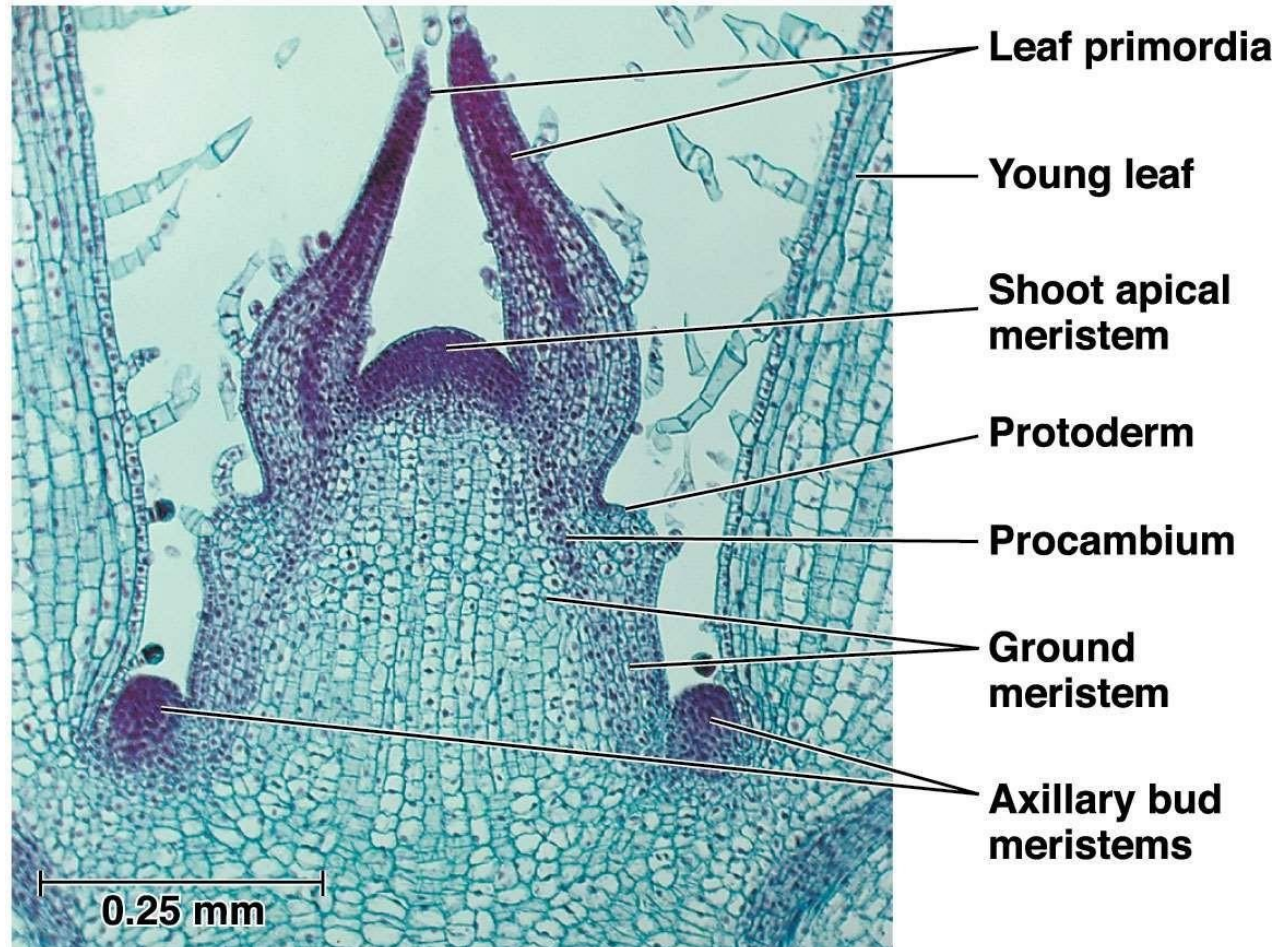


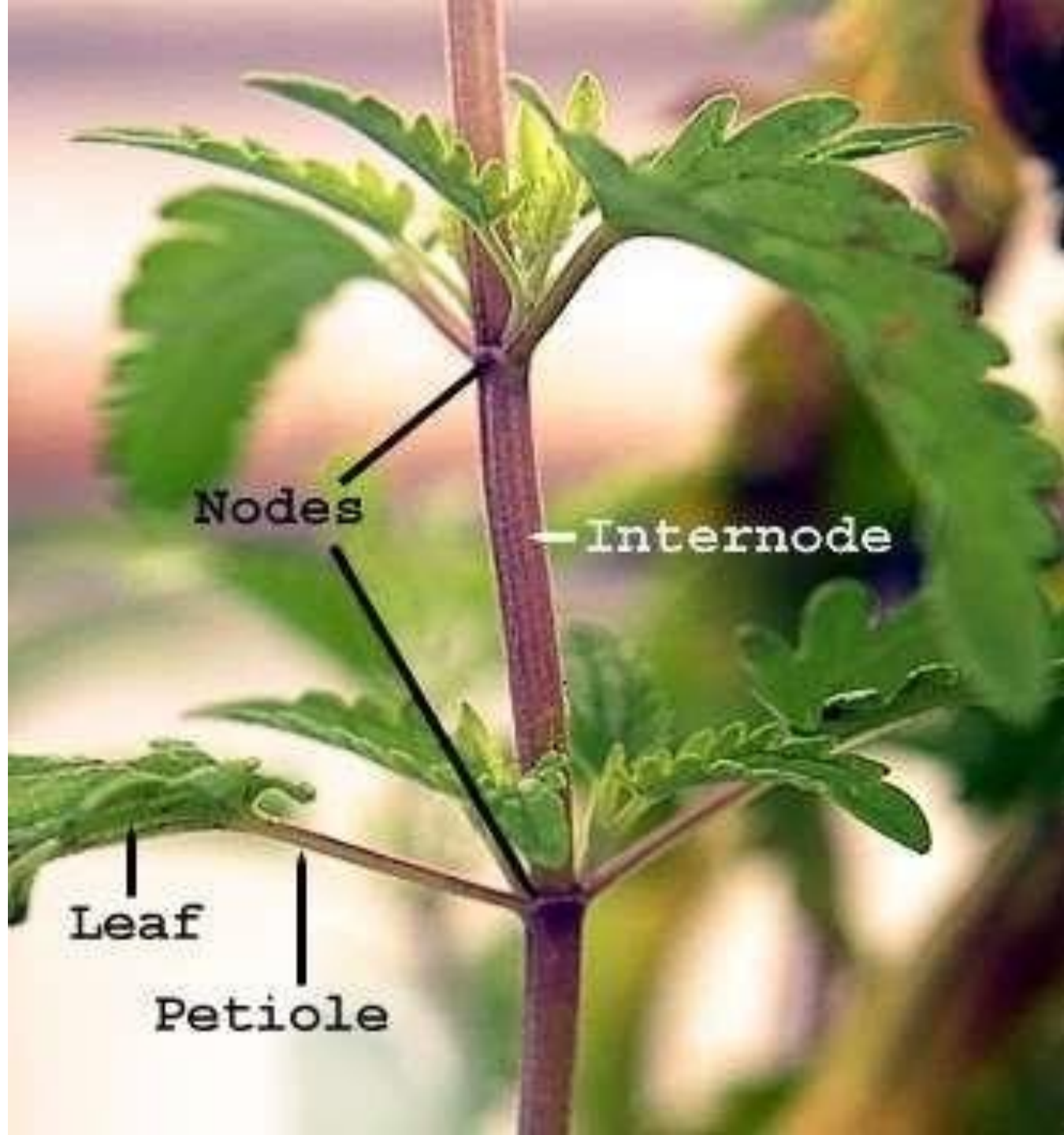
(a) Cross section of stem with vascular bundles forming a ring (typical of eudicots).

(b) Cross section of stem with scattered vascular bundles (typical of monocots).

Primary Growth of Shoots

- A shoot **apical meristem**
 - Is a **dome-shaped mass of dividing cells** at the tip of the terminal bud
 - Gives rise to a **repetition of internodes** and **leaf-bearing nodes**





Nodes

Internode

Leaf

Petiole

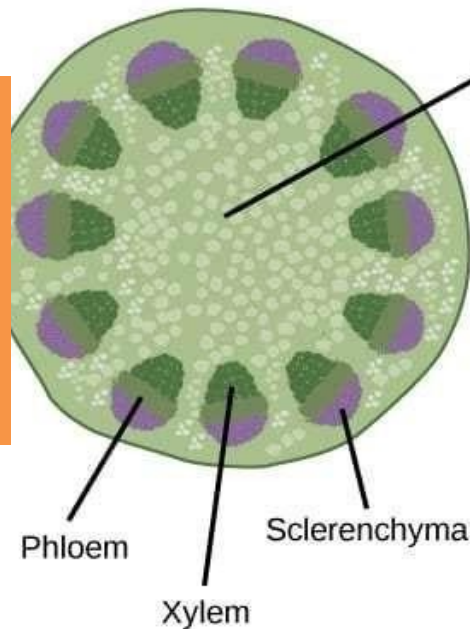
Secondary Growth Increases the Diameter of Stems and Roots in Woody Plants

Most land plants represent 2° growth – growth in thickness produced by lateral meristems

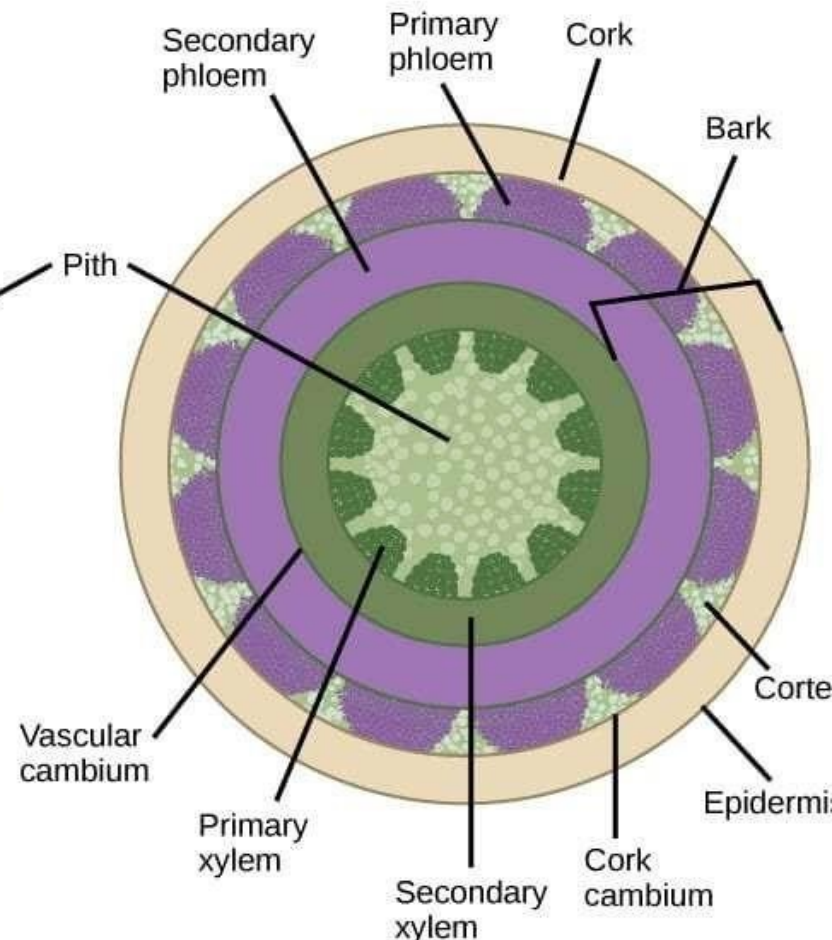
Vascular cambium : adds 2° xylem (wood) and 2° phloem, increases vascular flow + support to shoots

Cork cambium : produces **periderm** → tough, thick covering of waxy cells, protect stem from **water loss + invasion of insects, bacteria, fungi**

Primary growth



Secondary growth

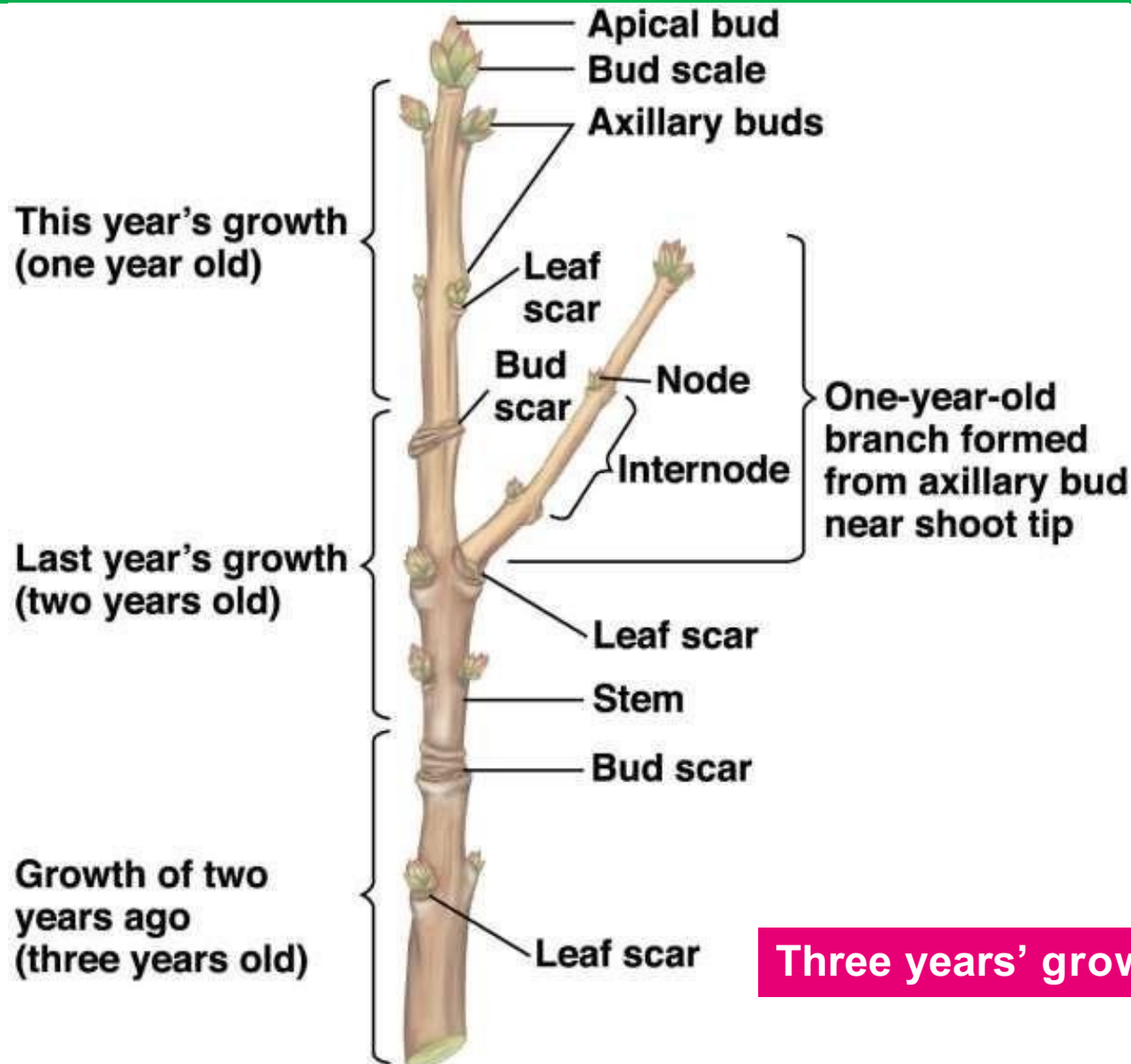


Secondary growth increases the diameter of stems and roots in woody plants

- Replaces epidermis with **2° dermal tissue (bark)**
- All **gymnosperms** (flowerless plants produce cones and seeds) + **many eudicots** undergo 2° growth, but unusual in monocots
- Occurs in **stems and roots of woody plants** (rarely in leaves)
- Wood: 2° xylem that accumulate over years

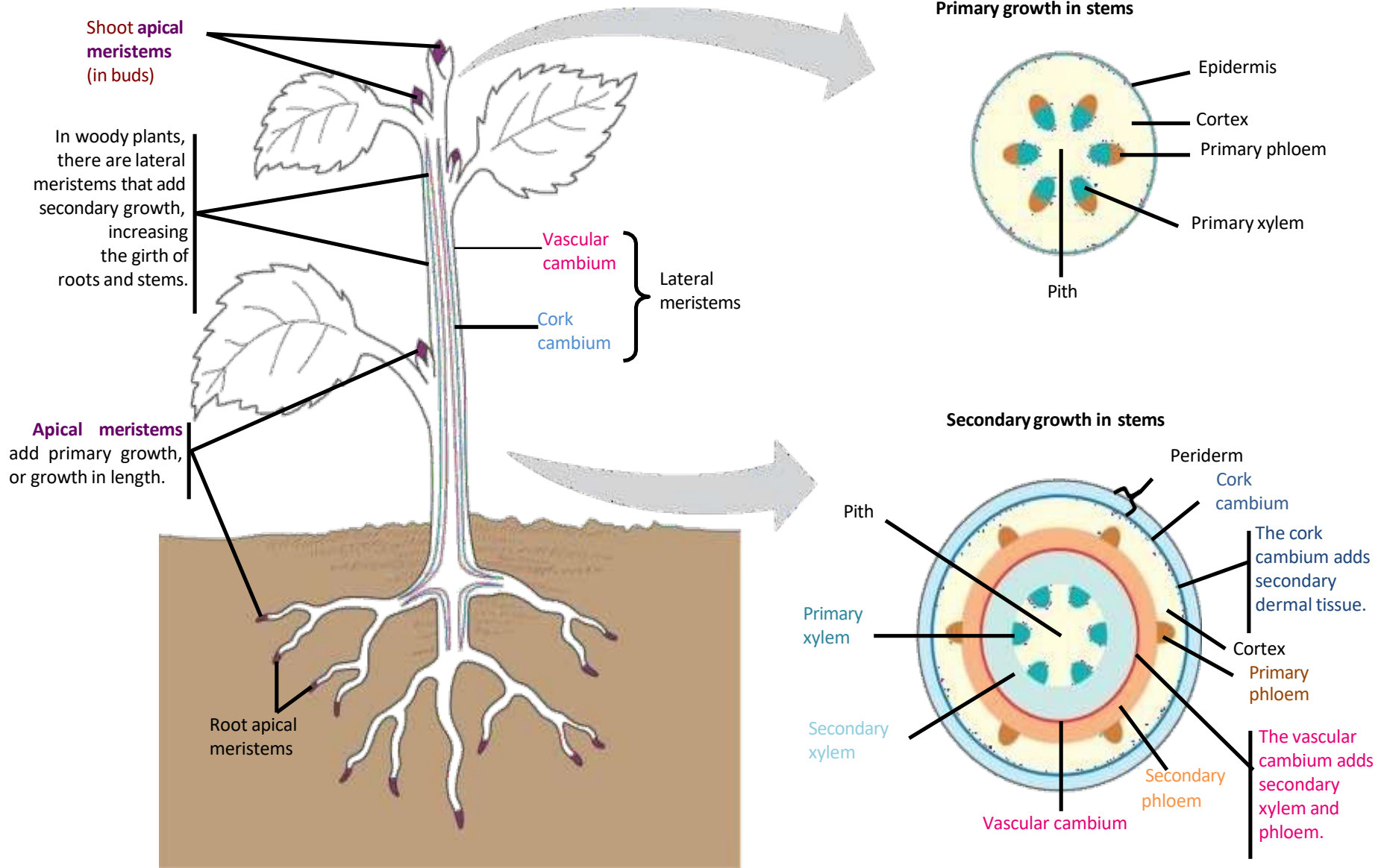


In woody plants, primary and secondary growth occur simultaneously but in different locations



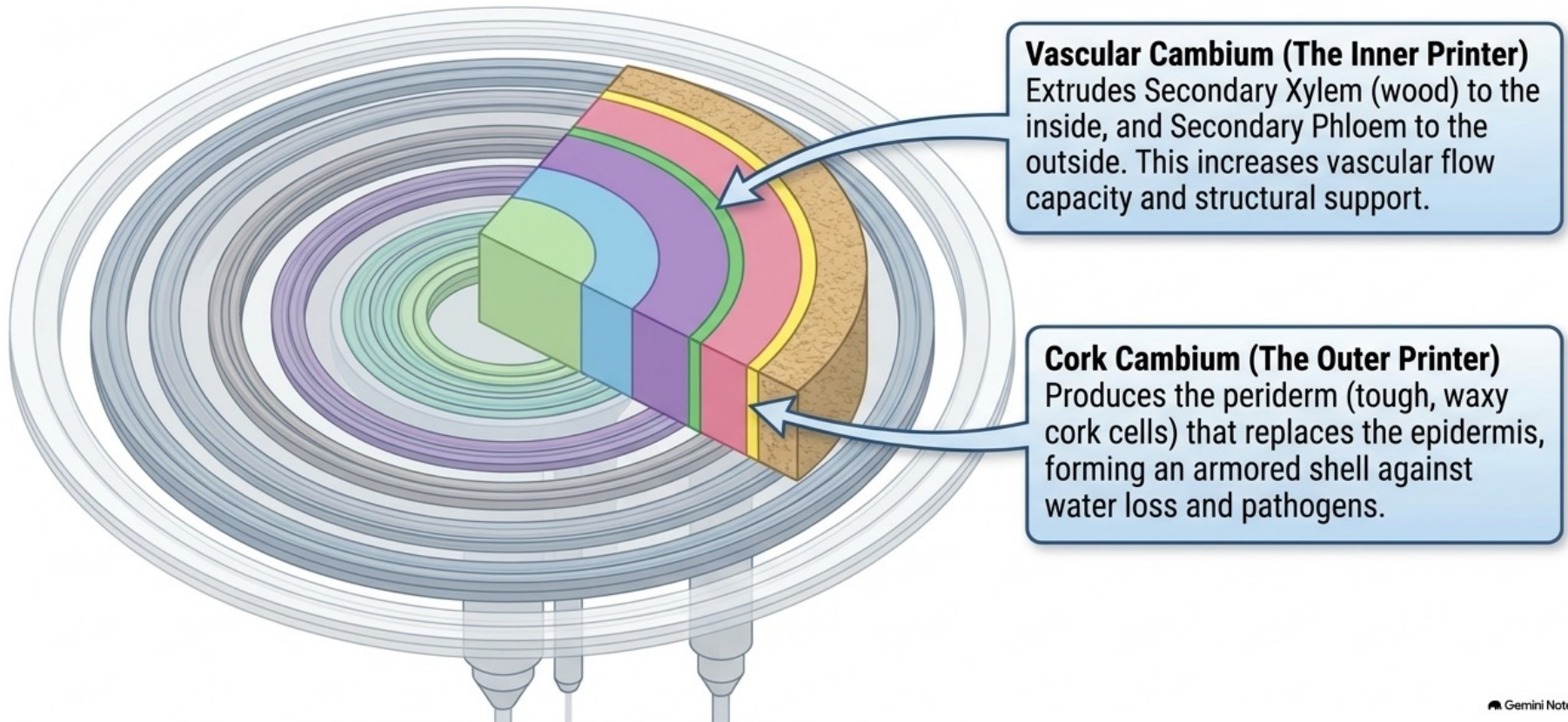
Three years' growth in a winter

An overview of primary and secondary growth

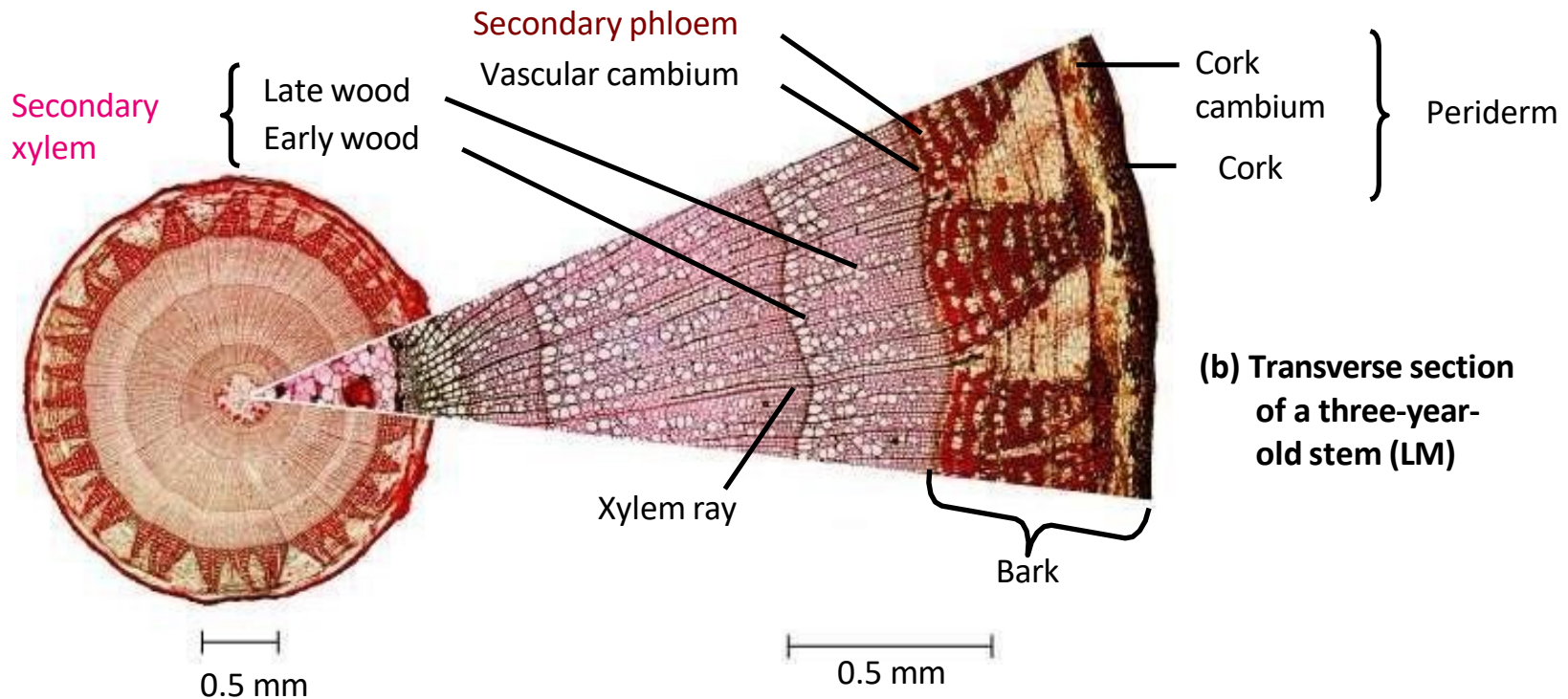


The 3D Printer of the Plant: Cambium Cylinders

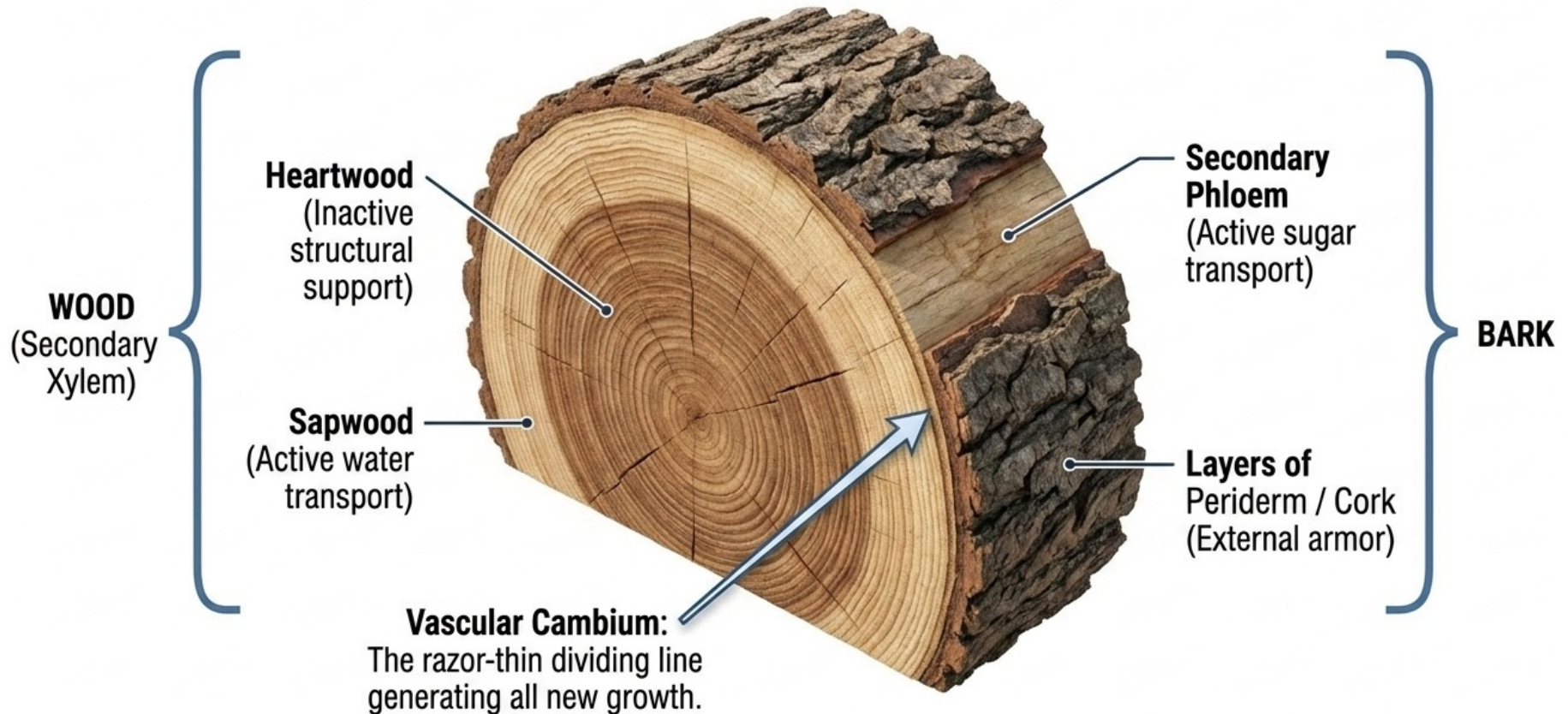
How lateral meristems extrude new tissues year over year



Secondary growth of a woody stem



Translating the Blueprint: Anatomy of a Tree Trunk



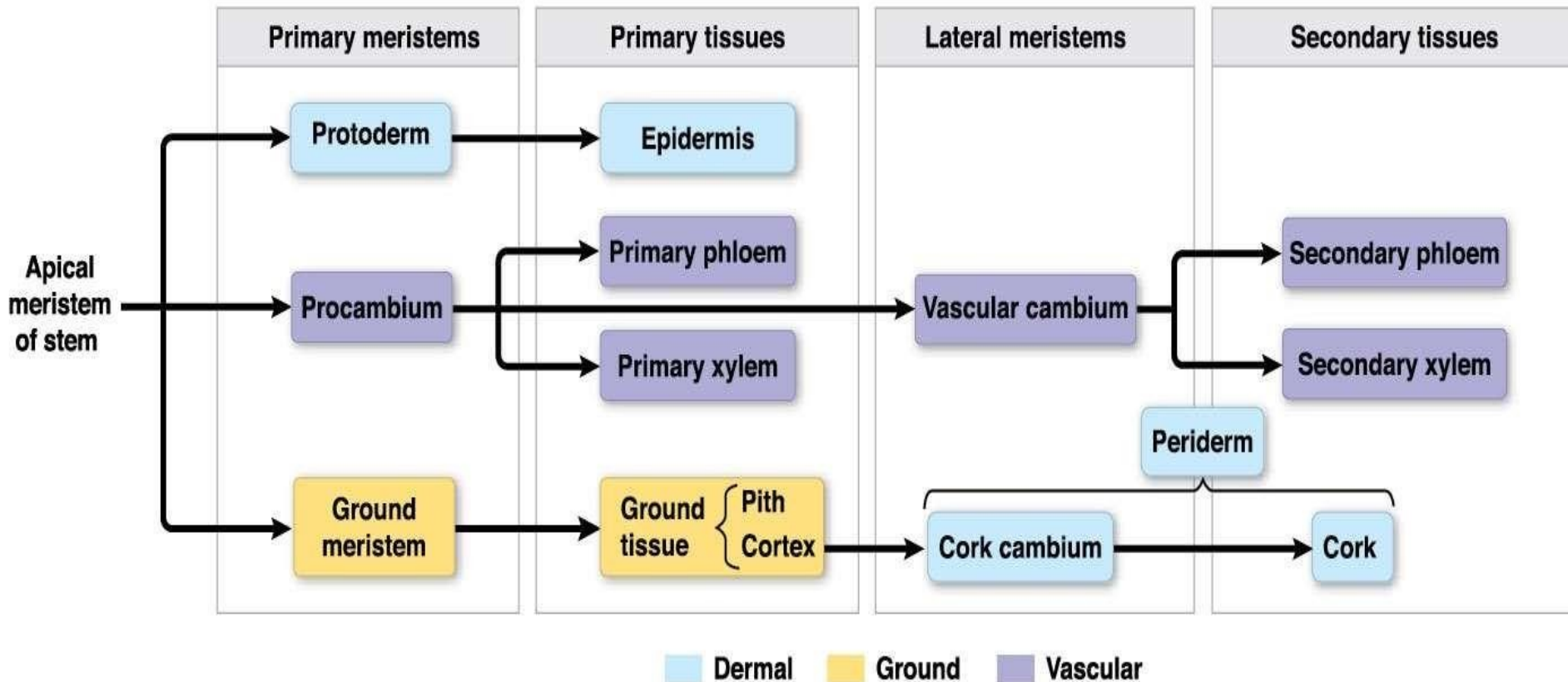
Is this tree living or dead?



- Wawona Sequoia tunnel in Yosemite National Park in California
- Was **cut in 1881** as a tourist attraction
- This giant sequoia (*Sequoiadendron giganteum*) **lived for another 88 years** before falling during a severe winter.
- 71.3 m tall and 2,100 years old
- **Trees can survive the excision of large portions of their heartwood.**



A summary of primary and secondary growth in a woody plant



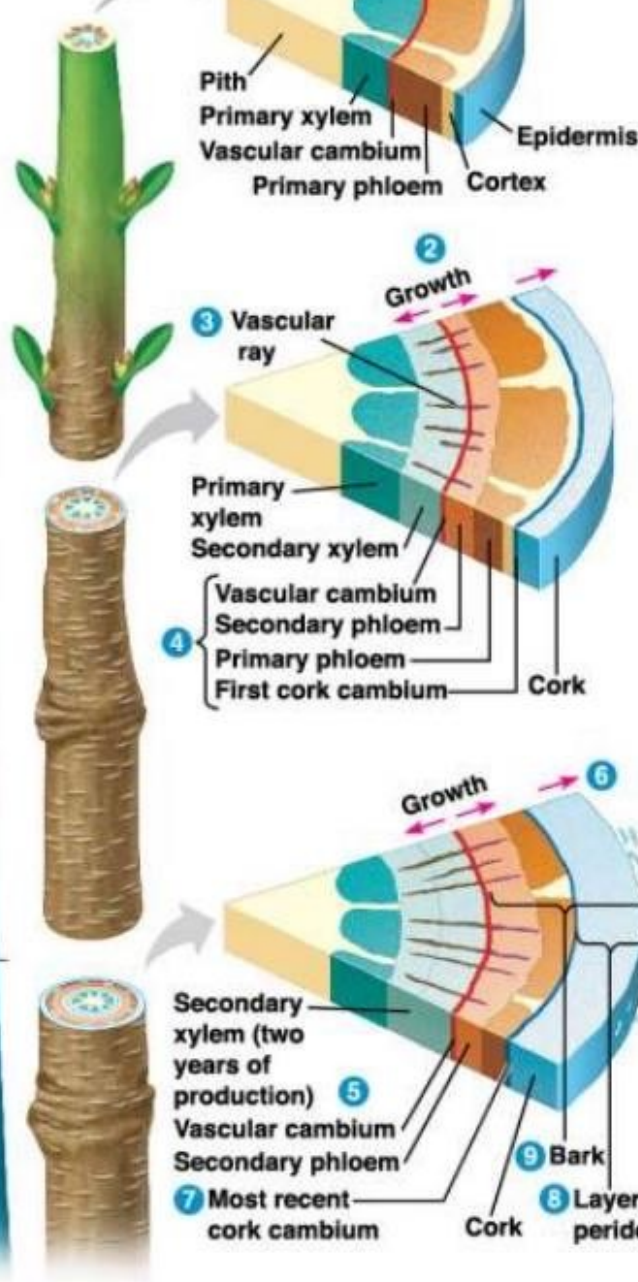
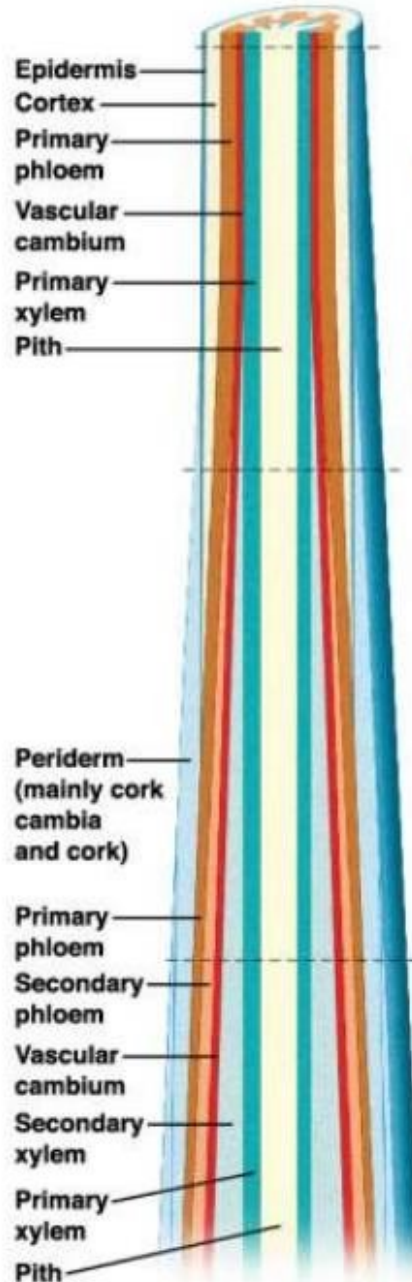
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- **Same meristem and tissues** are present in **woody roots**.
- But, ground tissue of a root is **not divided into pith and cortex**
- **Cortex cambium** arises from **pericycle**

Secondary growth

- **The cork cambium**
 - Gives rise to the secondary plant body's protective covering, or periderm
- **Periderm**
 - Consists of the cork cambium plus the layers of cork cells it produces
- **Bark**
 - Consists of all the tissues external to the vascular cambium, including secondary phloem and periderm

Secondary growth



- 1 Primary growth is nearing completion.
- 2 Secondary growth occurs in this region.
- 3 Some stem cells of the vascular cambium give rise to vascular rays.
- 4 Cork cambium develops from parenchyma cells in the cortex.
- 5 Year 2 of secondary growth. The vascular cambium produces more secondary xylem and phloem.
- 6 Outmost tissues rupture and are sloughed off.
- 7 Cork cambium re-forms deeper in the cortex.
- 8 Each cork cambium and the tissues it produces form a layer of periderm.
- 9 Bark consists of all tissues exterior to the vascular cambium.

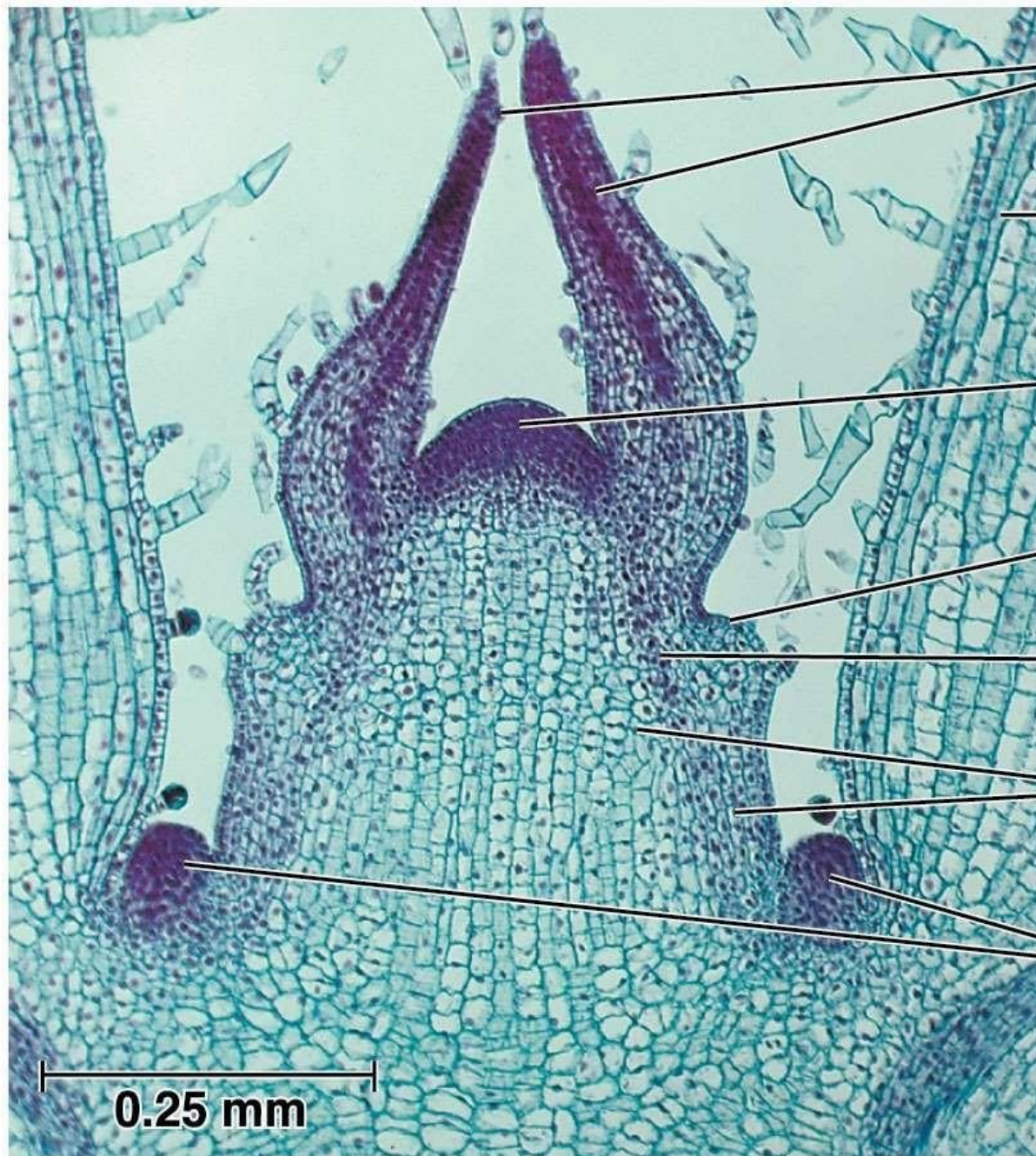
Video

- Secondary Growth of Woody Plants [Animation]
<https://www.youtube.com/watch?v=4KLFlz0SdIQ>
- Activity: Secondary growth: The vascular cambium
http://www.springer.com/cda/content/document/cda_downloaddocument/3401s.swf?SGWID=0-0-45-754415-0
- Tree Stories: How Tree Rings Reveal Extreme Weather Cycles
<https://www.youtube.com/watch?v=xmZO7aRgcW4>

Leaves



- Leaves developed from **leaf primordia** (projections emerged alongside shoot apical meristem)
- **Secondary growth** in leaves is **minor/nonexistent**
- **Epidermis** covered by **waxy cuticle**, except
- It is interrupted by **stomata** → allow **CO₂ and O₂ exchange** between the surrounding air and the photosynthetic cells inside a leaf
- The ground tissue in a leaf
 - **palisade mesophyll, spongy mesophyll**
 - Is sandwiched between the upper and lower epidermis
- The vascular tissue of each leaf
 - Is **continuous** with the vascular tissue of the stem



Leaf primordia

Young leaf

**Shoot apical
meristem**

Protoderm

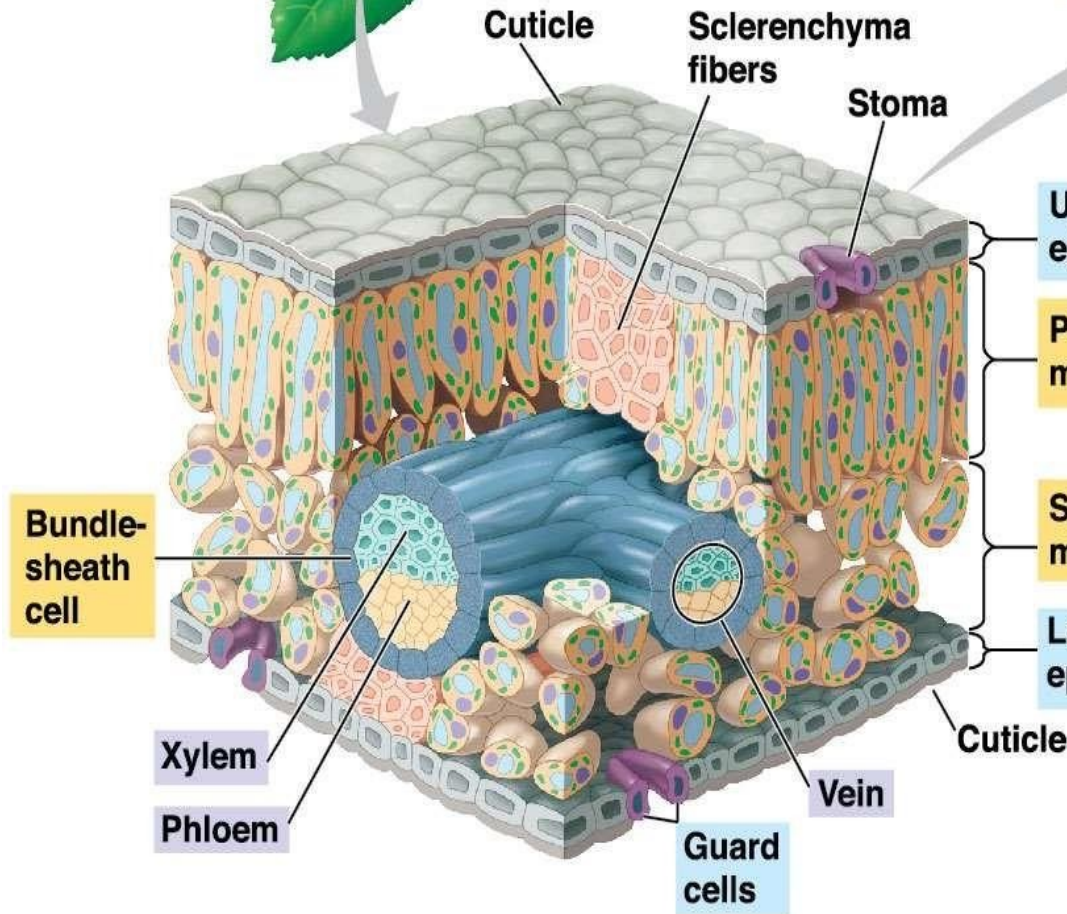
Procambium

**Ground
meristem**

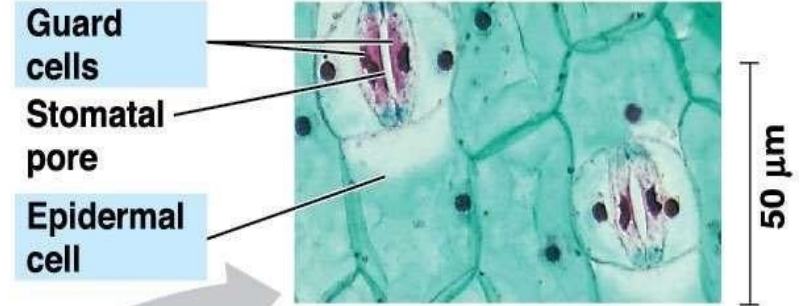
**Axillary bud
meristems**

0.25 mm

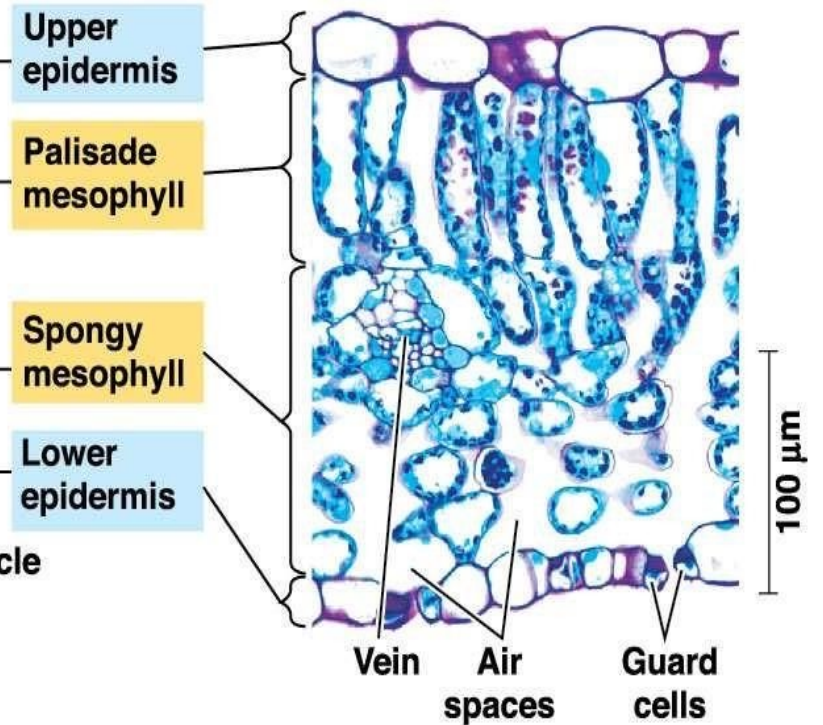
- Dermal
- Ground
- Vascular



(a) Cutaway drawing of leaf tissues



(b) Surface view of a spiderwort (*Tradescantia*) leaf (LM)



(c) Cross section of a lilac (*Syringa*) leaf (LM)

If a plant species has **vertically oriented leaves**, would you expect its mesophyll to be divided into spongy and palisade layers?

- No
- Because vertically oriented leaves such as maize, can **capture light equally well on both sides of leaf**.
- Have mesophyll **not differentiated into palisade and spongy** cells that are
- Vertically oriented leaves usually have **stomata on both leaf surfaces**.

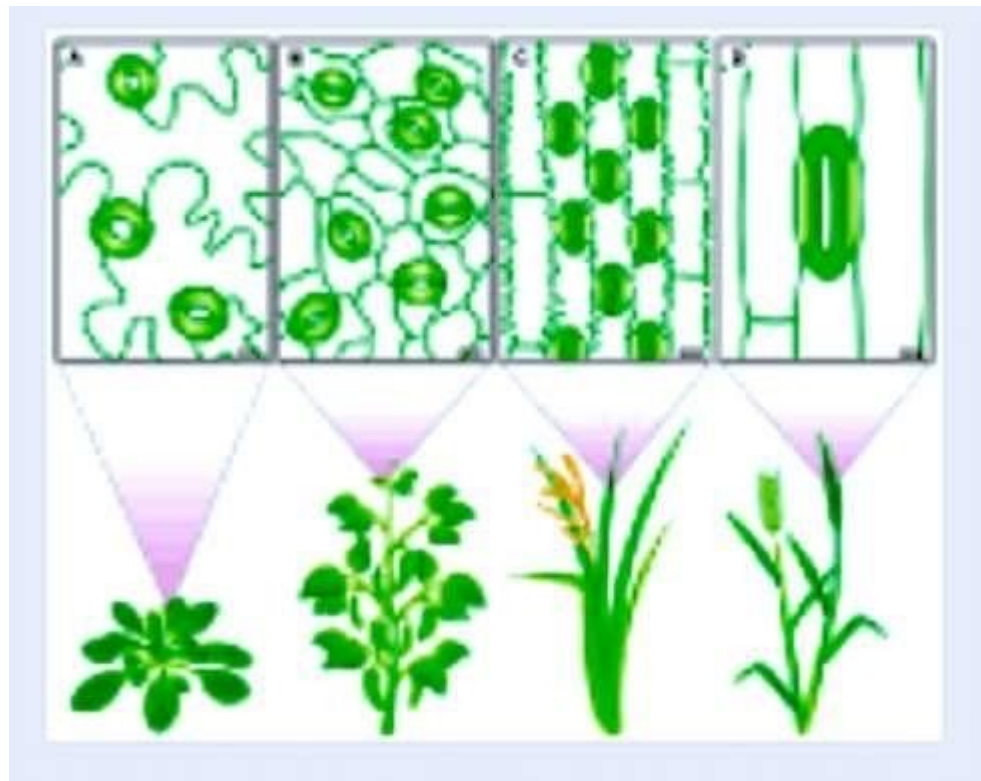


Stomata: Major Pathways for Water Loss

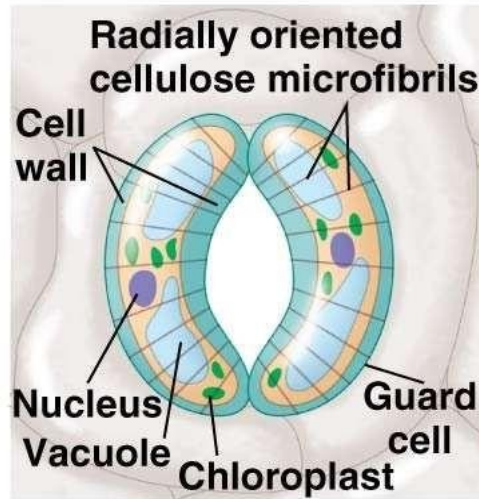
- About **95% of the water a plant escapes through stomata**
- Waxy cuticle limits water loss
- Each stoma is flanked by **a pair of guard cells**
- Guard cells **control the diameter of stoma by changing the shape – widening / narrowing the gap**
- Stomata help **regulate the rate of transpiration**



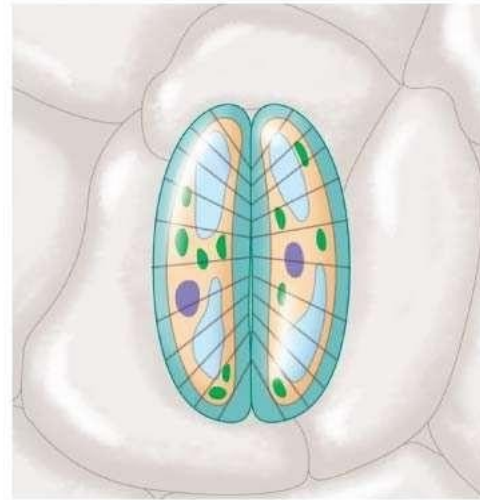
- Amount of **water lost** by a leaf depends on:
 - **Number of stomata**
 - **Average size of pores**
- **Desert plants** have **lower stomatal densities**, than marsh plants
- By measuring the **stomatal density of leaf fossils**, scientists gained idea into levels of **atmospheric CO₂ in past climates**.
 - Woodland species : Stomatal density **decreased** (since 1927), CO₂ **increased**



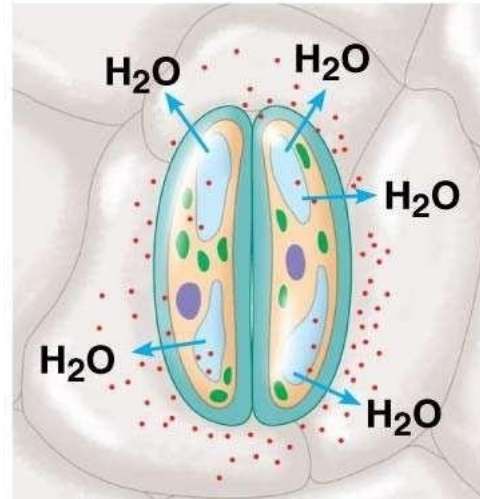
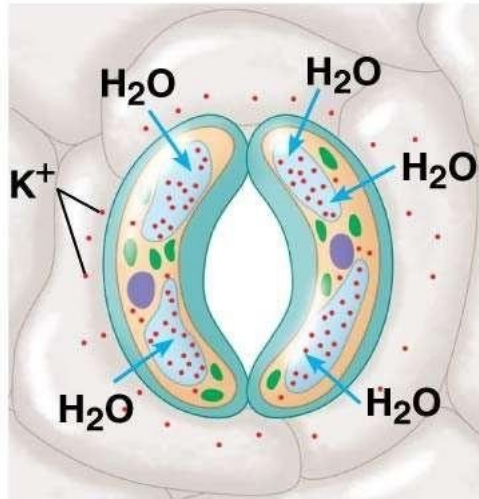
**Guard cells turgid/
Stoma open**



**Guard cells flaccid/
Stoma closed**

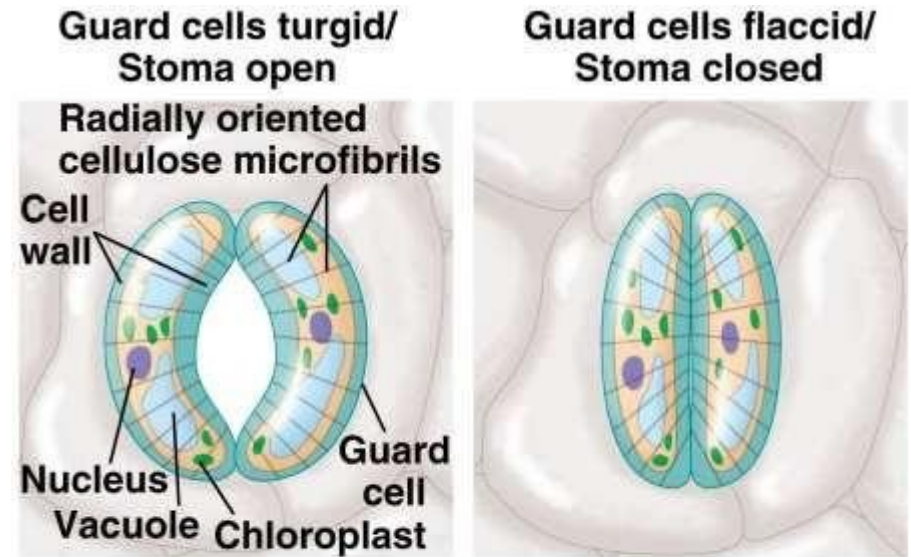


(a) Changes in guard cell shape and stomatal opening and closing (surface view)



(b) Role of potassium ions (K⁺) in stomatal opening and closing

Mechanisms of Stomatal Opening and Closing



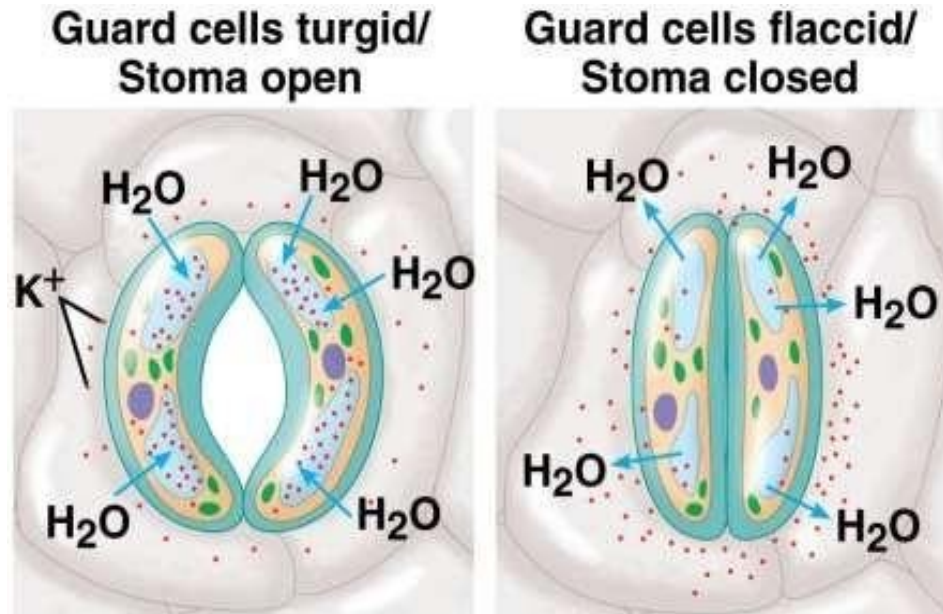
(a) Changes in guard cell shape and stomatal opening and closing (surface view)

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- When **guard cells take in water** from neighboring cells (osmosis), become more **turgid**
- **Bowing outward** ↑ **pore size**, cells **lose water**, becomes **flaccid**,
- In most angiosperms, cell walls of guard cells are
 - **uneven in thickness**
 - **Orientation of cellulose microfibrils** caused the guard cells to **bow outward** when turgid
 - Cells **lose water, flaccid** → **less bowed, pore closes**.

Turgor Pressure

- Reversible absorption and loss of K^+ $\longrightarrow$ Changes in turgor pressure in guard cells
- Guard cells actively accumulate K^+ from neighboring epidermal cells
- Flow of K^+ across plasma membrane of guard cells $\uparrow$ **membrane potential by proton pumps**



(b) Role of potassium ions (K^+) in stomatal opening and closing

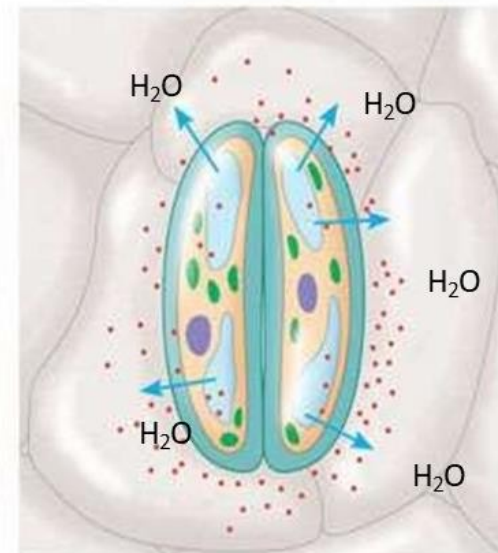
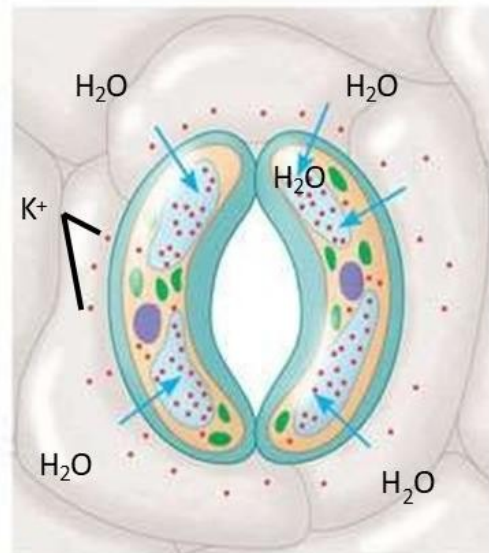
Factors for Opening & Closing of Stomates

- $\downarrow$ of CO_2 within air spaces of leaf during photosynthesis will open up stomata
- $\uparrow$ of potassium ions in the guard cells, lower ψ_w , H_2O enter guard cells then become turgid
- Active transport of H^+ out of the guard cells into surrounding epithelial cells
- $\uparrow$ photosynthesis rate, more glucose produced in guard cells

Role of K^+ ions in stomatal opening/closing

- **Guard cells actively accumulate K^+** from neighboring epidermal cells
- Flow of K^+ across membrane coupled with **proton pump** that alters **osmotic balance/membrane potential**
- **H^+ pumped out** of guard cell, **K^+ transported in**
- Accumulation of K^+ causes **Ψ_s to become more -ve** within guard cells.
- Ψ_s increases, H_2O flow into guard cells by osmosis.
- **$[K^+]$ in guard cells during opening > closing**

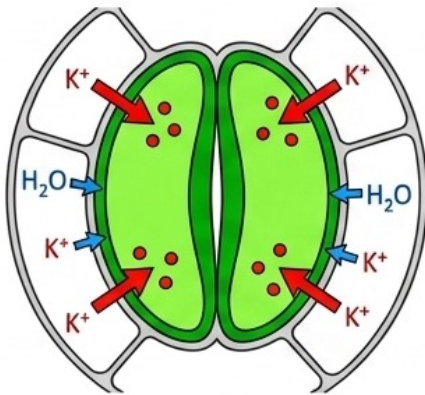
- The **transport of K^+** (potassium ions, symbolized here as red dots) across the **plasma membrane and vacuolar membrane** causes the **turgor** changes of guard cells.
- Also the **uptake of anions** (eg: malate and chloride ions)
- Contribute to guard cell swelling



The Stomatal Hydraulic Engine

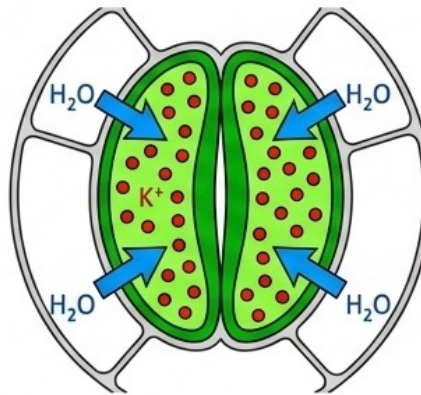
Converting chemical triggers into physical movement.

Step 1: The Trigger
(Active Transport)



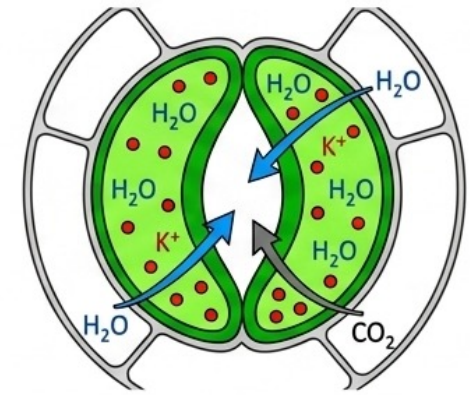
Guard cells actively pump Potassium ions (K^+) inside, increasing the membrane potential.

Step 2: The Fuel
(Osmosis)



The high concentration of K^+ causes water to rush into the guard cells from surrounding tissues via osmosis.

Step 3: The Action
(Turgor Pressure)



As cells fill with water, turgor pressure rises. Because the inner cell walls are thicker, the cells physically bow outward, opening the vent to allow CO_2 in and H_2O/O_2 out.

Video

The Mechanism of Opening and Closing the Stomatal Pore [21st century Education]

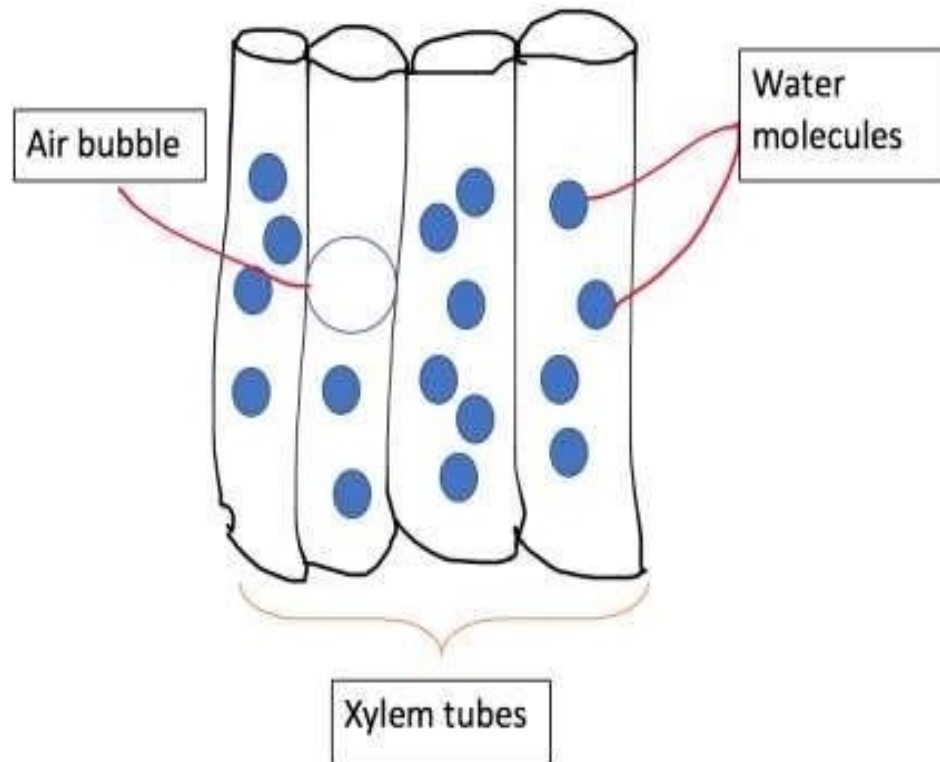
- https://www.youtube.com/watch?v=8DiVr0yn_QE

What if?

If you buy cut flowers, why might the florist recommend **cutting the stems underwater** and then transferring the flowers to a vase while the cut ends are still wet?



- After flowers are cut, **transpiration from any leaves and petals** (modified leaves) will **continue to draw water** up the xylem.
- If the cut flowers are **transferred directly to a vase**, **air pockets** in xylem **prevent delivery of water** from the vase to the flowers.
- Cutting stems again underwater (few centimeters from original cut) will **break off the xylem above the air pocket**.
- Water droplets prevent another air pocket to form while flowers are transferred to a vase.



Transport in Plants

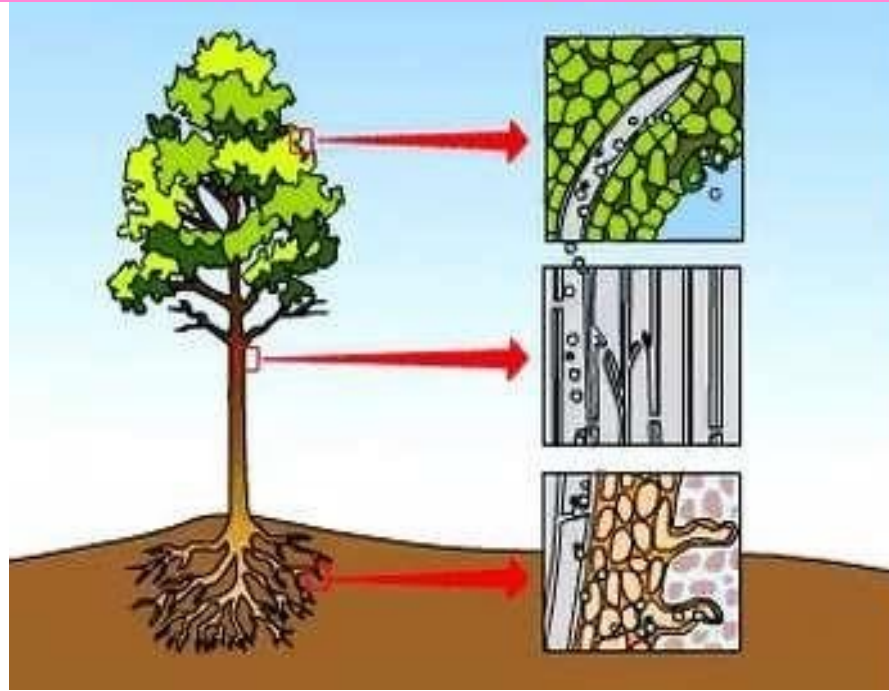


Video: Plant transport

- Plant Transport: Xylem and Phloem, Transpiration [3D Animation]

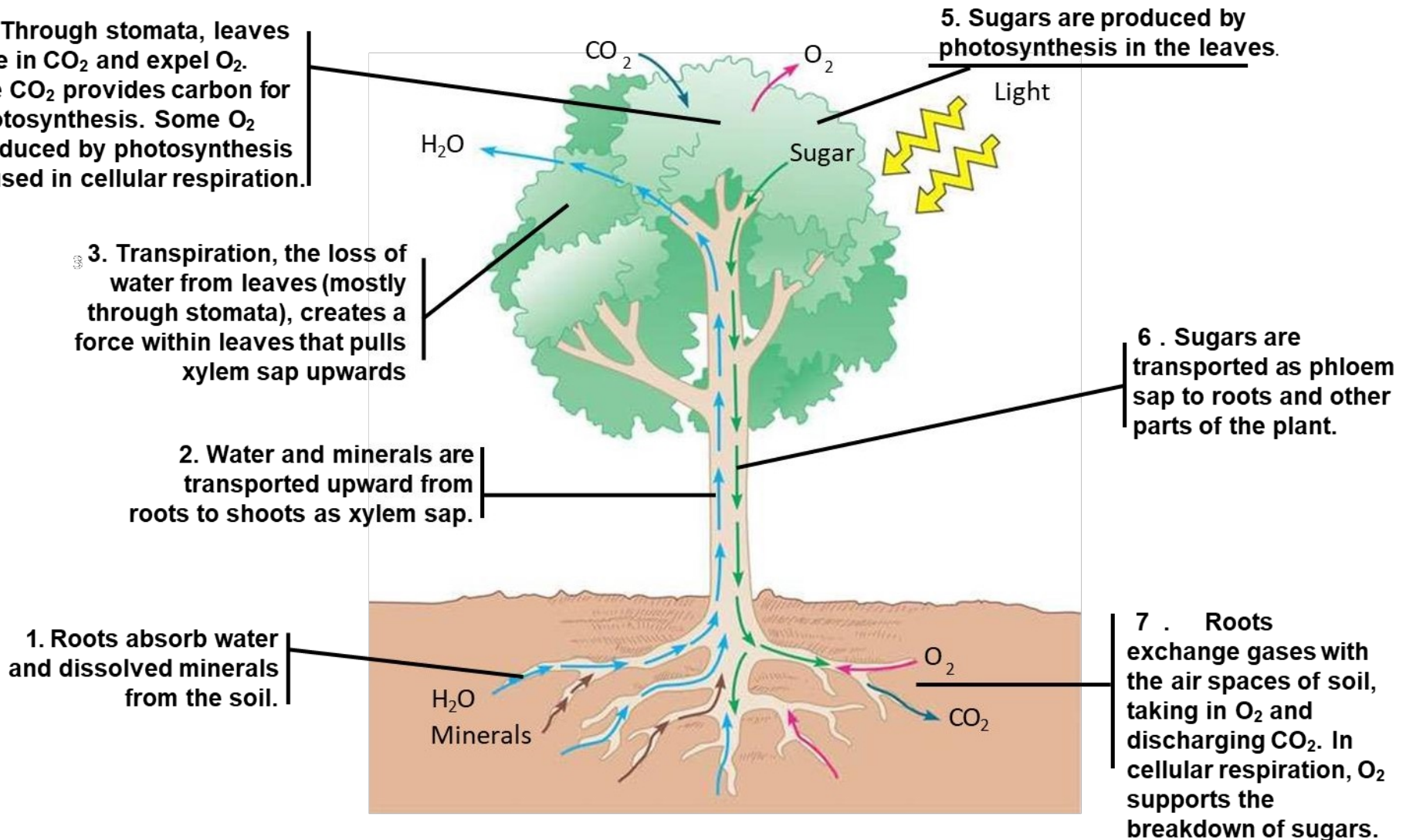
<http://www.youtube.com/watch?v=xGCnuXxbZGk>

Transport in Plants



- Transport in vascular plants occurs on **three scales**
 - Transport of water and solutes by **individual cells**, such as root hairs
 - **Short-distance transport** of substances from cell to cell at the levels of tissues and organs
 - **Long-distance transport** within xylem and phloem at the level of the whole plant

Transport in Vascular Plants



Ascent of Xylem Sap

- Transpiration pull
- Cohesion & adhesion
- Root pressure

Outside air Ψ
= -100.0 MPa

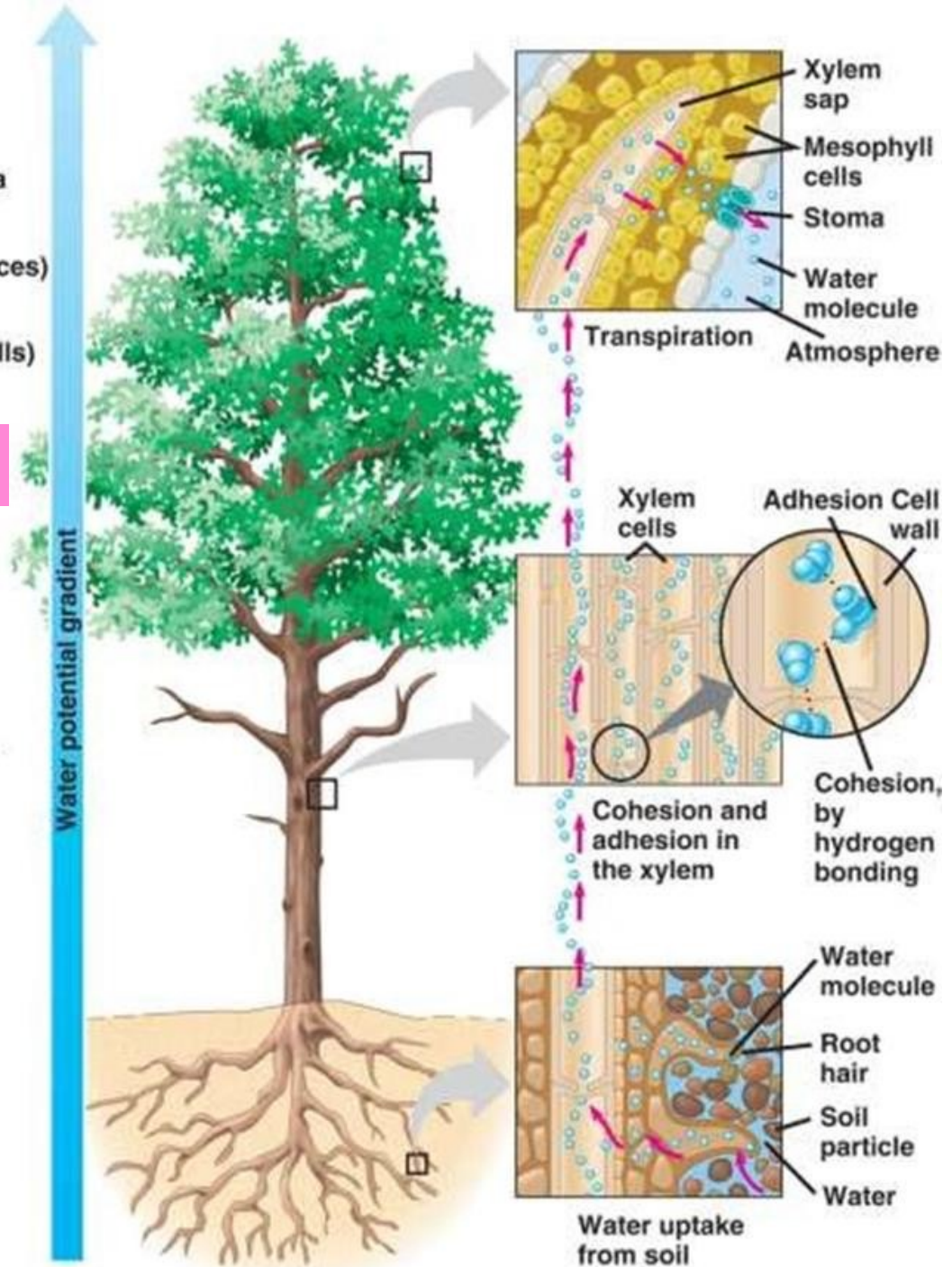
Leaf Ψ (air spaces)
= -7.0 MPa

Leaf Ψ (cell walls)
= -1.0 MPa

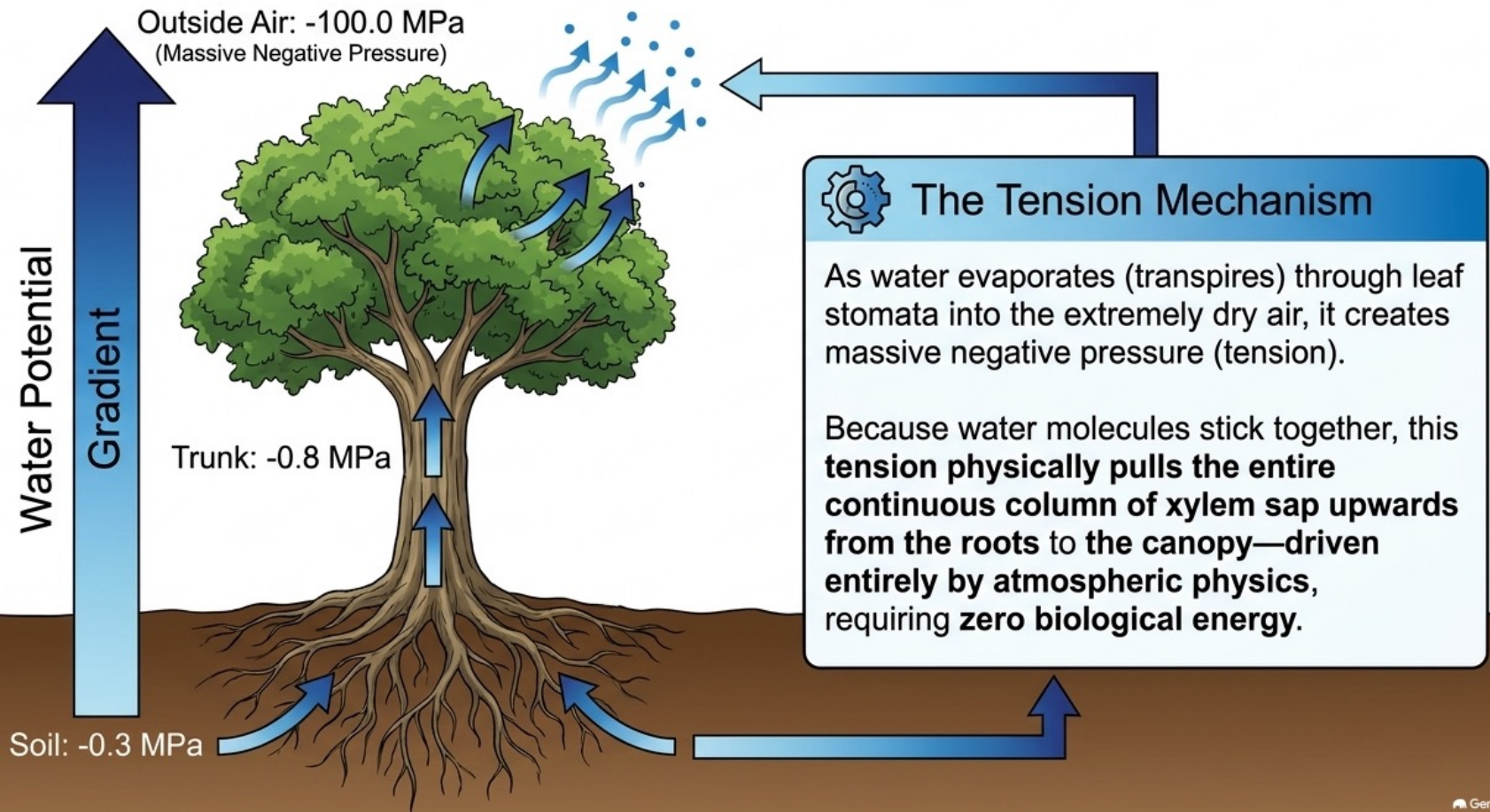
Trunk xylem Ψ
= -0.8 MPa

Root xylem Ψ
= -0.6 MPa

Soil Ψ
= -0.3 MPa

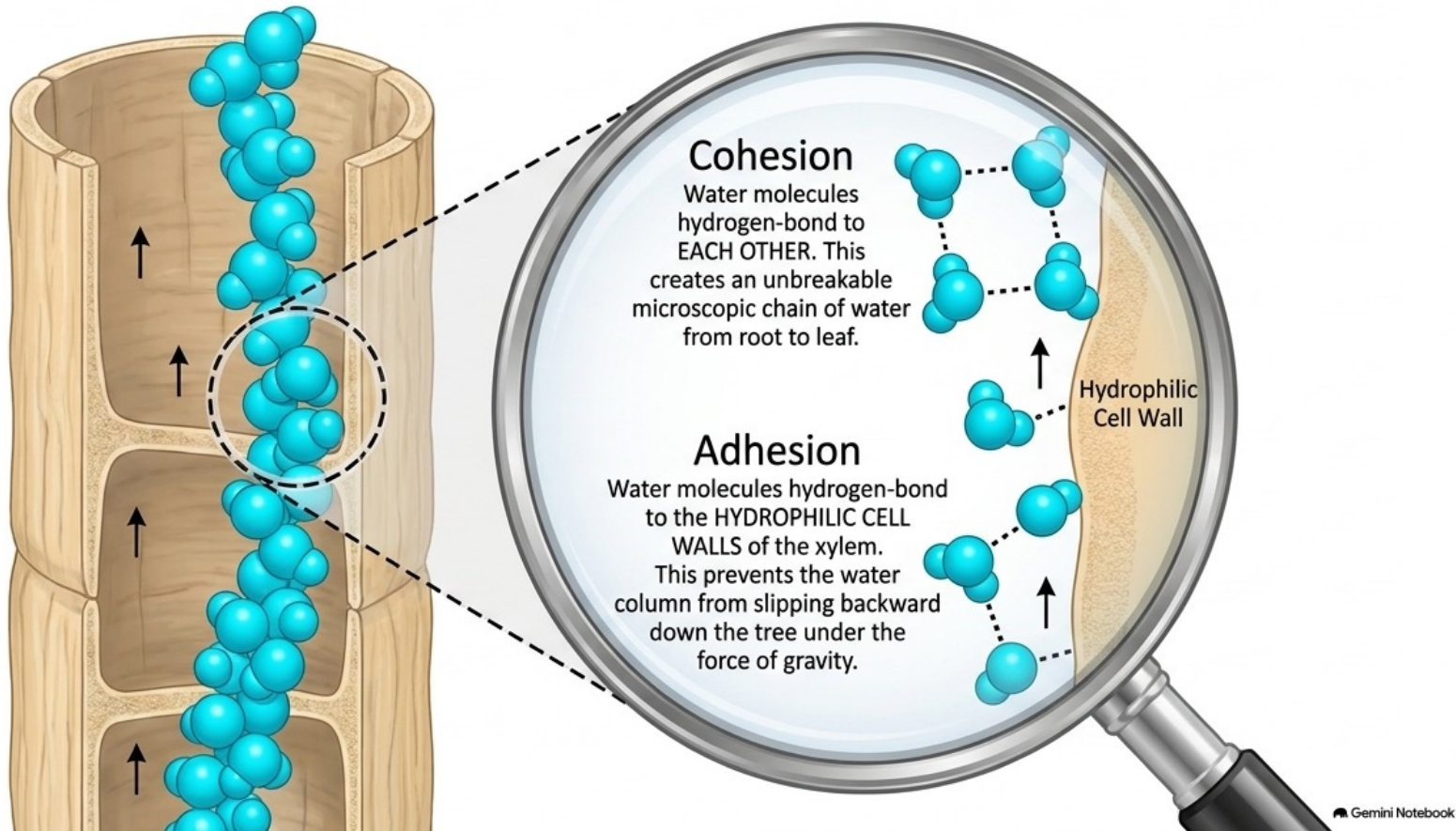


Transpiration Pull: Pumping Water Without a Heart

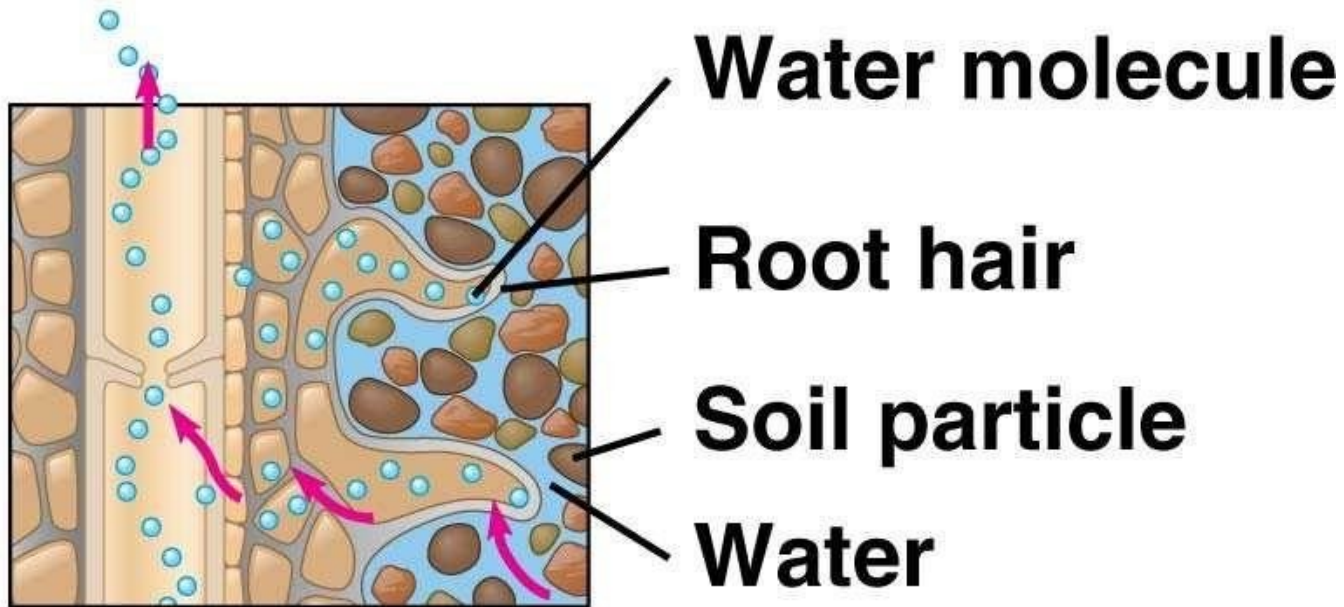


The Physics of the Xylem: Cohesion & Adhesion

Why the water column doesn't snap under 100-foot tension.



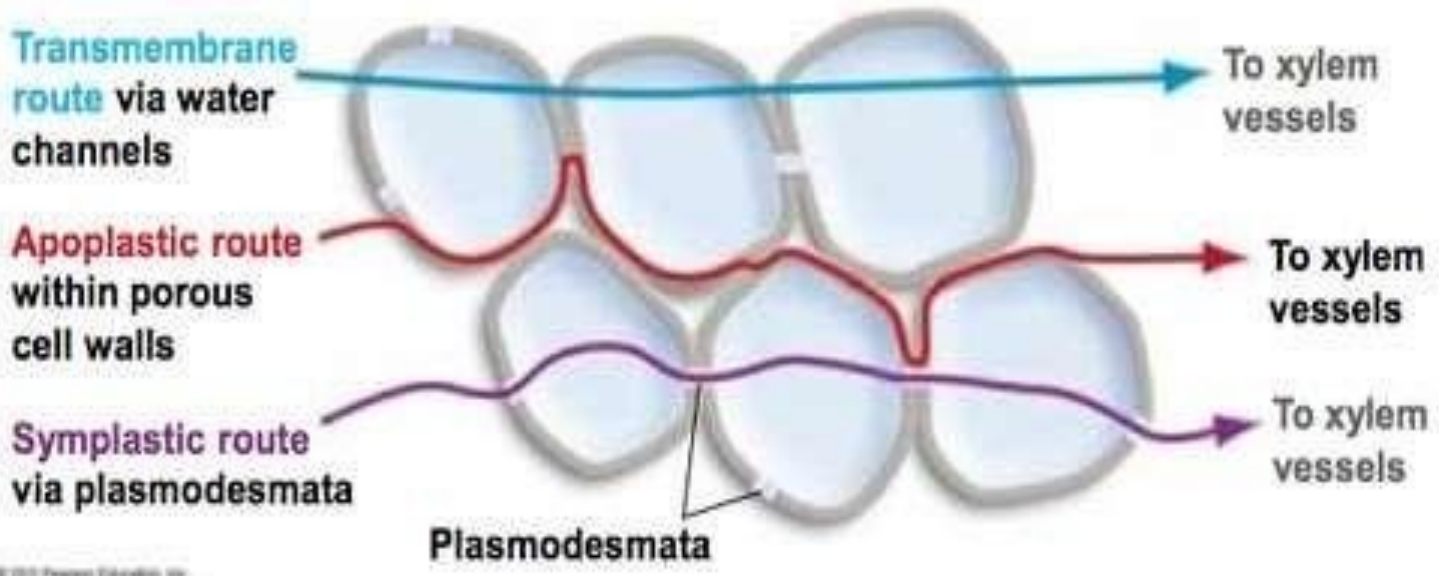
Root Pressure



Water uptake from soil

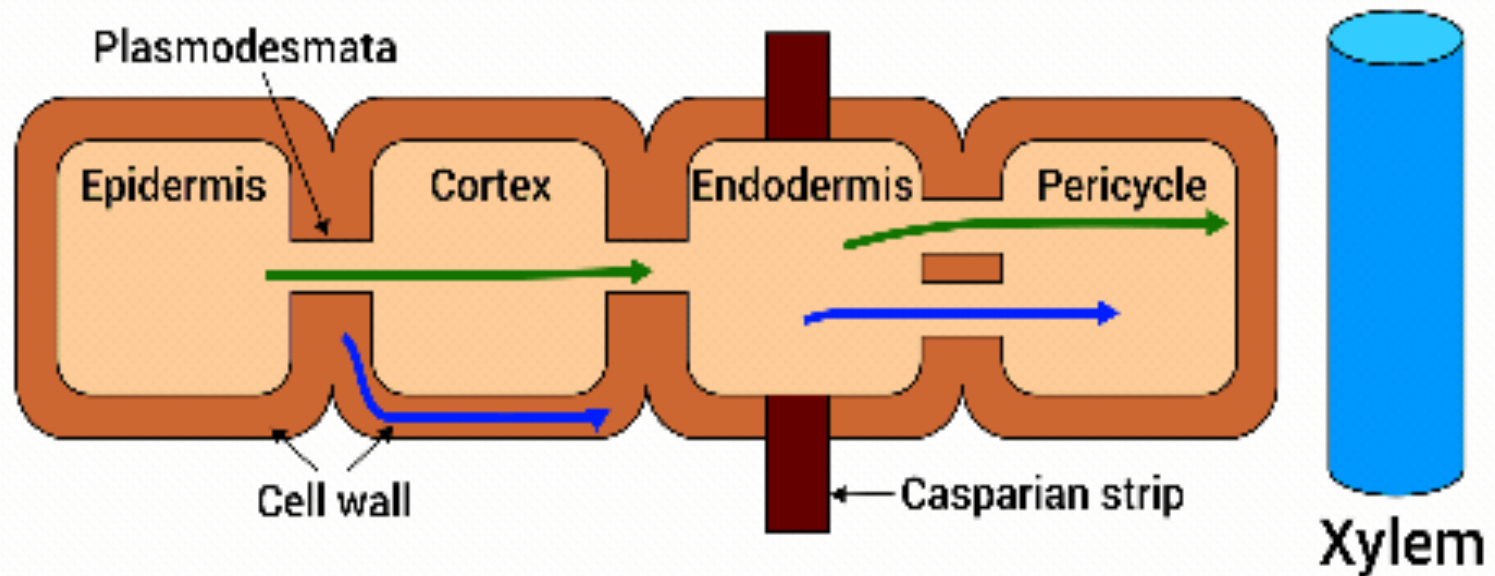
Routes of Water Transport

(a) Water travels from root hairs to xylem via three routes.



- Water and minerals can travel through a plant by one of three routes:
 - a) Transmembrane route:**
 - a) Out of one cell, across a cell wall, and into another cell
 - b) Via the **symplast**
 - a) Continuum of cytoplasm connected by plasmodesmata
 - c) Along the **apoplast**
 - a) Everything external to plasma membrane
 - Is the continuum of cell walls plus extracellular spaces

Movement of Water from Root Hair to Xylem



→ Symplastic pathway

→ Apoplastic pathway

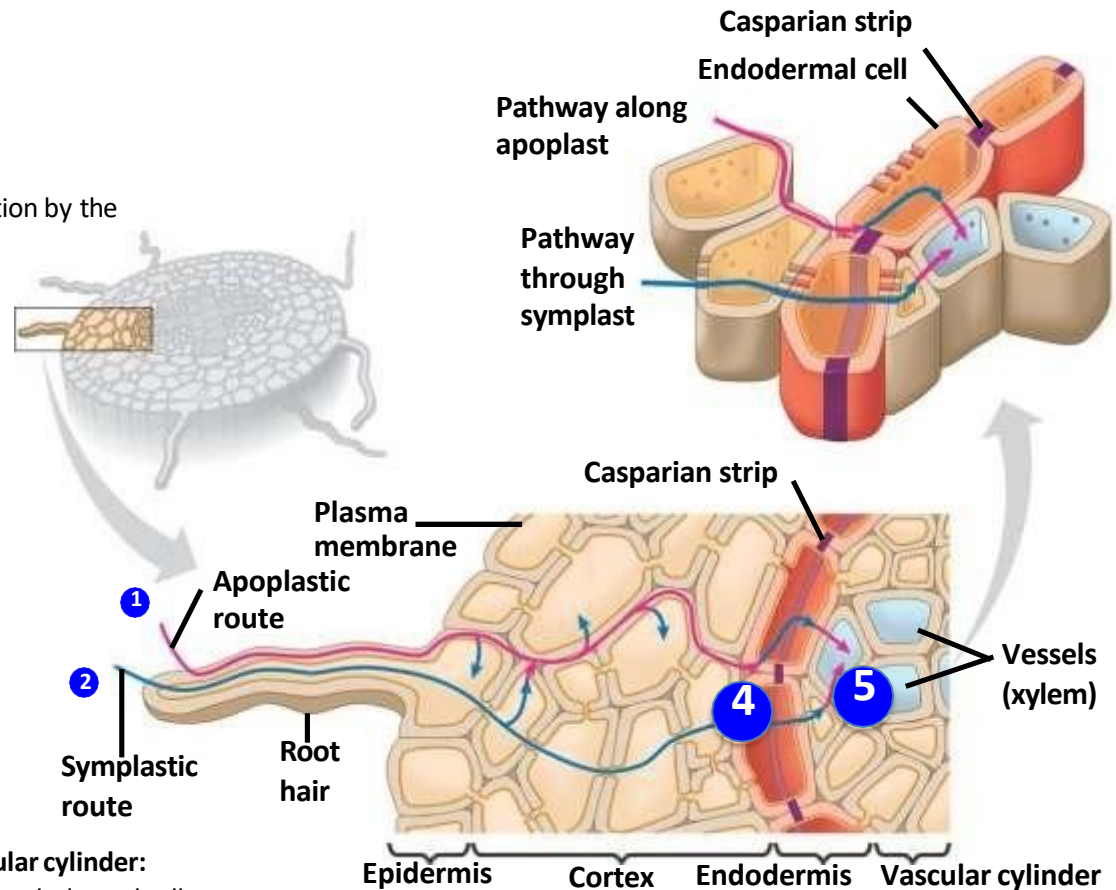
Transport of Water from Root Hair to Xylem

1 Apoplastic Route: Uptake of soil solution by the hydrophilic walls of root hairs provides access to the apoplast. Water and minerals can then soak into the cortex along this matrix of walls.

2 Symplastic route: Minerals and water that cross the plasma membranes of root hairs enter the symplast.

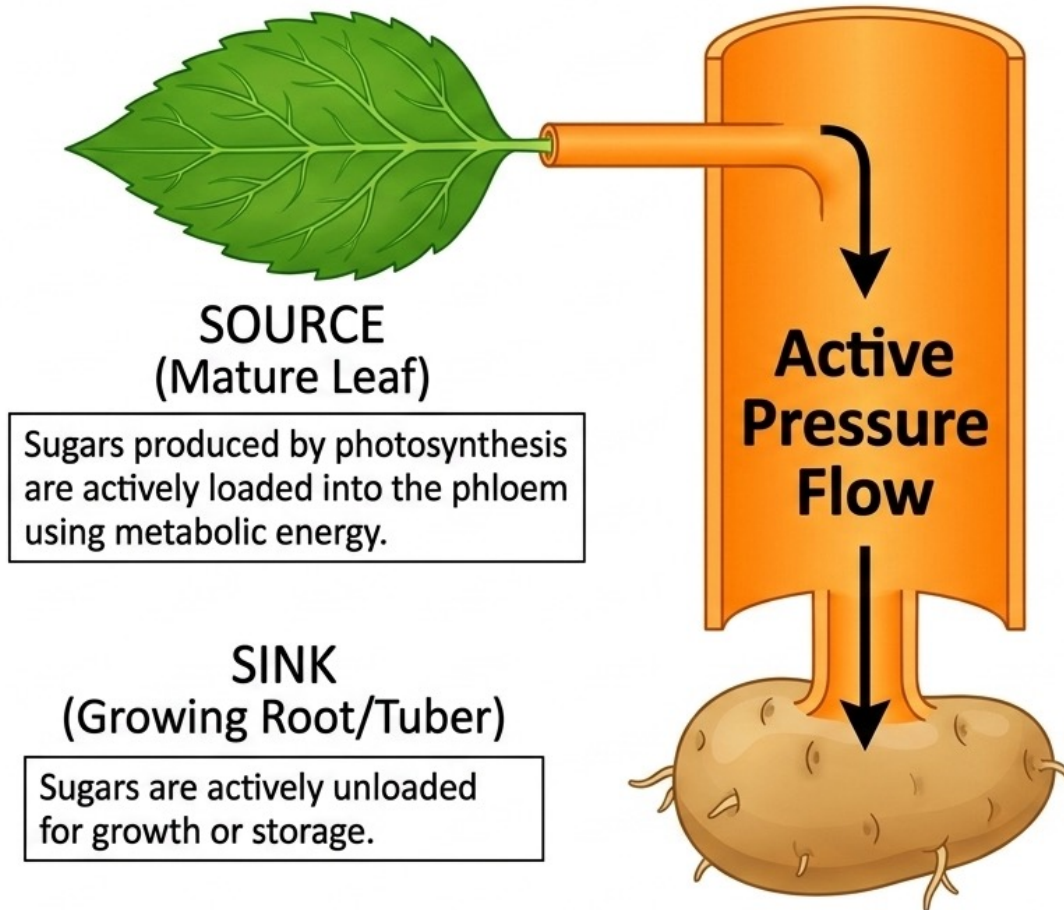
3 Transmembrane route: As soil solution moves along the apoplast, some water and minerals are transported into the protoplasts of cells of the epidermis and cortex and then move inward via the symplast.

4 The endodermis: controlled entry to the vascular cylinder: Within the transverse and radial walls of each endodermal cell is the Casparian strip, a belt of waxy material (purple band) that blocks the passage of water and dissolved minerals. Only minerals already in the symplast or entering that pathway by crossing the plasma membrane of an endodermal cell can detour around the Casparian strip and pass into the vascular cylinder.



5 Endodermal cells and also parenchyma cells within the vascular cylinder discharge water and minerals into their walls (apoplast). The xylem vessels transport the water and minerals upward into the shoot system.

Phloem Translocation: The Internal Economy

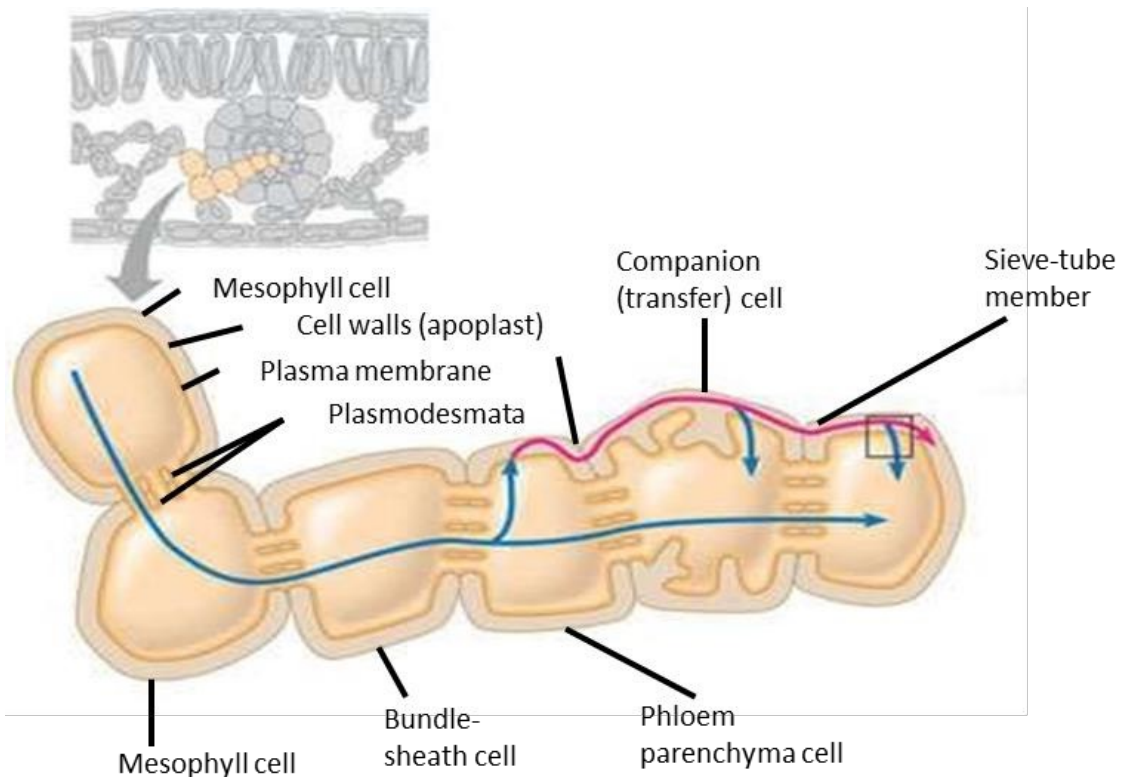


“ If the Xylem is a passive vacuum tube, the Phloem is an **active pressurized delivery network**. ”

Translocation

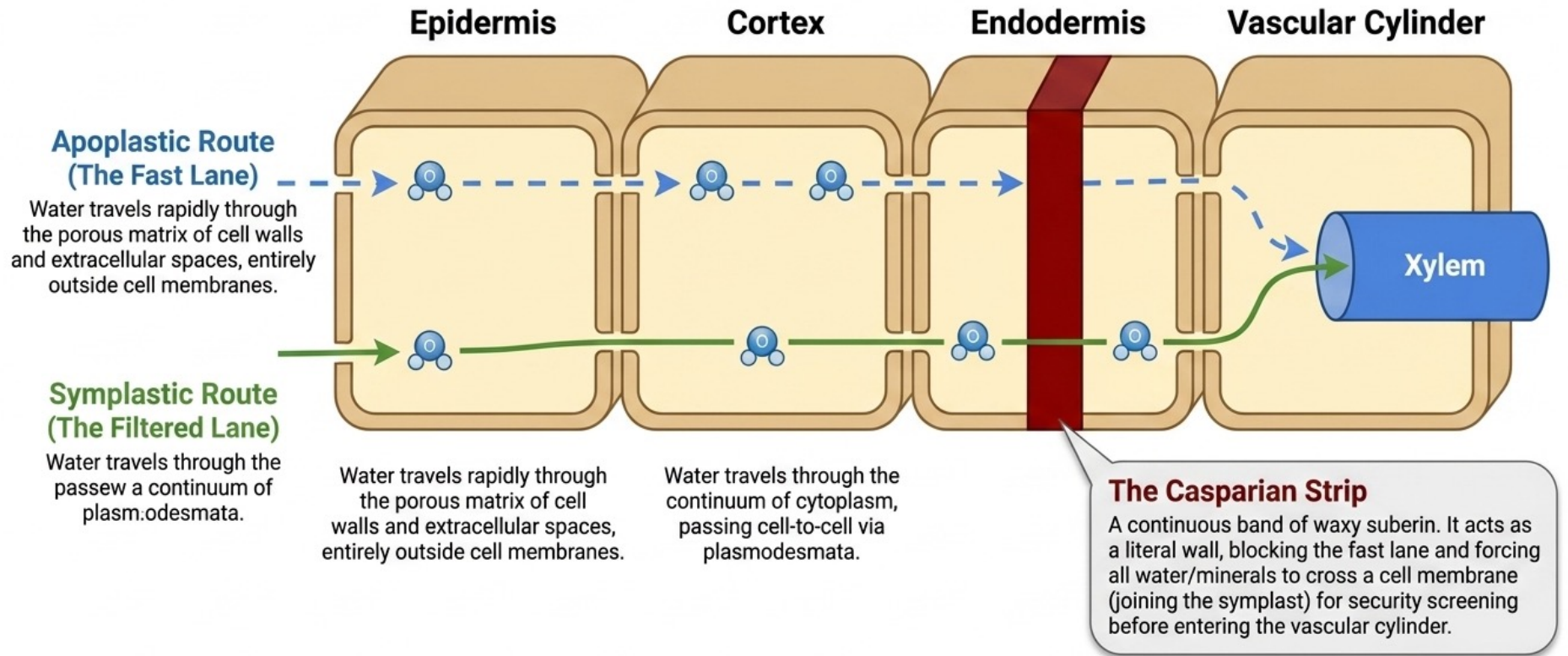
- Sugar must be loaded into sieve-tube members before being exposed to sinks
 - Active transport
- In many plant species, sugar moves by symplastic and apoplastic pathways

Sucrose manufactured in mesophyll cells can travel via the **symplast** (blue arrows) to sieve-tube members. In some species, sucrose exits the symplast (red arrow) near sieve tubes and is actively accumulated from the apoplast by sieve-tube members and their companion cells.



The Root Security Checkpoint

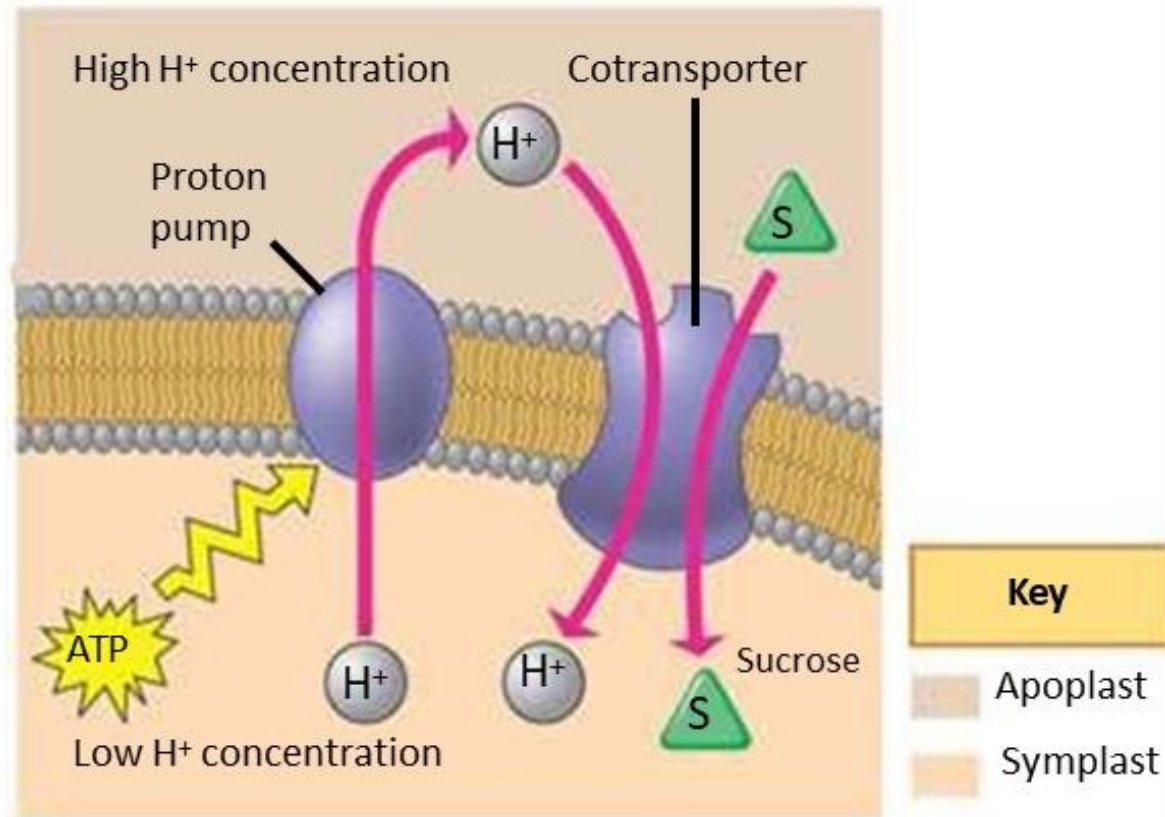
Controlling what enters the central plumbing.



Translocation

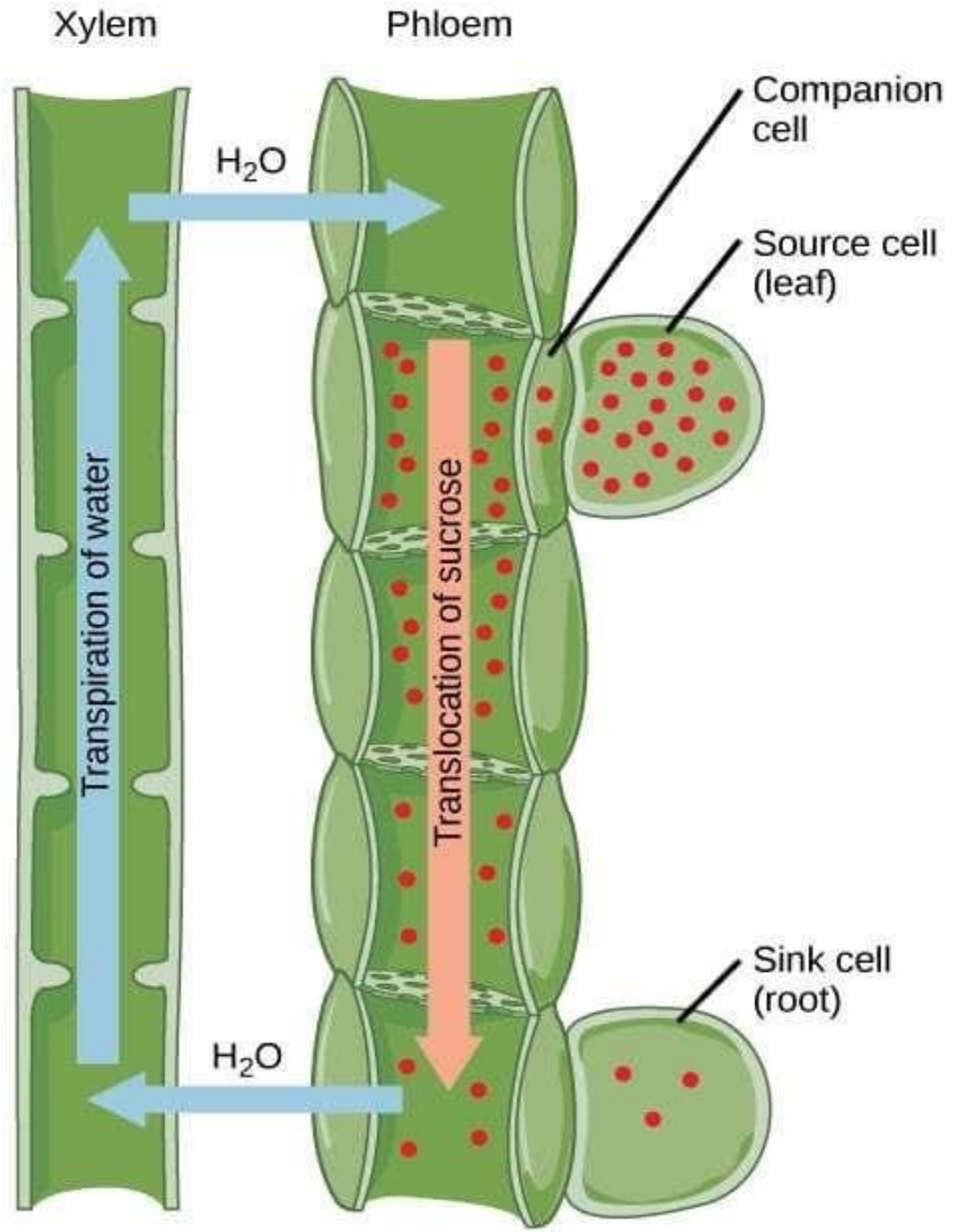
- In many plants
 - Phloem loading requires active transport
- Proton pumping and cotransport of sucrose and H^+
 - Enable the cells to accumulate sucrose

A chemiosmotic mechanism is responsible for the active transport of sucrose into companion cells and sieve-tube members. Proton pumps generate an H^+ gradient, which drives sucrose accumulation with the help of a cotransport protein that couples sucrose transport to the diffusion of H^+ back into the cell.



Translocation

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SUMMARY

- Plants have a hierarchy of organs, tissues and cells
- Primary growth lengthen roots and shoots
- Secondary growth increase the diameter of stems and roots in woody plants
- Different mechanism transport substances over short and long distances
- Transpiration transports water and mineral from roots to shoots via the xylem
- Rate of transpiration is controlled by the stomata
- Sugars are transported from sources to sinks via the phloem



THANK YOU :)